

DUALITY AND RECURSION IN DYNAMIC GAMES

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ABSTRACT. This paper develops dual recursive methods to study dynamic games with imperfect monitoring. It begins by defining two kinds of perfect public equilibrium that exploit communication differently, then formulates sequence problems that realize their welfare as linear programs. Their value functions are equated to maximal fixed points of corresponding functional equations defined by convex programs, with implied dual functional equations amenable to computation. In games with frequent actions, this yields welfare optimality (HJB) equations that relate the curvature of welfare to welfare itself. I apply these results to the repeated Prisoners' Dilemma, optimal dynamic contracting and approximate equilibrium.

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1. INTRODUCTION

Understanding dynamic behavior is essential in economics, with boundless applications spanning collusion, innovation, contracting and much beyond. To address such applications quantitatively, economists must resort to models of dynamic strategic behavior. However, even with a modicum of complexity, these models must be solved numerically, which requires reliable computational methods. Even though useful analytical and computational tools have been proposed in the literature, their applicability and robustness can be limited. A goal of this paper is to fill this gap.

In this paper, I make four contributions. First, I extend dynamic programming tools to games and cover a wider range of strategic settings in a unified way. Secondly, I derive a non-recursive foundation for these tools, which helps to clarify the structure of incentive dynamics. Thirdly, I propose and apply computational methods implied by this approach. Finally, I compare two computationally tractable equilibrium notions for dynamic games that differ in the nature of communication among players.

1.1. Overview of the Paper

The paper begins by defining two notions of equilibrium, perfect public randomization equilibrium (ppRE)—think pure strategy equilibrium with public randomization—and perfect public correlated equilibrium (ppCE)—where a mediator makes ephemerally secret recommendations. (Of course, for recommendations to be secret, they must also be random.) The paper’s first main result is to describe equilibrium outcomes of dynamic games as well-behaved planning problems. [Theorem 3.1](#) below characterizes these equilibria as linear programs called *primal sequence problems*.¹ The main tools employed in this paper are, of course, duality and recursion. From the primal, a *dual sequence problem* emerges naturally. [Theorem 3.2](#) shows that the values of the primal and this dual coincide. (We will return to this dual later.) The primal sequence problem yields a *primal functional equation* that exploits recursion. [Theorem 4.1](#) shows that its maximal fixed point coincides with the value function of the primal sequence problem.² The primal functional equation is a well-behaved finite-dimensional convex program, which naturally lends itself to duality.³ Thus, a *dual functional equation* emerges with the same value as its primal ([Theorem 4.2](#)).

¹Obtaining a linear programming formulation is not immediate; it requires careful consideration of off-path play and transversality conditions. For this reason, our preamble to the theorem is lengthy.

²The primal functional equation is not a contraction, and it generally has multiple fixed points.

³Obtaining a convex programming formulation is not immediate; it requires careful definition of choice variables to ensure that all nonlinearities are convex-like and due to recursion.

Next, the paper finds sufficient conditions for existence of solutions to the optimization problem defining the dual functional equation. Under these conditions, it is possible to ‘unfold’ the dual functional equation in order to construct candidate solution to the dual sequence problem. In general, though, this candidate is not feasible for the dual. At this point, a new duality relationship between variables and multipliers is determined that preserves duality while accommodating a larger space of multipliers that do include the candidate solution. In this larger space, [Theorem 4.3](#) shows that the dual functional equation’s unfolded solution solves the dual sequence problem.

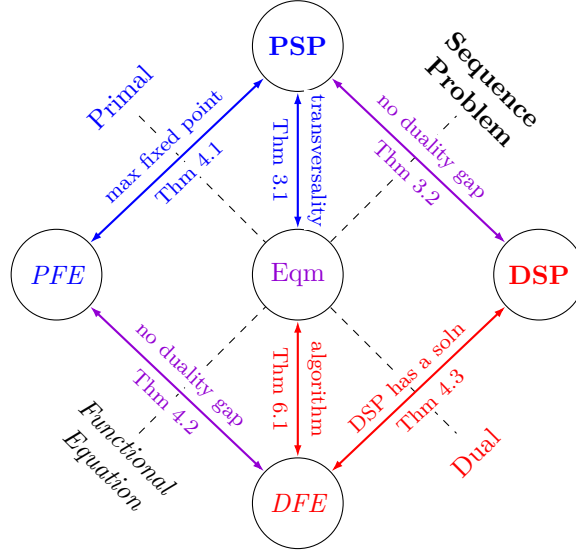


FIGURE 1. Roadmap of the paper

The paper then offers intuition for strategic and analytical differences between ppRE and ppCE. We also point out the prevalence of ‘kinks’ and ‘flats’ in the primal and dual value functions presented—equilibrium payoff sets and their support functions. Primal flats (and dual kinks) represent lotteries over continuation utility profiles—a concavification of equilibrium payoffs—whereas dual flats (and primal kinks) represent lotteries over continuation welfare weights affecting equilibrium dynamics.

The paper ends with applications and extensions. First, a computational algorithm is shown to converge to a solution ([Theorem 6.1](#)). Our first application concerns the repeated Prisoners’ Dilemma. There, we characterize the equilibrium payoff set for ppRE and ppCE with imperfect monitoring, with kinks and flats emerging in equilibrium. The primal flats reflect sudden drastic changes in players’ continuation values, arguably due to a lack of pairwise identifiability. However, these flats are captured by finding kinks in the dual functional equation. It is also seen how ppCE improves the scope for mutual cooperation.

Our next application is to optimal dynamic contracting. There, we compute the utility possibility frontier between a principal and a risk-averse agent with a dual functional equation. This allows us to automatically determine optimal lotteries for the principal via primal flats, which occur near endogenous retirement boundaries. We then extend the basic dynamic moral hazard model to include endogenous costly monitoring. This allows us to answer questions like when and how much an agent should be monitored. In our model, the agent is not monitored as punishment or to guarantee compliance. Rather, the agent is monitored closely to prevent him from getting fired because firing the agent is costly for the principal—monitoring reduces the probability of firing. The strategic difference between ppCE and ppRE in this economy is that the principal can monitor in secret—hence seldom—in ppCE, whereas she must monitor publicly in ppRE. In ppCE, monitoring can be more gradual to economize on monitoring costs.

Our last application uses [Theorem 4.3](#) to offer an equilibrium robustness result in terms of convergence of approximate equilibrium to equilibrium. A solution to the dual sequence problem implies that the primal value function has bounded steepness. Thus, welfare is Lipschitz in errors of incentive compatibility. The paper ends by extending the dual functional equation to games with frequent actions, games with private signals and stochastic games. Proofs are to be found in an online appendix.

1.2. Location in the Literature

This paper connects with several strands of literature. On the one hand, it extends duality techniques from work on recursive contracts ([Marcet and Marimon, 2019](#); [Pavoni et al., 2018](#); [Marimon and Werner, 2021](#); [Bloise and Siconolfi, 2022](#)) to the setting of dynamic games, in three main ways. First, the above recursive contracts papers work with contractive Bellman operators, inconsistent with many dynamic games. Secondly, in our model, the planning problem incorporates dynamic strategic lotteries as choice variables—in line with the spirit of [Phelan and Townsend \(1991\)](#). The recursive contracts approach uses deterministic allocations in (a version of) ℓ_∞ together with a norm-based Slater condition to find dual variables in ℓ_1 ,⁴ as opposed to the unruly ba (the norm dual of ℓ_∞). However, with dynamic lotteries, probabilities along the event tree deteriorate geometrically, and a uniform Slater condition cannot apply.⁵ Thirdly, we fill the void left by an inapplicable Slater condition with a dual solution in weighted ℓ_1 space ([Theorem 4.3](#))—still much less unruly than ba .

⁴This follows the logic of [Dechert \(1995\)](#), who extends the original approach by [Bewley \(1972\)](#).

⁵In part, this is because our primal sequence problem does not choose conditional probabilities directly. [Section 2.3](#) discusses why; [Section 5.3](#) explains how their Slater condition fails as a result.

This paper also builds on and extends existing recursive algorithms for dynamic games (Abreu et al., 1990; Fudenberg et al., 1994; Sannikov, 2007) by permitting the more general public correlated equilibrium, analyzing its sequence problem foundation, and developing a unified approach that applies to general games with perfect or imperfect monitoring, with or without full support. Also, the fundamental variable of choice in our dual approach is not so much continuation values (Spear and Srivastava, 1987), but rather welfare weights (Lucas and Stokey, 1984).

This paper also contributes to work on equilibrium computation. Abreu et al. (1990) characterized equilibrium payoffs as the maximal fixed point of a set-valued operator; Judd et al. (2003) implemented their algorithm numerically. Abreu et al. (2020) argued that this can be computationally intractable, but offered an improvement for pure-strategy equilibria of games with perfect monitoring inspired by Abreu and Sannikov (2014). This allowed them to describe equilibrium payoffs as the unique fixed point of a set-valued operator shown to be a contraction amenable to numerical implementation. Their algorithm does not apply to games with imperfect monitoring, or to non-pure-strategy equilibria. Ours does, and applies to non-contractions, too. The computational algorithm due to Sannikov (2007) for repeated games with Brownian information is also extended in this paper beyond games with product structure or pairwise identifiability, as well as by analyzing ppCE.

The notion of perfect public correlated equilibrium employed in this paper is based on and extends Tomala (2009) and Rahman (2014) as follows.⁶ In line with strongly sequential communication equilibrium due to Gerardi and Myerson (2007), we impose rational behavior after players' deviations but not after deviations by the mediator. In other words, rational play is imposed after otherwise impossible signals but not after impossible recommendations. This is a key insight that permits expressing the primal sequence problem of maximizing equilibrium welfare as a linear program.

Finally, this paper contributes to the literature via its other applications, too. Our dual approach to optimal contracting differs from and complements Sannikov (2008), as is the analysis of costly dynamic monitoring in relation to work such as Georgiadis and Szentes (2020), where monitoring must end, or Wong (2023), where wages are fixed. Our comparison of ppRE and ppCE can be viewed as a robustness check on communication protocols;⁷ Theorem 4.3 extends Fudenberg and Levine (1986) and Mailath et al. (2005) with new results on convergence of approximate equilibrium.

⁶See also related work by Rahman and Obara (2010) and Rahman (2012).

⁷Recently, Vong (2025) characterized optimal mediation dynamics in a stylized moral hazard game.

2. MODEL

In this section, we define a repeated game and two public equilibrium concepts.

2.1. Repeated Game

Let $\Gamma = (N, A, u)$ be a finite normal form game, with $N = \{1, \dots, n\}$ a finite set of players, $A = A_1 \times \dots \times A_n$ a finite space of action profiles and $u \in \mathbb{R}^{N \times A}$ mapping every action profile $a \in A$ to a payoff flow $u_i(a)$ for every player $i \in N$. Let Γ be the stage game of a repeated game with imperfect public monitoring. Periods are indexed by $t \in \mathbb{N}$. In every period t , an action profile $a_t \in A$ is played and a public signal z_t is drawn from a finite set Z with probability $\pi(z_t|a_t)$, where $\sum_z \pi(z|a) = 1$ for each $a \in A$. The public signal is IID over time conditional each action profile.

If $(a_1, a_2, \dots) \in A^{\mathbb{N}}$ is a sequence of action profiles and $a_t \in A$ is played in period t , each player i 's lifetime utility equals the normalized discounted sum of flow payoffs

$$(1 - \delta) \sum_{t \geq 1} \delta^{t-1} u_i(a_t)$$

for some discount factor $\delta \in [0, 1)$. Lotteries are evaluated using expected utility.

Every period t elapses over five stages: (i) a disinterested mediator recommends action a_{it} in A_i to each player i , (ii) each i simultaneously and independently chooses own behavior b_{it} in A_i , (iii) a public signal z_t realizes, (iv) the recommendation profile a_t is publicly announced, and, finally, (v) the period ends, giving way to the next period, $t + 1$. A *t-public history* is a list $h_{t-1} = (a_1, z_1, \dots, a_{t-1}, z_{t-1})$ of $t - 1$ recommendation profiles and signal realizations;⁸ a *public history* is a *t-public history* for some t . The spaces of all *t-public* and public histories are denoted by $H_{t-1} = (A \times Z)^{t-1}$ and $H = \bigcup_t H_{t-1}$. A *node* is any (a_t, h_{t-1}) such that $a_t \in A$ and $h_{t-1} \in H$. Each player i chooses whether or not to obey a recommendation at each node.

A *public correlated behavior strategy* is any $B \in (\Delta A)^H$,⁹ where $B(a_t|h_{t-1})$ is the probability that a_t is recommended given h_{t-1} . For any such B , the probability of node (a_t, h_{t-1}) entails multiplying conditional recommendation and signal probabilities:

$$P(a_t, h_{t-1}) = B(a_t|h_{t-1}) \prod_{s=1}^{t-1} \pi(z_s|a_s) B(a_s|h_{s-1}) = \prod_{s=1}^t B(a_s|h_{s-1}) \prod_{s=1}^{t-1} \pi(z_s|a_s),$$

where $h_{s-1} \leq h_{t-1}$ for all $s \leq t$ (i.e., $h_{t-1} = (h_{s-1}, a_s, z_s, \dots, a_{t-1}, z_{t-1})$).

⁸Despite minor notational abuse, we will label a *t-public history* h_{t-1} by both $(a_1, z_1, \dots, a_{t-1}, z_{t-1})$ and (a^{t-1}, z^{t-1}) . Let the 1-public history be null: $h_0 = (a^0, z^0) = (0, 0)$ and $H_0 = \{(0, 0)\}$.

⁹ $\Delta A = \{x \in \mathbb{R}_+^A : \sum_a x(a) = 1\}$ is the unit simplex; $(\Delta A)^H$ is the set of functions from H to ΔA .

Thus, the probability of each node decomposes into a product of endogenous (the B 's) and exogenous (the π 's) probabilities. Multiplying each $B(a_s|h_{s-1})$ along a public history into $C(a_t; h_{t-1}) = \prod_{s=1}^t B(a_s|h_{s-1})$ gives $P(a_t, h_{t-1}) = C(a_t; h_{t-1}) \prod_{s=1}^{t-1} \pi(z_s|a_s)$.¹⁰ Conversely, we may define $C(a_t; h_{t-1})$ as a primitive, independently of B , as follows.

DEFINITION 2.1. A *public correlated strategy* is any $C \in \mathbb{R}_+^{A \times H}$ with $\sum_a C(a; h_0) = 1$ and $\sum_a C(a; h_{t-1}, a_t, z_t) = C(a_t; h_{t-1})$ for all (h_{t-1}, a_t, z_t) .

The first condition in [Definition 2.1](#) requires that recommendation probabilities add up to one; the second that they stay consistent over time, so $\sum_{a_t} C(a_t; a^{t-1}, z^{t-1}) = 1$ for all z^{t-1} . When C equals the product of B 's as above, it clearly satisfies these two conditions, since $\sum_a B(a|h_{t-1}) = 1$ for all t . On the other hand, if $C(a_t; h_{t-1}) > 0$ then, by [Definition 2.1](#), $C(a_s; h_{s-1}) > 0$ for all $h_{s-1} \leq h_{t-1}$. Therefore, C induces the public correlated behavior strategy given by $B(a_{s+1}|h_s) = C(a_{s+1}; h_s)/C(a_s; h_{s-1})$ for all (a_{s+1}, h_s) . The key difference between B and C is that B defines behavior even conditional on recommendations with zero probability, whereas C does not.

The following notation will be useful. If $s \leq t$ and $h_{s-1} \leq h_{t-1} = (a^{t-1}, z^{t-1})$, let

$$P(a_t, h_{t-1}; h_{s-1}) = C(a_t; h_{t-1}) \prod_{r=s}^{t-1} \pi(z_r|a_r) = \frac{P(a_t, h_{t-1})}{\prod_{r=1}^{s-1} \pi(z_r|a_r)}$$

be the probability of (a_t, h_{t-1}) divided by the probability of z^{s-1} conditional on a^{s-1} .

2.2. Continuation Values

Given a public correlated strategy C , player i 's expected utility flow in period t equals

$$E[u_i(C_t)] = \sum_{a \in A} u_i(a) \sum_{h_{t-1}} P(a, h_{t-1}),$$

where C_t is the restriction of C to t -public histories. Define

$$E[u_i(C_t); h_{s-1}] = \sum_{a \in A} u_i(a) \sum_{h_{t-1} \geq h_{s-1}} P(a, h_{t-1}; h_{s-1})$$

to be player i 's expected utility flow at t , starting from h_{s-1} , divided by the probability of z^{s-1} conditional on a^{s-1} . To interpret this definition, observe that

$$E[u_i(C_t)|h_s] = \frac{E[u_i(C_t); h_s]}{C(a_s; h_{s-1})}$$

when $C(a_s; h_{s-1}) > 0$, where $E[u_i(C_t)|h_s]$ is the conditional expectation of $u_i(a_t)$ given h_s . If $s = t - 1$ then $E[u_i(C_t); h_{t-1}] = \sum_a u_i(a) P(a, h_{t-1}; h_{t-1}) = \sum_a u_i(a) C(a; h_{t-1})$.

¹⁰Therefore, $C(a_t; h_{t-1}) = P(a_t, h_{t-1}) / \prod_{s=1}^{t-1} \pi(z_s|a_s)$ whenever $\prod_{s=1}^{t-1} \pi(z_s|a_s) > 0$.

LEMMA 2.1. *If C is a public correlated strategy, player i 's lifetime utility $V_i(h_0)$ linearly decomposes into current flow payoffs and probability-weighted continuation values:*

$$V_i(h_0) = (1 - \delta) \sum_{t \geq 1} \delta^{t-1} \mathbb{E}[u_i(C_t)] = (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_0) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z),$$

where $V_i(a, z) = (1 - \delta) \sum_{t \geq 2} \delta^{t-2} \mathbb{E}[u_i(C_t); a, z]$.

According to Lemma 2.1, player i 's lifetime utility entails two terms. The first term, $(1 - \delta) \sum_a u_i(a) C(a; h_0)$, is i 's expected flow payoff from $C(\cdot; h_0)$ in period 1. The second, $\delta \sum_{(a, z)} \pi(z|a) V_i(a, z)$, is i 's discounted expected continuation value. Even though the probability $C(a; h_0)$ is not in view in this term, it is implicitly embedded in $V_i(a, z)$ via the equation $\mathbb{E}[u_i(C_t); a, z] = \mathbb{E}[u_i(C_t)|a, z] C(a; h_0)$ for all $t > 1$. Thus, $V_i(a, z)$ is i 's probability-weighted continuation value after the public history (a, z) .

Lemma 2.1 extends easily: for any public history h_{s-1} , player i 's probability-weighted lifetime payoff $V_i(h_{s-1})$ from h_{s-1} on decomposes similarly into

$$(1) \quad V_i(h_{s-1}) = (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_{s-1}) + \delta \sum_{(a, z)} \pi(z|a) V_i(h_{s-1}, a, z),$$

where $V_i(h_{s-1}, a, z) = (1 - \delta) \sum_{t \geq s+1} \delta^{t-s-1} \mathbb{E}[u_i(C_t); h_{s-1}, a, z]$. A useful implication of (1) is that each $V_i(h_{s-1})$ is *linear* in the public correlated strategy $C(a; h_{s-1})$ and subsequent probability-weighted continuation values $V_i(h_{s-1}, a, z)$.

2.3. Perfect Public Equilibrium

Next, we define two notions of perfect public equilibrium studied in this paper.

DEFINITION 2.2. Public correlated strategy C is a *perfect public correlated equilibrium* (ppCE) if for each player i , public history h_{s-1} , recommendation a_i and action b_i ,

$$(2) \quad (1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) C(a; h_{s-1}) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(h_{s-1}, a, z) \leq 0,$$

where $\Delta u_i(a, b_i) = u_i(b_i, a_{-i}) - u_i(a)$ and $\Delta \pi(z|a, b_i) = \pi(z|b_i, a_{-i}) - \pi(z|a)$.

Definition 2.2 introduces our main solution concept. In ppCE, obeying the mediator's recommendations is more profitable than disobeying, assuming recommendations are kept secret until stage (iv) of each period. This solution concept exploits the linearity of correlated equilibrium and the recursivity of perfect public equilibrium.¹¹ It is close to but different from equilibrium in Tomala (2009) and Rahman (2014).

¹¹Equilibrium is defined with one-shot deviations, not fully dynamic ones. By the one-shot deviation principle, this standard simplification incurs no loss of generality. Details are available on request.

Conditioning on z^{t-1} but not a^{t-1} when describing public correlated strategies ensures that incentive constraints are linear in C and V as well as valid after some impossible histories. There are two reasons for a public history h_t to occur that shouldn't, i.e., with $P(h_t) = 0$: (i) either the mediator made a recommendation with zero probability, or (ii) a public signal realized that shouldn't have (or both). Our formulation ignores (i), consistent with Gerardi and Myerson (2007). After a recommendation with zero probability, C turns to zero, hence so do subsequent continuation values, and incentive constraints become moot henceforth. On the other hand, C does not vanish after recommendations with positive probability and a signal with zero probability—precisely because $C(a_t; a^{t-1}, z^{t-1})$ conditions on the signal history.¹²

DEFINITION 2.3. C is a *perfect public randomization equilibrium* (ppRE) if for each player i , public history h_{s-1} , recommendation profile $a \in A$ and deviation $b_i \in A_i$,

$$(3) \quad (1 - \delta)\Delta u_i(a, b_i)C(a; h_{s-1}) + \delta \sum_{z \in Z} \Delta \pi(z|a, b_i)V_i(h_{s-1}, a, z) \leq 0.$$

Basically, ppRE are pure-strategy equilibria with a public randomization device. Clearly, every ppRE is a ppCE: a player, when asked to take some action a_i , does not know others' contemporaneous recommendations a_{-i} in (2) but does in (3).

3. DUALITY AND THE WELFARE POTENTIAL

In this section, we characterize the equilibria above via two 'sequence problems' that form mutually dual infinite-dimensional linear programs. We begin by introducing relevant spaces of primal and dual variables, then we derive the sequence problems.

3.1. Primal and Dual Processes

Let S be any finite set and recall that H is the set of public histories.

DEFINITION 3.1. Given $X \in \mathbb{R}^{S \times H}$, define 'primal semi-norms' $|X|_0 = \sum_{s \in S} |X(s; h_0)|$,

$$|X|_t = \sum_{a_1} \max_{z_1} \cdots \sum_{a_{t-1}} \max_{z_{t-1}} \sum_{a_t} \max_{z_t} \sum_{s \in S} |X(s; a^t, z^t)|$$

for all $t \in \mathbb{N}$, and the 'primal norm' $\|X\|_\infty = \sup_t |X|_{t-1}$, with corresponding space $\hat{\ell}_\infty^S = \{X \in \mathbb{R}^{S \times H} : \|X\|_\infty < +\infty\}$ of *primal processes*.¹³ (If $|S| = 1$, write simply $\hat{\ell}_\infty$.)

¹²Had the joint probability $P(a_t, h_{t-1})$ been the primitive variable, perfectness would have been lost. Had the conditional probability $P(a_t|h_{t-1})$ been the primitive instead, linearity would have been lost, since continuation values would multiply recommendation probabilities for incentive compatibility.

¹³It is easy to see that $|\cdot|_{t-1}$ is a semi-norm, $\|\cdot\|_\infty$ is a norm and $(\hat{\ell}_\infty^S, \|\cdot\|_\infty)$ is a Banach lattice.

To motivate [Definition 3.1](#), note first that $C \in \hat{\ell}_\infty^A$ whenever C is a public correlated strategy: $\sum_a C(a; h_0) = 1$ and $\sum_a C(a; h_{s-1}, a_s, z_s) = C(a_s; h_{s-1}) \geq 0$ for all s imply

$$|C|_{t-1} = \sum_{a_1} \max_{z_1} \cdots \sum_{a_{t-1}} \max_{z_{t-1}} \sum_{a \in A} |C(a; a^{t-1}, z^{t-1})| = 1$$

for all t , thus $\|C\|_\infty = \sup_t |C|_{t-1} = 1 < +\infty$.¹⁴ Moreover, $V \in \hat{\ell}_\infty^N$ acts as a transversality condition that prevents lifetime-utility bubbles, as the next result shows.

LEMMA 3.1. *If $V \in \mathbb{R}^{N \times H}$ satisfies (1) for some public correlated strategy C then*

$$V_i(h_{t-1}) = (1 - \delta) \sum_{\tau \geq t} \delta^{\tau-t} \mathbb{E}[u_i(C_\tau); h_{t-1}]$$

for all (i, h_{t-1}) if and only if $V \in \hat{\ell}_\infty^N$.

[Lemma 3.1](#) offers a key contribution of primal processes: $\hat{\ell}_\infty^N$ pins down just when recursively describing (probability-weighted) continuation values equals an expected discounted sum of utility flows without ‘bubble’ terms. Interestingly, spaces of primal processes are defined independently of u or π , so [Lemma 3.1](#) applies to every utility profile and signal distribution. Next, we define what will be dual variables.

DEFINITION 3.2. If $Y \in \mathbb{R}^{S \times H}$, define the ‘dual semi-norms’ $|Y|_0^* = \max_{s \in S} |Y(s; h_0)|$,

$$|Y|_t^* = \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} \max_{a_t} \sum_{z_t} \max_{s \in S} |Y(s; a^t, z^t)|$$

for all $t \in \mathbb{N}$, and the ‘dual norm’ $\|Y\|_1^* = \sum_t |Y|_{t-1}^*$, with its corresponding space $\hat{\ell}_1^{*S} = \{Y \in \mathbb{R}^{S \times H} : \|Y\|_1^* < +\infty\}$ of *dual processes*.¹⁵ (If $|S| = 1$, write simply $\hat{\ell}_1^*$.)

Primal and dual processes pair naturally with respect to the inner product

$$\langle X, Y \rangle = \sum_{t \in \mathbb{N}} \sum_{h_{t-1}} \sum_{s \in S} X(s; h_{t-1}) Y(s; h_{t-1}),$$

as [Lemma 3.2](#) below shows. E.g., $|\langle X, Y \rangle| < +\infty$ when X is a primal process and Y is a dual process, so we may interpret primal processes as legitimate dual variables, or shadow prices of constraints whose values are dual processes—and vice versa.

LEMMA 3.2. *Primal and dual processes form a dual pair with respect to $\langle \cdot, \cdot \rangle$,¹⁶ and the primal processes consist of all norm-continuous linear functionals on dual processes.*

¹⁴This primal norm naturally conditions on signals z_{s-1} but takes expectations of recommendations a_{s-1} . Adding z_{s-1} instead of maximizing would yield an unbounded norm for C , for instance.

¹⁵Again, it is easy to see that $|\cdot|_{t-1}^*$ is a semi-norm, $\|\cdot\|_1^*$ is a norm and $(\hat{\ell}_1^{*S}, \|\cdot\|_1^*)$ is a Banach lattice.

¹⁶Recall that two real vector spaces form a *dual pair* with respect to an inner product if it is real-valued and semi-norms induced by vectors from each space separate the points of the other space.

3.2. Primal Sequence Problem

The linear inequalities (2) and (3) characterize ppCE and ppRE. We lay them all out below for a sequence problem that is a linear program for each solution concept.

DEFINITION 3.3. *Primal sequence problem* for ppCE at welfare weights $\lambda \in \mathbb{R}^N$:

$$\begin{aligned} \underline{W}(\lambda) &= \sup_{C \geq 0, V} \sum_{i \in N} \lambda_i V_i(h_0) \quad \text{s.t.} \quad \sum_{a \in A} C(a; h_0) = 1, \\ &\sum_{a \in A} C(a; h_{t-1}, a_t, z_t) = C(a_t; h_{t-1}) \quad \forall (h_{t-1}, a_t, z_t), \\ V_i(h_{t-1}) &= (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_{t-1}) + \delta \sum_{(a, z)} \pi(z|a) V_i(h_{t-1}, a, z) \quad \forall (i, h_{t-1}), \\ (1 - \delta) \sum_{a-i} \Delta u_i(a, b_i) C(a; h_{t-1}) + \delta \sum_{(a-i, z)} \Delta \pi(z|a, b_i) V_i(h_{t-1}, a, z) &\leq 0 \quad \forall (i, h_{t-1}, a_i, b_i), \end{aligned}$$

where $C \in \hat{\ell}_\infty^A$ and $V \in \hat{\ell}_\infty^N$ (see Definition 3.1). The primal sequence problem for ppRE is the same except the inequalities are replaced with (3) indexed by (i, h_{t-1}, a, b_i) .

The sequence problems above maximize welfare by choosing public correlated strategies subject to being an equilibrium. Welfare is the weighted sum of lifetime utilities at the beginning of the game. The first two sets of constraints are precisely those of Definition 2.1, thus establishing that C is a public correlated strategy. The next two sets of constraints are precisely (1) and (2) for ppCE, (1) and (3) for ppRE. Finally, $C \in \hat{\ell}_\infty^A$ holds automatically and $V \in \hat{\ell}_\infty^N$ is a transversality condition (Lemma 3.1).

THEOREM 3.1. *(C, V) is feasible in the primal sequence problem for ppCE if and only if C is a ppCE and $V_i(h_{s-1}) = (1 - \delta) \sum_{t \geq s} \delta^{t-s} \mathbb{E}[u_i(C_t); h_{s-1}]$ for all (i, h_{s-1}) . The feasible set is non-empty, weak* compact and convex, hence the equilibrium payoff set is a compact, convex subset of $\mathbb{R}^N$. Likewise for ppRE, except for non-emptiness.*

Clearly, the value function $\underline{W}$ is positively homogeneous ($\underline{W}(x\lambda) = x\underline{W}(\lambda)$ if $x > 0$) and, by revealed preference, convex. By Theorem 3.1, $\underline{W}$ equals the support function of the equilibrium payoff set.¹⁷ If this set is empty, $\underline{W}(\lambda) = -\infty$ for all $\lambda \in \mathbb{R}^N$.

¹⁷To see this, let $F \subset \mathbb{R}^N$ be a closed, convex set. Its *indicator* function, defined as $\delta_F(x) = 0$ if $x \in F$ and $+\infty$ otherwise, is a closed, convex function, where ‘closed’ means its epigraph $\{(x, \mu) \in \mathbb{R}^N \times \mathbb{R} : \delta_F(x) \leq \mu\}$ is a closed set. The *support* function of F , denoted δ_F^* , is the conjugate of δ_F :

$$\delta_F^*(p) = \sup_{x \in \mathbb{R}^N} \{p \cdot x - \delta_F(x)\} = \sup_{x \in F} p \cdot x.$$

Clearly, $\delta_F^*(0) = 0$ if $F \neq \emptyset$ and $-\infty$ otherwise. If $\hat{x} \in F$ then $\delta_F^*(p) \geq p \cdot \hat{x}$ for every $p \in \mathbb{R}^N$ by definition, so $\hat{x} \in \partial \delta_F^*(0)$, and if $\hat{x} \notin F$ then, by the Strict Separating Hyperplane Theorem, there exists $p \in \mathbb{R}^N$ and $\varepsilon > 0$ such that $p \cdot \hat{x} > \varepsilon > p \cdot y$ for all $y \in F$, so $p \cdot \hat{x} > \delta_F^*(p)$ and $\hat{x} \notin \partial \delta_F^*(0)$.

Otherwise, $\underline{W}(0) = 0$ and the *subdifferential* of $\underline{W}$ at λ ,

$$\partial \underline{W}(\lambda) = \{v \in \mathbb{R}^N : \mu \cdot v - \underline{W}(\mu) \leq \lambda \cdot v - \underline{W}(\lambda) \quad \forall \mu \in \mathbb{R}^N\},$$

equals the set of equilibrium payoff profiles that maximize welfare in the direction λ . Moreover, when $\lambda = 0$, the subdifferential $\partial \underline{W}(0) = \{v \in \mathbb{R}^N : \mu \cdot v \leq \underline{W}(\mu) \quad \forall \mu \in \mathbb{R}^N\}$ equals the set of all equilibrium lifetime payoff profiles, and the directional derivative of $\underline{W}$ at 0 in any direction $\lambda \neq 0$ is the maximum equilibrium welfare in that direction.

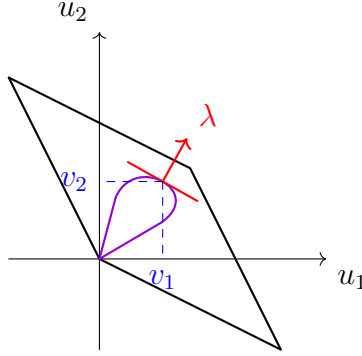


FIGURE 2. Welfare weights and extremal equilibrium payoffs

3.3. Dual Sequence Problem

This subsection derives the dual linear program of the primal sequence problem, offers an example to illustrate the dual, and establishes absence of a duality gap.

PROPOSITION 3.1. *The dual sequence problem for ppCE is the linear program below:*

$$\overline{W}(\lambda) = \inf_{Y \geq 0, \Lambda, W} W(h_0) \quad \text{s.t.} \quad \Lambda_i(h_0) = \lambda_i \quad \forall i \in N,$$

$$W(h_{t-1}) \geq (1 - \delta) \sum_{i \in N} [\Lambda_i(h_{t-1}) u_i(a) - Y_i(h_{t-1}, a_i) \Delta u_i(a)] + \sum_{z \in Z} W(h_{t-1}, a, z) \quad \forall (h_{t-1}, a),$$

$$\Lambda_i(h_{t-1}, a, z) = \delta [\Lambda_i(h_{t-1}) \pi(z|a) - Y_i(h_{t-1}, a_i) \Delta \pi_i(z|a)] \quad \forall (i, h_{t-1}, a, z),$$

where $W \in \hat{\ell}_1^*$, $\Lambda \in \hat{\ell}_1^{*N}$ and $Y \in \hat{\ell}_1^{* \Pi_i A_i \times A_i}$. For ppRE, $Y \in \hat{\ell}_1^{* \Pi_i A \times A_i}$ instead. Finally, the dual constraint set is non-empty, therefore $\overline{W}(\lambda) < +\infty$ for every $\lambda \in \mathbb{R}^N$.

The dual sequence problem picks dual processes to minimize welfare. The stochastic process of welfare, W , collects multipliers on public correlated strategy constraints. Those on value recursion, Λ , are the process of welfare weights with initial condition $\Lambda(h_0) = \lambda$. Multipliers on incentive constraints, Y , determine the law of motion for both welfare and welfare weights. The dual constraints on W have a public correlated strategy C as multipliers; the law of motion for welfare weights has V as multipliers.

EXAMPLE 3.1. Repeated Prisoners' Dilemma with two players, two signals:

	c_2	d_2		c_2	d_2
c_1	1, 1	-1, 2	c_1	1, 0	0, 1
d_1	2, -1	0, 0	d_1	0, 1	0, 1
Flow Payoffs			Signal Probabilities		

The bi-matrix on the left shows flow payoffs, u , and the right one (π) reveals two public signals, say g and b , such that g occurs if and only if both players cooperate, i.e., play $c = (c_1, c_2)$. Because defection is perfectly detectable, mutual cooperation is an equilibrium if players are sufficiently patient.¹⁸ Let us show this using the dual. Start with welfare weights $\lambda = (1, 1)$ and unfold the inequalities for welfare in the dual, focusing without loss on public histories of mutual cooperation and g signals:

$$\begin{aligned}
W(h_0) &\geq (1 - \delta) \sum_{(i,t)} \delta^{t-1} [\Lambda_i(c^{t-1}, g^{t-1}) u_i(c) - Y_i(c^{t-1}, g^{t-1}, c_i) \Delta u_i(c)] \\
&= (1 - \delta) [2 - Y_1(c_1) - Y_2(c_2) + \delta [2 + Y_1(c_1) + Y_2(c_2) - Y_1(c, g, c_1) - Y_2(c, g, c_2)] + \cdots] \\
&= (1 - \delta) \left[\frac{2}{1-\delta} + [Y_1(c_1) + Y_2(c_2)] \left(\frac{\delta}{1-\delta} - 1 \right) + [Y_1(c, g, c_1) + Y_2(c, g, c_2)] \delta \left(\frac{\delta}{1-\delta} - 1 \right) + \cdots \right] \\
&= 2 + (2\delta - 1) \sum_{(i,t)} \delta^{t-1} [Y_1(c^{t-1}, g^{t-1}, c_1) + Y_2(c^{t-1}, g^{t-1}, c_2)].
\end{aligned}$$

If $\delta \geq \frac{1}{2}$, $Y_i(c^{t-1}, g^{t-1}, c_i) = 0$ minimizes $W(h_0)$, so mutual cooperation obtains; if $\delta < \frac{1}{2}$, $Y_i(c^{t-1}, g^{t-1}, c_i)$ can be so large that this constraint is slack. By complementary slackness, its multiplier—the probability of mutual cooperation—equals zero.

The primal and dual sequence problems are infinite-dimensional linear programs, so absence of a duality gap is not guaranteed. Our next result establishes such absence.¹⁹ Its proof shows that the dual value function is subdifferentiable with respect to its right-hand side constraints and applies a classic result due to Gale (1967).

THEOREM 3.2 (NO DUALITY GAP). *There is no gap between the primal sequence problem and its dual: if the primal is infeasible, $\underline{W}(\lambda) = \overline{W}(\lambda) = -\infty$ for any welfare weights λ ; if the primal is feasible, for every optimal primal solution (C^*, V^*) ,*

$$\sum_{i \in N} \lambda_i V_i^*(h_0) = \underline{W}(\lambda) = \overline{W}(\lambda).$$

*Call this common value function $\underline{W} = \overline{W}$ the **welfare potential**.*

¹⁸A textbook demonstration of this is that a player's lifetime payoff from mutual cooperation is 1, whereas the payoff from a deviation followed by the worst possible punishment—mutual defection henceforth—pays $(1 - \delta)2 + \delta 0 = 2(1 - \delta)$. Cooperation is an equilibrium if $1 \geq 2(1 - \delta)$, or $\delta \geq \frac{1}{2}$.

¹⁹Weak duality—the value of the primal is bounded above by the value of the dual—follows easily.

4. RECURSIVE FORMULATION

We now turn from the sequence problems above to functional equations that characterize equilibrium recursively. This yields a computational approach via duality.

4.1. Primal Functional Equation

DEFINITION 4.1. Given welfare weights λ , the *primal functional equation* for ppCE is

$$\begin{aligned} W(\lambda) = \sup_{B \geq 0, V} \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] \quad \text{s.t.} \quad \sum_{a \in A} B(a) = 1, \\ (1 - \delta) \sum_{a \rightarrow i} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a \rightarrow i, z)} \Delta \pi(z|a, b_i) V_i(a, z) \leq 0 \quad \forall (i, a_i, b_i), \\ [B(a)W]^*(V(a, z)) \leq 0 \quad \forall (a, z), \end{aligned}$$

where $B \in \mathbb{R}^A$, $V \in \mathbb{R}^{N \times A \times Z}$ and

$$[B(a)W]^*(v) = \sup_{\mu \in \mathbb{R}^N} \{ \mu \cdot v - B(a)W(\mu) \}$$

is the convex conjugate of the function $B(a)W(\mu)$ of μ . The corresponding primal functional equation for ppRE is exactly the same as for ppCE except that the incentive constraints (2), indexed by (i, a_i, b_i) , are replaced with (3), indexed by (i, a, b_i) .

Definition 4.1 defines a family of welfare maximization problems subject to incentive compatibility and promise-keeping. $W(\lambda)$ is the maximal λ -weighted sum of payoffs from correlated strategies B and probability-weighted continuation values V that form a correlated equilibrium and satisfy the conjugate constraint above, equivalently

$$\sum_{i \in N} \mu_i V_i(a, z) \leq B(a)W(\mu) \quad \forall a \in A, z \in Z, \mu \in \mathbb{R}^N.$$

These constraints express the principle of optimality for games: any promised welfare must be attainable. They also ensure that $V_i(a, z) = 0$ when $B(a) = 0$, so incentive compatibility is void after a recommendation that isn't made. The maximization problem above is a well-behaved convex program, since $[B(a)W]^*(v)$ is a convex function of $(B(a), v)$,²⁰ and, as will be seen below, at a fixed point of the functional equation the constraint set is compact and convex, so *the supremum is attained*.

²⁰Indeed, if $p \in [0, 1]$, $B_0, B_1 \in \Delta(A)$, $v_0, v_1 \in \mathbb{R}^N$, $B_p = pB_1 + (1 - p)B_0$ and $v_p = pv_1 + (1 - p)v_0$,
 $[B_p(a)W]^*(v_p) = \sup_{\mu \in \mathbb{R}^N} \{ \mu \cdot v_p - B_p(a)W(\mu) \} = \sup_{\mu \in \mathbb{R}^N} \{ p[\mu \cdot v_1 - B_1(a)W(\mu)] + (1 - p)[\mu \cdot v_0 - B_0(a)W(\mu)] \}$
 $\leq p[B_1(a)W]^*(v_1) + (1 - p)[B_0(a)W]^*(v_0).$

To obtain such a well-behaved convex program we rely on three key ingredients: (i) probability-weighted continuation values, (ii) correlated equilibrium as a solution concept, and (iii) following [Rahman \(2014\)](#), recommendation- and signal-contingent continuation values. Had we used just signal-contingent continuation values, as in [Fudenberg et al. \(1994\)](#), linearity of incentive constraints would have failed. This program thus sweeps all nonlinearities under the rug of conjugate constraints.

The functional equation above is solved by finding the fixed points of its Bellman operator. This operator is defined on the space $\mathcal{P}_U$ of positively homogeneous, convex functions bounded above by δ_U^* ,²¹ the support function of the set of possible payoff profiles. Alas, this Bellman operator is not generally a contraction. It is monotone, though, so, by Tarski's Theorem, it has a maximal, if not unique, fixed point.²² As we show next, this maximal fixed point characterizes the set of equilibrium payoffs.

THEOREM 4.1. *The primal functional equation attains its supremum at a proper W in $\mathcal{P}_U$.²³ The welfare potential $\underline{W} = \overline{W}$ is the maximal fixed point for ppCE and ppRE.*

[Theorem 4.1](#) reconciles the sequence problems with the primal functional equation. Its proof relies on elementary convex analysis. [Theorem 4.1](#) generalizes previous results by [Judd et al. \(2003\)](#), [Tomala \(2009\)](#), [Rahman \(2014\)](#) and [Abreu et al. \(2020\)](#), and will help to deliver an algorithm that characterizes equilibrium in richer games.²⁴

4.2. Dual Functional Equation

A computational algorithm for the maximal fixed point of a primal functional equation proved impractical in our numerical experiments. However, an alternative that avoids computing conjugate functions suggests itself in the dual functional equation below.

PROPOSITION 4.1. *The dual of the primal functional equation for ppCE is given by*

$$W(\lambda) = \inf_{y \geq 0} \max_{a \in A} (1 - \delta) \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a) + \delta \sum_{z \in Z} W(\mu(a, z)) \quad s.t.$$

$$\mu_i(a, z) = \lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a) \quad \forall (i, a, z),$$

$y_i(a_i) = (y_i(a_i, b_i))_{b_i \in A_i}$, $\Delta \pi_i(z|a) = (\Delta \pi(z|a, b_i))_{b_i \in A_i}$, and $\Delta u_i(a) = (\Delta u_i(a, b_i))_{b_i \in A_i}$. The dual problem for ppRE is the same as above except that $y_i(a) = (y_i(a, b_i))_{b_i \in A_i}$ may depend on a_{-i} in addition to (i, a_i, b_i) . Call these problems dual functional equations.

²¹In contrast with the usual space of bounded, measurable functions used in dynamic programming.

²²Non-uniqueness is easy to see. For example, $\delta_\emptyset^*(\lambda) = -\infty$ is clearly always a fixed point. Moreover, in the Prisoners' Dilemma ([Example 3.1](#)), $\delta_{\{(0,0)\}}^*(\lambda) = 0$ is clearly a fixed point, too.

²³Recall that a convex function W is *proper* if $W(\lambda) > -\infty$ for all λ and $W(\lambda) < +\infty$ for some λ .

²⁴Further generalizations of [Theorem 4.1](#) to, e.g., stochastic games, are straightforward ([Section 7](#)).

The dual functional equation above solves a saddle problem that minimizes maximum welfare. Dual variables labeled y are multipliers on primal incentive constraints; those labeled with $\mu > 0$ correspond to (probability-weighted) welfare weights where the promise-keeping constraint binds. The dual constraints describe a law of motion for welfare weights depending on recommendations and signals. Weights corresponding to binding conjugate constraints are normal vectors to the equilibrium payoff set at continuation values that solve the primal functional equation at those welfare weights. Interestingly, this Bellman equation no longer digests a value function's conjugate, but, rather, the value function itself. Finally, since W is positively homogeneous and convex, the (right-hand side of the) dual functional equation is a well-behaved finite-dimensional convex program; this facilitates the computational algorithm below.

Intuitively, the dual functional equation determines incentive costs by trading off current and future welfare weights. Current weights are distorted by deviation gains, future weights are distorted by detection. Thus, if $y_i(a_i, b_i) > 0$, higher deviation gains $\Delta u_i(a, b_i)$ reduce player i 's effective welfare weight in the current period. Moreover, next period's welfare weight is subject to risk depending on the realization of (a, z) . This risk is exploited by the convexity of W , which increases with a mean-preserving spread of welfare weights. When incentive constraints don't bind, welfare weights, hence payoffs, do not change, hence they reach a steady state. Conversely, the welfare risk necessary to provide players with incentives is exploited dynamically. First, for intuition, consider a simple corollary of [Proposition 4.1](#) when π has full support.

COROLLARY 4.1. *If $\pi(z|a) > 0$ for all (a, z) , the dual functional equation for ppCE is*

$$W(\lambda) = \inf_{y \geq 0} \max_{a \in A} (1 - \delta) \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a) + \delta \sum_{z \in Z} \pi(z|a) W(\lambda'(a, z)) \text{ s.t.}$$

$$\lambda'_i(a, z) = \lambda_i - y_i(a_i) \cdot \frac{\Delta \pi_i(z|a)}{\pi(z|a)} \quad \forall (i, a, z).$$

The dual for ppRE is the same, except $y_i(a)$ may also depend on a_{-i} . In both cases, (i) welfare weights are martingales, i.e., $E[\lambda'] = \lambda$, and (ii) when incentive constraints are slack, i.e., $y \equiv 0$, welfare stays the same over time.

Three noteworthy observations are apparent here. First, the dynamic program above looks like a standard recursion with a simple, intuitive law of motion. Yet, just like its primal counterpart, this Bellman equation is not a contraction.²⁵ Nevertheless, this dual functional equation is amenable to numerical analysis, as discussed in [Section 6](#)

²⁵As the proof of [Proposition 4.1](#) shows, this is due to the domain chosen for the Bellman operator.

below. Secondly, the martingale property of welfare weights in [Corollary 4.1](#) naturally generalizes the standard inverse Euler equation of [Rogerson \(1985\)](#) to the multi-agent setting with imperfect monitoring and ppCE. (For details, see [Section 6.3.1](#).) Finally, it is interesting to note that differences between ppCE and ppRE express themselves only through the dimension of y , the vector of multipliers on incentive constraints. Such apparently minor difference has significant implications for the characterizing properties of equilibrium; see [Section 5.1](#) for this discussion.

4.3. Strong Duality and Dual Attainment

Strong duality for functional equations means that the value of the primal equals that of the dual, and dual attainment that the dual has a solution. In this subsection, we find sufficient conditions on the parameters of the game to guarantee both for each vector of welfare weights λ , and then uniformly on compact sets of λ .

We begin by finding no duality gap between primal and dual functional equations. To this end, given W in $\mathcal{P}_U$ and welfare weights λ , let $\underline{\mathbf{B}}(W)(\lambda)$ be the right-hand side of the primal functional equation and $\overline{\mathbf{B}}(W)(\lambda)$ the right-hand side of its dual counterpart. Since both problems are well-behaved convex programs, the no-gap result below follows from a straightforward application of the Minimax Theorem.

THEOREM 4.2 (NO DUALITY GAP). *There is no gap between the values of the primal and dual functional equations: for any $W \in \mathcal{P}_U$ and $\lambda \in \mathbb{R}^N$,*

$$\underline{\mathbf{B}}(W)(\lambda) = \overline{\mathbf{B}}(W)(\lambda).$$

By [Theorem 4.2](#), the value of the primal equals the value of the dual. Since the primal already has a solution, strong duality requires just that the dual have a solution, too. We now find sufficient conditions for this. First, some preliminary definitions.

For any player i and $\eta_i \in \mathbb{R}_+^{A_i}$, define the *change* in η_i relative to action a_i by

$$\Delta\eta_i(a_i) = \sum_{b_i \neq a_i} \eta_i(b_i),$$

and the *impact* of η_i at action profile a by

$$\|\eta_i \cdot \Delta\pi_i(a)\| = \sum_{z \in Z} \left| \sum_{b_i \in A_i} \eta_i(b_i) [\pi(z|a_{-i}, b_i) - \pi(z|a)] \right|.$$

Intuitively, the change $\Delta\eta_i(a_i)$ is the mass that η_i puts on actions different from a_i , whereas the impact $\|\eta_i \cdot \Delta\pi_i(a)\|$ is the sum of absolute values of signal probability changes induced by η_i at a . These notions are used in the next definitions.

DEFINITION 4.2. *Strong detectability* holds if for every player i and action profile a ,

$$\inf_{\Delta\eta_i(a_i)>0} \frac{\|\eta_i \cdot \Delta\pi_i(a)\|}{\Delta\eta_i(a_i)} \geq K > 0,$$

so the infimum is taken with respect to $\eta_i \in \mathbb{R}_+^{A_i}$ whose change from a_i is positive.

Weak detectability holds if, for every player i and action profile a ,

$$\inf_{\Delta\eta_i(a_i)>0} \sup_{a'_{-i}} \frac{\|\eta_i \cdot \Delta\pi_i(a_i, a'_{-i})\|}{\Delta\eta_i(a_i)} \geq K > 0,$$

where the supremum is taken with respect to a'_{-i} in A_{-i} .

Weak and strong detectability both mean that every vector of positive change has positive impact. Strong detectability requires positive impact at every action profile; weak only requires it at some action profile of a player's opponents. Such conditions hold, e.g., under individual identifiability/full rank; see [Rahman \(2012\)](#) for details.²⁶

ASSUMPTION 4.1. (i) At some discount factor $\underline{\delta} < 1$, the equilibrium payoff set has non-empty interior, and (ii) strong detectability holds under ppRE, weak under ppCE.

[Assumption 4.1](#)(i) requires that the equilibrium payoff set have nonempty interior at some $\underline{\delta} < 1$.²⁷ For example, this condition holds if the set of feasible, individually rational payoffs has non-empty interior²⁸ and deviations are individual- and pairwise-identifiable (see, e.g., [Fudenberg et al., 1994](#)). [Assumption 4.1](#)(ii) entails detectability. Under ppRE, it says that at every action profile a , every unilateral deviation changes signal probabilities. Under ppCE, it says that every deviation by any player i changes signal probabilities for some action profile a'_{-i} of the others. As [Rahman \(2012\)](#) shows, [Assumption 4.1](#)(ii) holds generically under mild conditions.²⁹

PROPOSITION 4.2 (DUAL ATTAINMENT). *If [Assumption 4.1](#) holds and W is the welfare potential, for any vector of welfare weights λ there exists $\delta_\lambda < 1$ such that the infimum on the right-hand side of the dual functional equation is attained for all $\delta \in [\delta_\lambda, 1)$.*

[Proposition 4.2](#) uses [Assumption 4.1](#)—a ‘coerciveness’ condition—to establish that the minimization problem in $\overline{\mathbf{B}}(W)(\lambda)$ has a solution. However, it doesn't take a stand on how optimal dual solutions depend on λ . A simple extension fills this gap.

²⁶In [Rahman \(2012\)](#), strong detectability corresponds to ‘every $\{a\}$ -deviation is $\{a\}$ -detectable’ for all a ; weak to ‘every A -deviation is A -detectable.’ Finiteness of Z gives the Lipschitz bound.

²⁷This kind of assumption has precedent (e.g., [Fudenberg and Levine, 1994](#); [Sannikov, 2007](#)).

²⁸The set of feasible, individually rational payoffs is $U_+ = U \cap L_{\underline{V}}$, where $L_{\underline{V}} = \{v \in \mathbb{R}^N : v \geq \underline{V}\}$ and $\underline{V}_i = \min_B \max_{b_i} \{\sum_{a_{-i}} u_i(b_i, a_{-i}) B(a_{-i}) : B \geq 0, \sum_{a_{-i}} B(a_{-i}) = 1\}$ is each i 's minmax payoff.

²⁹[Assumption 4.1](#)(ii) can be generalized in line with [Rahman \(2012\)](#) while preserving strong duality.

PROPOSITION 4.3 (UNIFORM BOUNDEDNESS). *Given Proposition 4.2's assumptions and a compact set of weights λ , there exists $\bar{\delta} < 1$ such that, for all $\delta \in [\bar{\delta}, 1)$, solutions of the right-hand side of the dual functional equation are uniformly bounded in λ .*

On the one hand, Proposition 4.2 yields dual attainment, so the primal value function is subdifferentiable with respect to its right-hand side constraints at each λ . In principle, though, this subdifferential could behave poorly near λ . Proposition 4.3 shows that the primal value function is ‘doubly’ subdifferentiable: as λ changes slightly, the shadow cost of relaxing primal constraints varies in a uniformly bounded way.

4.4. Back to the Dual Sequence Problem

Even though there is no duality gap between the primal sequence problem and its dual, we have yet to establish whether or not the dual has a solution. In this subsection, we ‘close the loop’ by forging a link between the dual sequence problem and the dual functional equation, in effect solving the former with a solution of the latter. Specifically, we use Proposition 4.3 to ‘unfold’ solutions of the dual functional equation into a candidate solution to the dual sequence problem. As will be seen, this candidate does not generally belong to our previously defined space of dual processes; rather it belongs to a modified space of ‘weighted’ processes, defined next.

DEFINITION 4.3. For any growth rate $g \in (0, 1]$, let the g -weighted dual norm be given by $\|Z\|_{1,g}^* = \sum_t g^{t-1} |Z|_{t-1}^*$, with space $\hat{\ell}_{1,g}^{*S} = \{Z \in \mathbb{R}^{S \times H} : \|Z\|_{1,g}^* < +\infty\}$ of g -dual processes. The g -inner product between primal and g -dual processes X and Z is

$$\langle X, Z \rangle_g := \sum_{t \in \mathbb{N}} \sum_{h_{t-1}} g^{t-1} X(h_{t-1}) Z(h_{t-1}).$$

Thus, semi-norms of g -dual processes may grow more slowly than g^{-t} , although this growth is ‘neutralized’ by the g -inner product. Primal and g -dual processes form a dual pair under $\langle \cdot, \cdot \rangle_g$: clearly, $|\langle X, Z \rangle_g| \leq \|X\|_\infty \|Z\|_{1,g}^* < +\infty$ when $X \in \hat{\ell}_\infty^S$ and $Z \in \hat{\ell}_{1,g}^{*S}$, and these two spaces separate each other’s points just as in Lemma 3.2.

Our next result uses Proposition 4.3 to bound the growth of solutions of the dual functional equation, thus solving a ‘weighted’ dual sequence problem. We begin by unfolding the dual functional equation. Given initial welfare weights λ , let $y^*(\lambda)$ solve the minimization in the dual functional equation. A candidate solution of the dual sequence problem obtains iteratively as follows. Let $\Lambda^*(h_0) = \lambda$ and $Y^*(h_0) = y^*(\lambda)$, with $W^*(h_0) = \underline{W}(\lambda)$. Now define recursively the processes after each history h_{t-1} by $\Lambda_i^*(h_{t-1}, a, z) = \delta[\Lambda_i^*(h_{t-1})\pi(z|a) - Y_i^*(h_{t-1}, a_i) \cdot \Delta\pi_i(z|a)]$ as in Proposition 3.1, and $Y^*(h_{t-1}, a, z) = y^*(\Lambda^*(h_{t-1}, a, z))$, $W^*(h_{t-1}, a, z) = \underline{W}(\Lambda^*(h_{t-1}, a, z))$.

By construction, $D^* = (W^*, \Lambda^*, Y^*)$ attains the value $\underline{W}(\lambda)$ of the primal sequence problem and satisfies all of the dual sequence problem's constraints. Still, D^* is not guaranteed to be feasible for the (unweighted) dual. Specifically, D^* need not belong to $\hat{\ell}_1^* \times \hat{\ell}_1^{*N} \times \hat{\ell}_1^* \Pi_i A_i \times A_i$ because the norms $\|W^*\|_1^*$, $\|\Lambda^*\|_1^*$ and $\|Y^*\|_1^*$ may be infinite. On the other hand, some g -weighted norms are bounded, as the next result shows.

LEMMA 4.1. *If Assumption 4.1 holds, there exists $\bar{\delta} < 1$ such that for all $\delta \in [\bar{\delta}, 1)$ there is a growth rate $g \in (0, 1]$ for which, regardless of λ ,*

$$\|W^*\|_{1,g}^*, \|\Lambda^*\|_{1,g}^*, \|Y^*\|_{1,g}^* < +\infty.$$

By Lemma 4.1, ‘unfolded’ solutions of the dual functional equation live in a space of g -dual processes. Clearly, the case $g = 1$ yields dual attainment.

COROLLARY 4.2. *If the growth rate in Lemma 4.1 is $g = 1$ then (W^*, Λ^*, Y^*) is an optimal solution of the dual sequence problem.*

This result relates to the contractions of Pavoni et al. (2018) and Abreu et al. (2020). Generally, though, $g = 1$ is not guaranteed. In this case, we obtain a dual solution by inserting D^* into a modified version of the dual sequence problem, defined next.

DEFINITION 4.4. The g -dual sequence problem for ppCE is the linear program below:

$$\begin{aligned} \overline{W}_g(\lambda) &= \inf_{Y \geq 0, \Lambda, W} W(h_0) \quad \text{s.t.} \quad \Lambda_i(h_0) = \lambda_i \quad \forall i \in N, \\ W(h) &\geq (1 - \delta) \sum_{i \in N} [\Lambda_i(h) u_i(a) - Y_i(h, a_i) \Delta u_i(a)] + g \sum_{z \in Z} W(h, a, z) \quad \forall (h, a), \\ \Lambda_i(h, a, z) &= (\delta/g) [\Lambda_i(h) \pi(z|a) - Y_i(h, a_i) \Delta \pi_i(z|a)] \quad \forall (i, h, a, z), \end{aligned}$$

where $(W, \Lambda, Y) \in \hat{\ell}_{1,g}^* \times \hat{\ell}_{1,g}^{*N} \times \hat{\ell}_{1,g}^* \Pi_i A_i \times A_i$. For ppRE, $Y \in \hat{\ell}_{1,g}^* \Pi_i A \times A_i$ instead.

The g -dual sequence problem above modifies the original of Proposition 3.1 as follows: the spaces of dual variables and right-hand side constraints are now g -weighted, and the dual constraints now incorporate the growth rate g explicitly. As we show next,³⁰ these minor modifications of the dual sequence problem yield the ‘trifecta’ of duality.

THEOREM 4.3. *Assume the hypotheses of Lemma 4.1, with growth rate g . The primal and g -dual sequence problems above are mutually $\langle \cdot, \cdot \rangle_g$ -dual linear programs with no duality gap. If the primal has a solution then D^* above is a solution of the g -dual.*

Theorem 4.3 completes the path from the primal sequence problem to the primal functional equation, to the dual functional equation and finally the g -dual sequence problem. An application and comparison with the literature appear in Section 5.

³⁰The proof of Theorem 4.3 in Appendix A derives the g -dual sequence problem in detail.

5. DISCUSSION

In this section, we make three points. First, we discuss key differences between ppRE and ppCE. Next, we discuss interpretation of the welfare potential, especially in relation to lotteries and dynamic incentives. Finally, we discuss how our formal structure compares with the relevant existing literature.

5.1. ppRE versus ppCE

The two solution concepts, ppRE and ppCE, differ in significant ways. In the primal, their key difference is that in ppRE each player is informed of others' behavior when told by the mediator to take some action, whereas in ppCE this information may be withheld from a player until each player takes the current period's action. As will be seen in [Section 6](#), this difference is substantial in important examples—even those when the length of a period is arbitrarily small, so the length of time during which a player is uninformed of others' actions becomes negligible.

In the dual, the difference between ppRE and ppCE is apparent: when minimizing the dual objective, ppRE has more choice variables than ppCE, so ppRE achieves a lower value than ppCE, and by strong duality a lower primal value, too. As [Section 6.3.2](#) suggests, this distinction can be interpreted as the difference between monitoring publicly versus secretly. Of course, the advantage of monitoring secretly is being able to monitor *seldom* without relinquishing the threat of monitoring, thus economizing monitoring costs. There is ample empirical evidence for this. For instance, according to [Hexner \(1943, p. 95\)](#), in a steel manufacturing cartel “[...] *more often than one might suppose, infringements of cartel regulations were reported by competing distributions within a few hours* [...]” This suggests that cartel members were secretly monitored, but not all the time, in line with ppCE's predictions.

At a technical level, it is interesting to note how the dual functional equations differ across solution concepts. In ppCE, the dual choice variables are $y_i(a_i, b_i)$, whereas in ppRE they are $y_i(a, b_i)$. Considering the saddle problems to which they belong, clearly in ppRE the order of min and max is interchangeable, whereas in ppCE it is not.³¹ This implies that, in ppRE, y can be chosen to satisfy optimality conditions action profile by action profile. Thus, when applicable, y can satisfy first-order conditions at each possible action profile, after which the best possible actions given contingently optimal choices y may be selected. This aligns with [Sannikov \(2007\)](#), for instance.

³¹This is because, for simplicity, only pure strategy profiles are chosen in the dual recursive problems.

On the other hand, in ppCE the order matters, and such simple first-order-condition, or action-profile-by-action-profile characterization of optimality no longer applies. Shadow costs of incentives are now indexed as $y_i(a_i, b_i)$, after which an objective is maximized with respect to a . In general, the optimal choices of y will have repercussions across action profiles such that, instead of a first-order condition, optimal shadow-cost levels y will equate objective values *across* action profiles.

An interesting benchmark to keep in mind is when ppRE and ppCE coincide. From the discussion above, in ppRE the order of min and max in the dual functional equation doesn't matter. When is this the case for ppCE, too? Our next result offers intuitive sufficient conditions, illustrated in Figure 6 below in an application to the repeated Prisoners' Dilemma with imperfect monitoring.

A game has *product structure* if players have independent signals of own actions, i.e., $z = (z_1, \dots, z_n)$ and $\pi(z|a) = \pi_1(z_1|a_1) \cdots \pi_n(z_n|a_n)$, where each π_i is a probability on some set Z_i conditional on each a_i in A_i . Thus, player i only affects the i th signal. A game has *sum structure* if $\Delta u_i(a, b_i)$ does not depend on a_{-i} for each i .

PROPOSITION 5.1. *In games with both product structure and sum structure, the set of payoff profiles in ppCE and ppRE coincide.*

5.2. Kinks and Flats

The welfare potential $\underline{W}$ is the support function of the set of equilibrium payoff profiles; call this set E and its indicator function δ_E . As Footnote 17 shows, $\underline{W} = \delta_E^*$ is the conjugate of δ_E , and since δ_E is a closed convex function, by the Biconjugate Theorem (e.g., Rockafellar, 1970), $\underline{W}^* = \delta_E^{**} = \delta_E$, that is, the conjugate of the conjugate is the original function. Thus, $\underline{W}$ and δ_E are mutually dual functions. These primal (δ_E) and dual ($\underline{W}$) functions share interesting phenomena—*kinks* and *flats*.³² Because one is the conjugate of the other, a primal kink corresponds to a dual flat and a primal flat to a dual kink.³³ Primal flats may be interpreted as lotteries over continuation utility profiles, and dual flats as lotteries over welfare weights. This resonates with the convexifying effect of large numbers in general equilibrium as well as Bayesian persuasion in the design of incentives.

³²A kink in a function is a point where the function's subdifferential is not a singleton. (For example, $\max\{x, 1\}$ has a kink at $x = 1$.) A flat is a region of the function's domain where the function is linear. (For example, the concavification of $-x^2$ and $-(x-1)^2$ has a flat along $x \in [0, 1]$.)

³³If f is a proper closed convex function and f^* its conjugate, f has a kink at v when $\partial f(v)$ is not a singleton. Now, $f^*(\lambda) = \lambda \cdot v - f(v)$ if $\lambda \in \partial f(v)$, so $f^*(\lambda) - f^*(\lambda') = (\lambda - \lambda') \cdot v$ for all $\lambda, \lambda' \in \partial f(v)$, i.e., f^* has a flat. Replacing f with f^* and f^* with $f^{**} = f$, a dual kink becomes a primal flat.

Our dynamic context uses lotteries via randomization made public ex ante in ppRE and with delay in ppCE, thus offering two different approaches to convexification or, depending on the point of view, concavification. Since primal flats and dual kinks coincide, whenever they occur, in ppRE they follow equilibrium behavior switching from one action profile to another—in between, public randomization may take place. In ppCE, dual kinks follow changes in the support of the correlated strategy being played. This is illustrated vividly in our applications of [Section 6](#).

As for primal kinks, from the dual functional equation it can be seen that the dual flats they induce reduce dynamic incentives. Indeed, convexity of the welfare potential implies that the shadow cost y introduces a mean-preserving spread in welfare weights, which increases the dual objective via promised continuation welfare. This trades off with the diminution of value from current deviation gains. In the presence of a dual flat, the mean-preserving spread loses impetus. At the extreme, if $\underline{W}$ were linear throughout the range of μ then $\sum_z \underline{W}(\mu(a, z))$ would simply equal $\underline{W}(\lambda)$, and thus continuation welfare would not depend on y . As such, current play would follow a static equilibrium. Nevertheless, the y 's would still stipulate future welfare weights, so even if play were stuck temporarily at some static equilibrium outcome, it may not be stuck there forever, depending on the realizations of z over time.

One case where static equilibrium behavior can become absorbing is when $y = 0$ is an optimal solution to the right-hand side of the dual functional equation. This is the case, for instance, in the Prisoners' Dilemma, where the static equilibrium is on the boundary of the feasible set. In this case, maximizing welfare in the direction $(-1, -1)$ coincides with or without incentive constraints, so $y = 0$, hence continuation welfare weights $\mu(a, z)$ and future play remain unchanged henceforth. Again, see [Section 6](#).

For games in continuous time, [Sannikov \(2007\)](#) made assumptions about the impact of players' behavior on the distribution of signals in order to avoid primal flats.³⁴ In either of these two cases, extreme equilibria of his continuous-time repeated game with Brownian signals feature continuation values that are Brownian motions evolving along the boundary of the equilibrium payoff set which entail no public randomization, hence no primal flats. Here, however, signals need neither be Brownian nor consistent with pairwise identifiability/product structure, as [Section 6](#) illustrates.

³⁴Specifically, [Sannikov \(2007\)](#) assumed either (i) unique static best responses and pairwise identifiability, or (ii) signal distributions with product structure. Intuitively, pairwise identifiability means that, for every pair of players and any profile of deviations from their prescribed behavior, it is possible to statistically identify the deviator. See [Rahman and Obara \(2010\)](#) for a discussion.

5.3. Ubi Est Slater

We end this section with a discussion of [Theorem 4.3](#) in relation to the literature. [Marcet and Marimon \(2019\)](#) and [Pavoni et al. \(2018\)](#) develop a dual approach to dynamic programming with recursive contracts assuming deterministic consumption plans. In their sequence problems, the commodity space is (like) ℓ_∞ and shadow prices belong to its dual space ba of finitely additive measures. Because ba is arguably unruly, these papers make further assumptions to ensure that shadow prices belong to the simpler space ℓ_1 .³⁵ Following [Dechert \(1995\)](#), assuming a uniform Slater condition and two relatively friendly continuity conditions yields well-behaved shadow prices.³⁶

Intuitively, given a dual solution Y^* in ba , a Slater point $\hat{X}$ in ℓ_∞ can be used to create a non-zero pairing $\langle \hat{X}, Y_s^* \rangle$ with just the singular part Y_s^* of Y^* , contradicting weak duality. Hence, $Y_s^* = 0$, i.e., dual solutions belong to ℓ_1 . In the context of the present paper, the continuity conditions are not controversial, but the Slater condition becomes problematic and the above argument does not apply, as we discuss next.

In dynamic games, it is natural to consider strategic lotteries, as we do in this paper. Because our primal sequence problem uses public correlated strategies, it is purposely defined in terms of probabilities that are unconditional with respect to behavior, as discussed in [Section 2.3](#). Therefore, as long as there is any persistent mixing, a uniform slack on incentive constraints is impossible to sustain. Of course, persistent mixing is necessary to make all incentive constraints slack. Moreover, persistently mixed behavior is generally necessary for detection in ppCE, so the left-hand side of incentive constraints implodes over time due to endogenous unconditional probabilities.

Despite not being able to resort to a uniform Slater condition for a solution of the dual sequence problem, [Theorem 4.3](#) uses [Assumption 4.1](#)—detectability and interiority of equilibrium payoffs—to translate a sequence of solutions of the dual functional equation into a solution of the dual sequence problem. Although this solution does not belong to ℓ_1 (or, rather, $\hat{\ell}_1$), with the help of [Assumption 4.1](#) we are able to confine a dual solution to a weighted ℓ_1 space. This weight reflects the fact that multipliers on promise-keeping constraints may accumulate over time enough to grow and escape summability. This dual solution is still reasonably interpretable, see [Section 6.4](#).

³⁵[Bewley \(1972\)](#), who studies existence of general equilibrium in ℓ_∞ -type commodity spaces with prices in ℓ_1 -type spaces, is a leading starting point for this approach.

³⁶Slater condition: there is a feasible solution satisfying all inequality constraints with a uniformly bounded strict slack. Continuity conditions: constraints in the distant future are not much affected by current-period variables, and variables in the distant future do not much affect current constraints.

6. APPLICATIONS

We now present a computational algorithm and offer three applications. The first application is an instance of the repeated Prisoners' Dilemma with imperfect public monitoring. We compute its set of ppRE and ppCE payoffs, and discuss equilibrium and computational phenomena. The next application is optimal dynamic contracting. There, we find how our dual approach naturally incorporates lotteries. We also explore optimal monitoring, with important differences between ppRE and ppCE. Our third, more theoretical, application derives convergence rates for approximate equilibrium.

6.1. Computation

In this subsection we present our basic computational algorithm, consisting of two stages: constrained optimization given a value function and value function iteration. Its goal is to numerically compute the largest fixed point of the dual functional equation to find the welfare potential. This algorithm, coded in Python, solves for any repeated two-person two-action game with public monitoring and a finite number of signals with any conditional probabilities, for both ppCE and ppRE.

Computationally, the algorithm proceeds as follows. A number of welfare directions is specified in advance, say $\alpha = 100$. Value function iteration commences with an initial guess given by maximal welfare from flow payoff profiles of the stage game, i.e.,

$$W_0(\lambda) = \delta_U^*(\lambda) = \sup_{B \in \Delta A} \sum_{(i,a)} \lambda_i u_i(a) B(a).$$

With this initial W_0 , the optimization problem on the right-hand side of the dual functional equation in [Proposition 4.1](#) is solved along each of the α welfare directions. This generates a function W_1 evaluated at the α given directions. Next, the function W_1 is extended to every vector of welfare weights λ by cubic spline interpolation of the α directions already calculated and positive homogeneity. This completes the first stage of value function iteration. This calculation is then iterated with W_n instead of W_{n-1} to obtain W_{n+1} , for each $n \in \mathbb{N}$, until the process of value function iteration converges, that is, until a stage n is reached with $\|W_n - W_{n-1}\|$ sufficiently small. Our model is amenable to this approach, as the next result shows.³⁷ Its proof recalls and applies Kleene's Theorem ([Kleene, 1952](#))—an old extension of Tarski's.

THEOREM 6.1. *The sequence $\{W_n\}$ converges monotonically to the largest fixed point of the primal sequence problem, hence, by [Theorem 4.1](#), to the welfare potential $\underline{W}$.*

³⁷This result extends Theorem 5 of [Abreu et al. \(1990\)](#) to ppCE; [Sargent and Stachurski \(2025\)](#) study this convergence issue structurally, in the context of abstract dynamic programming.

6.2. Repeated Prisoners' Dilemma

Consider a two-player repeated Prisoners' Dilemma with imperfect public monitoring and two signals, whose specific flow payoffs and signal probabilities are the following:

	c_2	d_2		c_2	d_2
c_1	1, 1	-1, 2	c_1	$\frac{3}{4}, \frac{1}{4}$	$\frac{1}{2}, \frac{1}{2}$
d_1	2, -1	0, 0	d_1	$\frac{1}{2}, \frac{1}{2}$	$\frac{1}{4}, \frac{3}{4}$
Flow Payoffs			Signal Probabilities		

6.2.1. *ppRE* versus *ppCE*

Solving the dual functional equation numerically for each solution concept, ppRE and ppCE, produces the equilibrium payoff sets shown in Figure 3 below with $\delta = 0.975$.

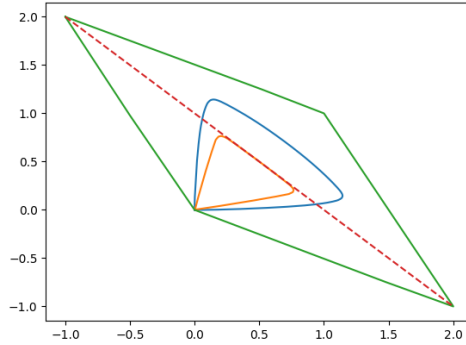


FIGURE 3. Boundaries of ppCE (blue) and ppRE (orange) payoff profiles in the above repeated Prisoners' Dilemma assuming $\delta = .975$

Figure 3 has player 1's payoffs on the x axis and 2's on the y axis. The green polygon is the boundary of feasible payoffs, the diagonal red dotted line segment links the payoffs from (c_1, d_2) and (c_2, d_1) , and the blue and orange curves depict the boundaries of the ppCE and ppRE equilibrium payoff sets, respectively. In ppRE, the equilibrium payoff set is bounded by the red diagonal,³⁸ an expression of the fact that mutual cooperation is untenable in ppRE. The claim below formalizes this observation.

CLAIM 6.1. *The sum of players' payoffs in any ppRE is less than or equal to 1.*

Intuitively, mutual cooperation fails because, after a deviation, it could have been any player, so both players must be punished after a 'bad' signal, which leads to a welfare loss that does not compensate for the gain from one more period of mutual cooperation given the game's parameters. Still, Figure 3 also shows that (i) unilateral cooperation is possible in ppRE, and (ii) so is mutual cooperation in ppCE.

³⁸This aligns with Fudenberg et al. (1994, Example 1), where pairwise identifiability fails.

6.2.2. Value Function Iteration

Figure 4 below shows the result of value function iteration, beginning with δ_U^* and ending with the support functions of the solution sets depicted in Figure 3 above.

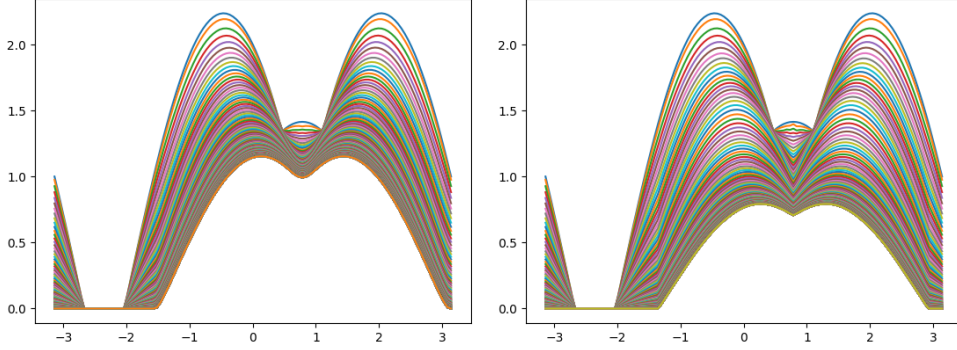


FIGURE 4. Value function iteration for ppCE (left) and ppRE (right)

To understand Figure 4, note that in our two-person game a vector of welfare weights λ is two-dimensional, and on the unit circle it can be parametrized with the single parameter $\theta \in [-\pi, \pi]$, where $\lambda = (\cos(\theta), \sin(\theta))$. The x axis of Figure 4 corresponds to precisely this θ .³⁹ The y axis corresponds to values of support functions at each stage of value function iteration. The top curve depicts the support function of the green polygon in Figure 3, i.e., δ_U^* . The bottom curve on the left (right) panel is the support function of the set whose boundary is the blue (orange) curve in Figure 3. It is apparent from Figure 4 that value function iteration converges monotonically.

There are many ways to improve this algorithm. For example, ‘policy iteration’—where iterations that do not update y^* are interspersed—significantly reduces the number of time-costly iterations, as shown in Figure 5 below.

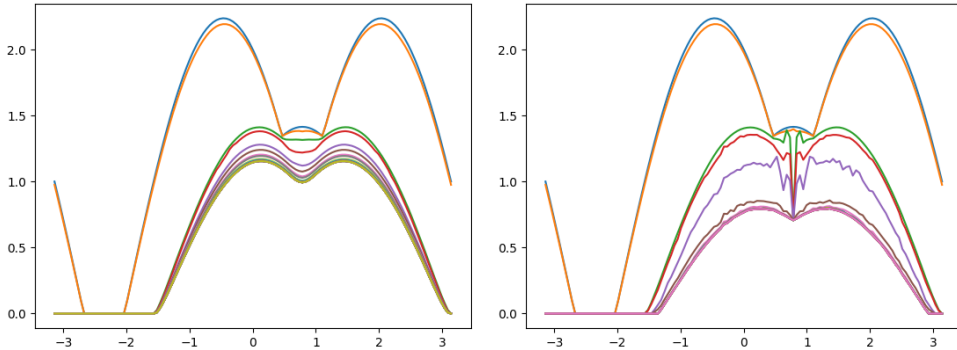


FIGURE 5. Policy iteration for ppCE (left) and ppRE (right)

³⁹To see where the ‘mustaches’ come from, consider the support function of the single point $\{(0, 1)\}$, namely $\delta_{\{(0,1)\}}^*(\lambda) = \sup_{v \in \{(0,1)\}} \lambda \cdot v = \lambda \cdot (0, 1)$. Therefore, if $\lambda = (\cos(\theta), \sin(\theta))$ belongs to the unit circle then $\delta_{\{(0,1)\}}^*(\lambda) = \sin(\theta)$. Basically, linear support functions are sinusoidal functions of θ .

6.2.3. Kinks and Flats

Figures 3.1 and 4 point to the following interesting phenomena: the equilibrium payoff sets and welfare potentials have kink and flats. As discussed in Section 5, primal kinks correspond to dual flats and primal flats to dual kinks.

In ppRE, the equilibrium payoff set has a (primal) kink at $(0,0)$ and three (primal) flats: one on the diagonal with outward normal vector $(1,1)$, and one on each side of $(0,0)$. Figure 4 shows that the welfare potential has one (dual) flat, where its value equals 0, and three (dual) kinks, at $\theta \approx -1.3, 0.8, 2.9$. At the dual flat, $y^* = 0$ and $\sum_z W(\mu(a, z)) = W(\lambda)$, so play follows a static equilibrium. Of course, this corresponds to mutual defection with payoff profile $(0,0)$. The dual kinks represent transition points where players switch their equilibrium behavior from one action profile to another. Primal flats correspond to lotteries that may lead to drastic changes in continuation values. Thus, even if welfare weights change gradually over time, continuation payoffs may suddenly jump from one end of a flat to the other.

In ppCE, the primal flat at $\theta \approx 0.8$ disappears—as confirmed by a lack of dual kink in the welfare potential there—and the two other dual kinks become less pronounced. This is verified in Figure 3, where the equilibrium boundary is curved around the outward normal vector $(1,1)$, and the flats around $(0,0)$ are shorter than in ppRE.

The Prisoners' Dilemma has sum structure, i.e., $\Delta u_i(a, b_i)$ doesn't depend on a_{-i} , so by Proposition 5.1 product structure equates payoffs in ppRE and ppCE. In addition, primal flats disappear, in line with Sannikov (2007), who solves a dual HJB equation assuming there are no dual kinks. Figure 6 below makes this point by showing the primal and dual value functions when signals have product structure.⁴⁰

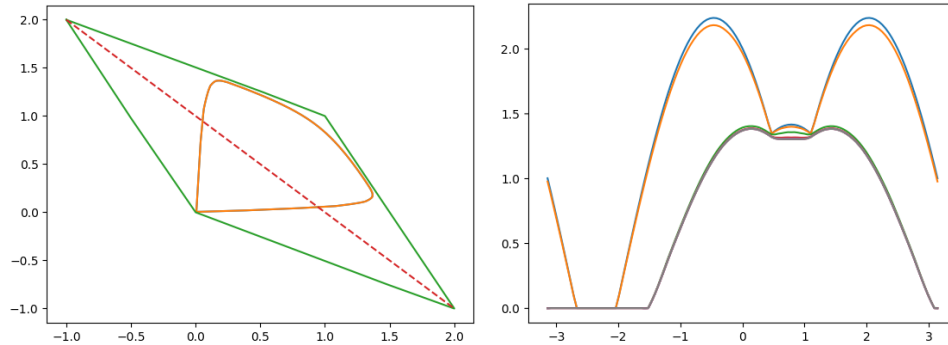


FIGURE 6. Primal and dual value functions assuming product structure

⁴⁰Each i 's actions have independent binary signals, with probabilities $(\frac{3}{4}, \frac{1}{4})$ given c_i , $(\frac{1}{2}, \frac{1}{2})$ for d_i .

6.3. Optimal Dynamic Contracting

The martingale property of welfare weights from [Corollary 4.1](#) naturally generalizes the standard inverse Euler equation of [Rogerson \(1985\)](#) to the multi-agent setting. Even in the single-agent case, the dual approach above provides valuable insight, though. This subsection illustrates that by analyzing and generalizing an otherwise standard dynamic principal-agent problem.

6.3.1. Principal-Agent Problem

Consider an economy with one principal (she) and one agent (he). In every period t , the agent may work ($a_t = 1$) or shirk ($a_t = 0$). The agent's cost of effort is $h > 0$. The agent receives a wage $c_t \geq 0$ in units of a consumption good from the principal. The agent's utility flow over c and a is $u(c) - ha$, where u is continuously differentiable, strictly increasing, strictly concave, $u'(c) \rightarrow 0$ as $c \rightarrow +\infty$, $u'(c) \rightarrow +\infty$ as $c \rightarrow 0$, and $u(0) = 0$. The principal is risk neutral. Both parties share a discount factor $\delta = e^{-r\Delta} < 1$, where r is a discount rate and Δ is the length of time per period.

Every period, output z_t depends stochastically on the agent's effort: $z_t = \sigma/\sqrt{\Delta}$ with probability $p = \frac{1}{2}[1 + (\mu/\sigma)\sqrt{\Delta}]$ if the agent works (and probability $\frac{1}{2}$ if he shirks) and $z_t = -\sigma/\sqrt{\Delta}$ with probability $q = 1 - p$ if the agent works (and $\frac{1}{2}$ if he shirks), where μ and σ are positive and Δ is small enough for p and q to be probabilities. The principal (i) receives output and shares it with the agent at will, and (ii) makes effort recommendations to the agent observing output but not effort. Let U be the lifetime utility of the agent and V that of the principal. The standard approach to optimal contracting in this environment uses the agent's promised utility as a state variable. Optimally, the principal will retire the agent if cumulative output becomes too low or too high, this means that the agent receives a constant wage and exerts no effort. Let $V_0(U) = -u^{-1}(U)$ be the cost to the principal in units of consumption of retiring the agent with lifetime utility U . In this case, the principal's problem can be written recursively as follows:

$$V(U) = \max_{c, U_{\pm} \geq 0} \max \{ (1 - \delta)(\mu - c) + \delta[pV(U_+) + qV(U_-)], V_0(U) \} \quad \text{s.t.}$$

$$(1 - \delta)h \leq \delta(p - \tfrac{1}{2})(U_+ - U_-),$$

$$U = (1 - \delta)[u(c) - h] + \delta(pU_+ + qU_-).$$

The principal chooses the agent's effort, wage, and output-contingent continuation utility (U_+ after high output and U_- after low) to maximize expected net revenue, subject to incentive compatibility and promise-keeping.

The inverse Euler equation follows easily: $V'(U) = 1/u'(c) = pV'(U_+) + qV'(U_-)$. Consider instead the dual version of this problem, where the principal's welfare weight is normalized to 1 and the agent's is denoted by λ . It is convenient to switch signs in conjugate functions below: let $f^*(\lambda) = \sup_{x \geq 0} \{f(x) + \lambda x\}$ for any function f .

PROPOSITION 6.1. *The dual functional equation for the principal-agent problem is*

$$V^*(\lambda) = \min_{y \geq 0} \max \left\{ (1 - \delta)[\mu + \lambda u^*(-1/\lambda) - (\lambda + y)h] \right. \\ \left. + \delta \left[pV^*(\lambda + y[1 - \frac{1}{2p}]) + qV^*(\lambda - y[\frac{1}{2q} - 1]) \right], \hat{V}_0^*(\lambda) \right\},$$

where $\hat{V}_0^*(\lambda) = \sup \{V_0(U) + \lambda U : U \in \partial V^*(\lambda)\}$.

Proposition 6.1 presents a dual functional equation that is relatively simple to solve computationally; the dual retirement function generalizes the usual smooth pasting condition that applies even if V^* is not differentiable. By **Proposition 6.1**, the agent's welfare weight evolves over time as follows: if output is high, it grows by $y[1 - \frac{1}{2p}]$, and if output is low it declines by $y[\frac{1}{2q} - 1]$.

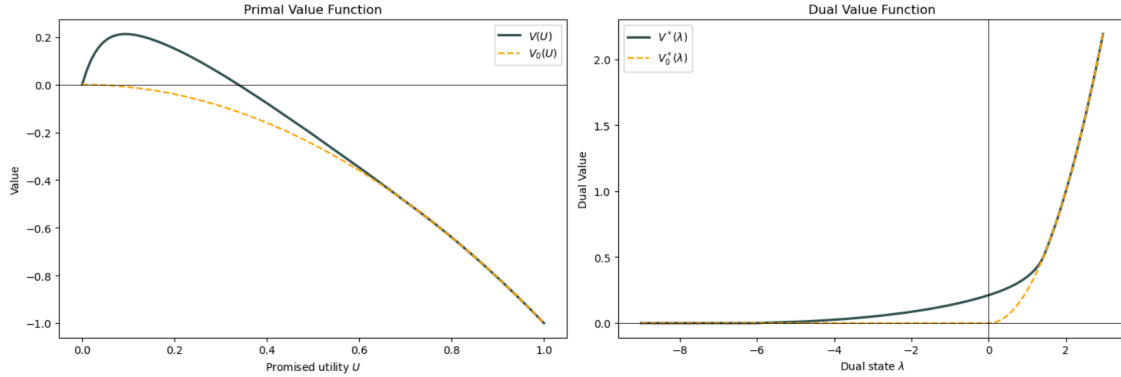


FIGURE 7. Primal (left) and dual (right) value functions for the principal-agent problem assuming $u(c) = \sqrt{c}$, $h = 0.9$, $\mu = 1$, $\sigma = 1$, $r = 0.1$, $\Delta = 0.01$

Figure 7 above displays the results of numerically computing the principal's value and conjugate functions. The left panel shows the principal's value as a function of the agent's promised utility, U , as well as the retirement value. The right panel shows the principal's conjugate value function and the conjugate retirement value as functions of the agent's welfare weight, λ , or, when the envelope condition applies, $-V'(U)$. The primal value function follows the standard depiction in [Sannikov \(2008\)](#). The optimal contract pays nothing but promises while the value function has non-negative slope, then consumption gradually grows with promised utility until it becomes profitable for the principal to retire the agent with a high wage. Finally, if promised utility deteriorates enough (to zero), the agent is retired with no wage payments.

Figure 7 reveals the following noteworthy nuance: the conjugate value function has a kink at $\lambda \approx 1.2$.⁴¹ As discussed in Section 5, this corresponds to a flat in the primal. This flat encompasses the region of promised utilities where the primal value function ‘pastes smoothly’ onto the retirement value. Because the conjugate is not differentiable at the kink, smooth pasting occurs at a region instead of a single point. Economically, this flat represents a lottery in the optimal contract that concavifies the maximum between the primal value function and the retirement value.⁴² By computing the dual value function directly, this primal lottery is subsumed into the dual kink and incorporated in the primal value of the optimal dynamic contract.⁴³ This observation showcases the practical value of our dual approach and ppRE.

6.3.2. Monitoring the Agent

Next, let us exemplify the distinction between ppRE and ppCE in this dynamic contracting environment. To do so, we expand the model above by allowing the principal to make a binary monitoring decision in every period, choosing between σ_h at flow cost $k_h = 0$ and σ_ℓ at flow cost $k_\ell > 0$, where $\sigma_h > \sigma_\ell > 0$. We interpret this as a form of costly monitoring in that higher variance decreases detectability of deviations.⁴⁴ The principal is able to choose the variance of the output process as part of her contract. Write $p_j = \frac{1}{2}[1 + (\mu/\sigma_j)\sqrt{\Delta}]$, where $j \in \{\ell, h\}$, for the implied probability of high output. In ppRE, the agent knows the principal’s monitoring decision before making an effort choice, but not in ppCE. As will be seen, when the principal can monitor secretly, she can monitor seldom, and thus save on monitoring costs. In the dual, the shadow cost of incentives depends on the principal’s monitoring choices for ppRE, but not for PPCE. Thus, the dual functional equation for ppRE is

$$V^*(\lambda) = \min_{y \geq 0} \max_{j \in \{\ell, h\}} \max \left\{ (1 - \delta)[\mu - k_j + \lambda u^*(-1/\lambda) - (\lambda + y_j)h] \right. \\ \left. + \delta \left[p_j V^*(\lambda + y_j[1 - \frac{1}{2p_j}]) + q_j V^*(\lambda - y_j[\frac{1}{2q_j} - 1]) \right], \hat{V}_0^*(\lambda) \right\}.$$

Figure 8 below depicts a numerical solution to this recursive characterization of ppRE. It displays the primal (left panel) and dual (right panel) value functions (left axes) together with optimal monitoring choices (right axes).

⁴¹It also has a flat to the left of $\lambda \approx -4.3$. This corresponds to the primal kink at $U = 0$, in the sense that—by convention—the primal value function equals $-\infty$ to the left of zero.

⁴²Zeng (2025) and Zeng and Rajan (2025) find that lotteries can supplement the standard contracting environment of Sannikov (2008) and produce higher value for the principal under some assumptions.

⁴³The primal value function is derived as the conjugate of the computed conjugate function.

⁴⁴Another way of introducing costly monitoring is via the acquisition of additional signals of effort above and beyond output. This idea is pursued in Birkan and Rahman (2025).

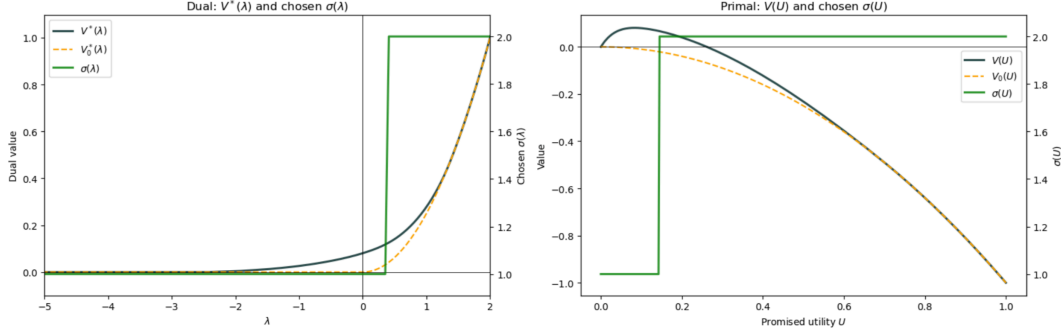


FIGURE 8. Primal (left) and dual (right) value functions with optimal monitoring in ppRE ($\sigma_h = 2$, $\sigma_\ell = 1$, $k_\ell = 0.5$; all other parameters unchanged)

As can be gleaned from Figure 8, the principal monitors the agent more intensely (i.e., she invests in lowering output variance) when the agent's continuation value is low, equivalently when the negative of the slope of the primal value function or the welfare weight on the agent is low. Economically, this can be interpreted as follows. The principal chooses to monitor when the agent's continuation value is very low in order to avoid the looming bad absorbing state of the agent being fired. Indeed, as Figure 9 below illustrates, the principal gains significantly, especially at low continuation values, from monitoring the agent relative to the previous benchmark of $\sigma_\ell = \sigma_h = 1$. Interestingly, Figure 9 shows that monitoring can reduce the level of promised utility that maximizes the principal's payoff.

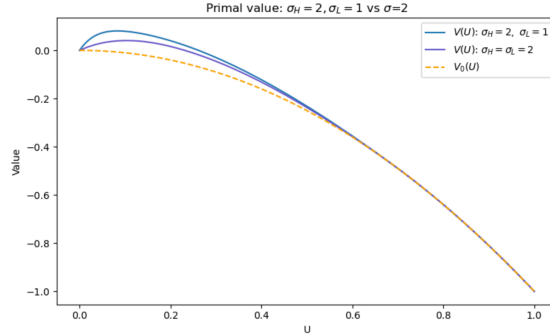


FIGURE 9. Comparison of value functions with and without monitoring

In ppCE, the principal's dual functional equation becomes

$$V^*(\lambda) = \min_{y \geq 0} \max_{j \in \{\ell, h\}} \max \left\{ (1 - \delta)[\mu - k_j + \lambda u^*(-1/\lambda) - (\lambda + y)h] \right. \\ \left. + \delta \left[p_j V^*(\lambda + y[1 - \frac{1}{2p_j}]) + q_j V^*(\lambda - y[\frac{1}{2q_j} - 1]) \right], \hat{V}_0^*(\lambda) \right\}.$$

The only difference between this problem and the previous one is that now shadow costs are constant across monitoring choices ($y_\ell = y_h = y$). Since shadow costs are chosen to minimize, the value from ppCE is at least as high as that from ppRE.

Monitoring in ppCE means that the principal can choose σ without the agent knowing at the time that he decides whether or not to exert effort, even though he learns the principal's decision immediately afterwards. Figure 10 illustrates key differences between ppRE and ppCE in terms of strategic behavior and overall payoffs.

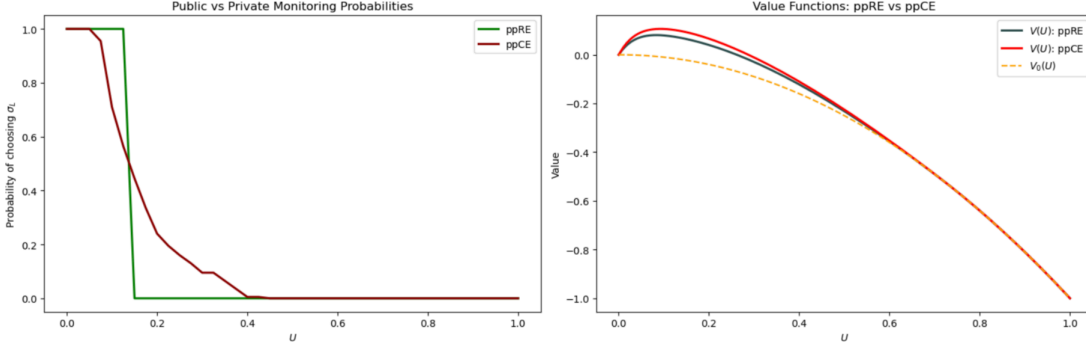


FIGURE 10. Public versus secret monitoring strategies and values

The right panel of Figure 10 confirms that the utility possibility set is larger under ppCE than ppRE. The left panel shows, as a function of the agent's promised utility, the probability that the principal monitors the agent (i.e., plays σ_ℓ) under ppRE and ppCE. The blue step function corresponds to the previous ppRE regime, where the probability is either zero or one. The orange curve depicts the principal's secret monitoring probability. It can be seen from this panel that the principal smooths out her monitoring strategy in ppCE across the agent's utility space relative to that in ppRE: she secretly monitors the agent less when promised utility is moderately low and more when it is moderately high.

6.4. Approximate Equilibrium

We end this section with a more theoretical application of previous results: bounding convergence rates of approximate equilibrium. Most existing work on approximate equilibrium typically culminates with establishing continuity of ε -Nash equilibrium (e.g., Fudenberg and Levine, 1986; Mailath et al., 2005) with respect to $\varepsilon > 0$. Such continuity is important because it renders equilibrium robust to possible errors or miscalculations by players (and game theorists) in the evaluation of profitable deviations, for instance. On the other hand, continuity is a minimal robustness requirement, in the sense that it says nothing about the convergence rate of approximate equilibrium. Thus, at any approximation, the rate of convergence could in principle be arbitrarily slow without further information. A stronger requirement like Lipschitz continuity would therefore be desirable, since it could assert that convergence occurs at some uniformly bounded rate. We pursue such a result below.

To motivate, consider first a finite normal form game. The problem of maximizing welfare for any given welfare weights by choosing a correlated equilibrium is a well-understood finite-dimensional linear program (e.g., [Myerson, 1997](#)): there is strong duality, the dual has a solution and the primal value function is subdifferentiable with respect to the right-hand side of correlated equilibrium's incentive constraints. Therefore, the welfare potential of a finite normal form game is Lipschitz in ε .

In our dynamic game, the basic apparatus of finite-dimensional linear programming no longer applies. Even though the primal sequence problem is a well-behaved linear program, as [Theorem 3.1](#) attests, its value function need not be Lipschitz anymore. What guarantees a Lipschitz value function is strong duality and dual attainment, as shown next. Our primal sequence problem suggests the following ε -equilibrium.

DEFINITION 6.1. For $\varepsilon \in \hat{\ell}_{\infty+}^{\prod_i A_i \times A_i}$, let the *perturbed sequence problem* for ppCE be

$$\begin{aligned} \underline{W}(\lambda; \varepsilon) &= \sup_{C \geq 0, V} \sum_{i \in N} \lambda_i V_i(h_0) \quad \text{s.t.} \quad \sum_{a \in A} C(a; h_0) = 1, \\ &\quad \sum_{a \in A} C(a; h_{t-1}, a_t, z_t) = C(a_t; h_{t-1}), \\ V_i(h_{t-1}) &= (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_{t-1}) + \delta \sum_{(a, z)} \pi(z|a) V_i(h_{t-1}, a, z), \\ (1 - \delta) \sum_{a-i} \Delta u_i(a, b_i) C(a; h_{t-1}) &+ \delta \sum_{(a-i, z)} \Delta \pi(z|a, b_i) V_i(h_{t-1}, a, z) \leq \varepsilon_i(h_{t-1}, a_i, b_i), \end{aligned}$$

where $C \in \hat{\ell}_{\infty}^A$ and $V \in \hat{\ell}_{\infty}^N$. The version for ppRE takes $\varepsilon \in \hat{\ell}_{\infty+}^{\prod_i A \times A_i}$ instead.

Note that ε is a primal process, so if ε has full support, $\sum_{(i, a_i, b_i)} |\varepsilon_i(a^t, z^t, a_i, b_i)|$ must spread across a^t for each t so that its primal semi-norms remain bounded.

LEMMA 6.1. *The following statements are equivalent: (i) both the dual has an optimal solution and there is no duality gap, so $\underline{W}(\lambda) = \overline{W}(\lambda)$, (ii) $\underline{W}(\lambda; \varepsilon)$ is subdifferentiable with respect to ε at $\varepsilon = 0$, (iii) $\underline{W}(\lambda; \varepsilon)$ is Lipschitz continuous in ε .*

This result constitutes basic duality theory (see [Gale, 1967](#), and [Gretsky et al., 2002](#)). By [Theorem 3.2](#), there is no duality gap, and [Theorem 4.3](#) finds conditions under which the dual also has a solution, in which case welfare is Lipschitz.

PROPOSITION 6.2. *If the hypotheses of [Theorem 4.3](#) hold with growth rate g and dual solution (W^*, Λ^*, Y^*) , the welfare potential is Lipschitz, with steepness bound*

$$\frac{\underline{W}(\lambda; \varepsilon) - \underline{W}(\lambda)}{\|\varepsilon\|_{\infty}} \leq \|Y^*\|_{1, g}^*.$$

7. EXTENSIONS

The results presented thus far were confined to a repeated game. In this section we discuss some extensions of this basic framework to more general dynamic settings: games with frequent actions, stochastic games, and private information. As will be seen, the main methodological message of the paper prevails.

7.1. Games with Frequent Actions

Our first extension considers games with frequent actions. To this end, let $\delta = e^{-r dt}$, where $r > 0$ is a discount rate and $dt > 0$ is the calendar length of time per period. We will derive dual functional equations in the limit as $dt \rightarrow 0$ for asymptotically Brownian and Poisson signals. First, we describe the stochastic framework for Brownian motion to make comparisons more amenable with [Sannikov \(2007\)](#).

7.1.1. Brownian Signals

Consider the following binomial random walk representation of d -dimensional Brownian motion. Let $Z = \{\pm 1\}^d$, with typical element $z = (z_1, \dots, z_d)$ and conditional probability $\pi(z|a) = 2^{-d}[1 + \sum_k (\gamma_k(a)/\sigma_k^2) z_k dh_k]$, where $dh_k = \sigma_k \sqrt{dt}$ and $\sigma_k > 0$.⁴⁵ Let $S_{k0} = 0$, and for each $t \geq dt$, let $dS_{kt}(z) = z_k dh_k$, where z has probability mass given by $\pi(z|a_t)$ such that a_t is the action profile played in period $\lfloor t/dt \rfloor \in \mathbb{N}$. These realizations are independent across periods conditional on past action profiles. It is a standard exercise to verify that, as $dt \rightarrow 0$, each stock of signals $S_{kt} = \sum_{m=1}^{\lfloor t/dt \rfloor} dS_{kmdt}$ converges to a Brownian motion such that $S = (S_1, \dots, S_d)$ obeys the joint law

$$dS_t = \gamma(a_t)dt + \sigma dX_t,$$

where $X = (X_1, \dots, X_d)$ is a d -dimensional vector of mutually independent standard Brownian motions (i.e., with zero drift and unit volatility), $\gamma(a_t) = (\gamma_1(a_t), \dots, \gamma_d(a_t))$ is the drift vector and $\sigma = \text{diag}(\sigma_1, \dots, \sigma_d) \in \mathbb{R}^{d \times d}$ is the diagonal matrix of volatilities. Two immediate examples are $d = 1$, so there is only one signal, and $d = n$, so there is one signal per player. Introducing d -dimensional Brownian motion allows us to consider signal structures where pairwise identifiability fails (e.g., $d = 1 < n$) and ones where it holds. This has implications for properties of the boundary of the set of equilibrium payoff profiles, as argued by [Sannikov \(2007\)](#).

⁴⁵For simplicity, we assume that increments z_k and z_ℓ , where $k \neq \ell$, are statistically uncorrelated, though they are still related by the drifts $\gamma_k(a)$ and $\gamma_\ell(a)$ through the influence of action profile a . There are many ways of introducing correlation among increments. One way of doing so is replacing $\pi(z|a)$ with $2^{-d}[1 + \sum_k (\gamma_k(a)/\sigma_k^2) z_k dh_k + \sum_{k < \ell} \rho_{k\ell} z_k z_\ell]$, with correlation coefficient $\rho_{k\ell} \in (-1, 1)$ between dS_k and dS_ℓ . In this case, the derivations below would be similar, but more cluttered.

We now take the frequent actions limit (i.e., $dt \rightarrow 0$) of the dual functional equation of [Proposition 4.1](#) to derive a Hamilton-Jacobi-Bellman (HJB) equation for welfare and a law of motion for welfare weights. In preparation, we introduce the following notation: for each player $i \in N$, action profile $a \in A$, deviation $b_i \in A_i$ by i , and signal dimension $k \in \{1, \dots, d\}$, let $\Delta\gamma_k(a, b_i) = \gamma_k(b_i, a_{-i}) - \gamma_k(a)$ be the change in drift of the k th signal process when player i plays b_i instead of a_i and everyone else plays a_{-i} . Write $\Delta\gamma_{ik}(a) = (\Delta\gamma_k(a, b_i))_{b_i \in A_i} \in \mathbb{R}^{A_i}$ for the (column) vector that collects all these changes across player i 's deviations for the k th signal process. Finally, let

$$\Delta\gamma_i(a) = [\Delta\gamma_{i1}(a) \ \cdots \ \Delta\gamma_{id}(a)]$$

denote the matrix in $\mathbb{R}^{A_i \times d}$ of all drift changes across public signals for player i .

PROPOSITION 7.1. *The dual functional equation for ppCE with Brownian signals converges to the following welfare optimality equation as $dt \rightarrow 0$:*

$$\begin{aligned} rW(\lambda) = \inf_{y \geq 0} \max_{a \in A} \ & r \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \Delta u_i(a) \\ & + \frac{1}{2} \sum_{(i,j)} W_{ij}(\lambda) y_i(a_i) \Delta\gamma_i(a) \sigma^{-2} (y_j(a_j) \Delta\gamma_j(a))^\top, \end{aligned}$$

where $W_{ij}(\lambda) = \frac{\partial^2 W(\lambda)}{\partial \lambda_i \partial \lambda_j}$ and $^\top$ means transpose. Welfare weights λ follow the law

$$d\lambda_i(a_t) = -y_i(a_{it}) \Delta\gamma_i(a_t) \sigma^{-2} [dS_t - \gamma(a_t)dt].$$

The same equations apply to ppRE with $y_i(a)$ instead of $y_i(a_i)$.

The HJB equations of [Proposition 7.1](#) for ppCE and ppRE above are linear-quadratic in the shadow costs y of incentive constraints. They relate the Hessian matrix of W with deviation gains Δu_i and detection $\Delta\gamma_i$. Since W is a support function, it is convex, so the Hessian of W is a positive semi-definite matrix. Therefore, the second summation in the HJB equation is always non-negative. To see this, let $Y_k(a) = (y_i(a_i) \Delta\gamma_{ik}(a) / \sigma_k)_{i \in N} \in \mathbb{R}^N$ denote the (row) vector corresponding to the volatility contributions of the k th signal process to each player's welfare weight. In this case, the welfare optimality equation can be expressed more succinctly as

$$rW(\lambda) = \inf_{y \geq 0} \max_{a \in A} \ & r \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \Delta u_i(a) + \frac{1}{2} \sum_{k=1}^d Y_k(a) H(\lambda) Y_k(a)^\top,$$

where $H(\lambda)$ is the Hessian matrix of W at λ . As players become impatient, i.e., $r \rightarrow \infty$, or signals become uninformative, i.e., $\sigma^2 \rightarrow \infty$, the second term disappears from the HJB equations and the remaining terms correspond to the dual formulation

of maximal welfare in correlated equilibrium (for ppCE) and pure-strategy equilibrium (for ppRE) of the one-shot game. More interestingly, as in [Sannikov \(2007\)](#), the HJB equation shows how curvature of W provides incentives for players to cooperate on actions that may be suboptimal from a static perspective, thus adding to welfare.

To better understand the HJB equations above, consider first the version for ppRE. With ppRE, the choice variables $y_i(a, b_i)$ may vary with each $a \in A$, therefore the order of inf and max can be reversed without any loss of generality. As a result, each $y_i(a, b_i)$ can be found as the result of simple linear Kuhn-Tucker conditions:

$$\sum_{j \in N} W_{ij}(\lambda) \Delta \gamma_i(a) \sigma^{-2} (y_j(a) \Delta \gamma_j(a))^\top \geq r \Delta u_i(a) \quad \text{and} \quad y_i(a) \geq 0$$

with (elementwise) complementary slackness for all (i, a) . If there is no solution, say because $\Delta u_i(a) > 0$ but $\Delta \gamma_i(a) = 0$, for instance, then there is no bounded solution to the quadratic in y above and the right-hand side of the welfare optimality equation excludes a as an argument in its maximization, i.e., action profile a is not played. In games with product structure, the spans of $\Delta \gamma_i(a) \sigma^{-1}$ and $\Delta \gamma_j(a) \sigma^{-1}$ are orthogonal for $i \neq j$, so $\Delta \gamma_i(a) \sigma^{-2} (\Delta \gamma_j(a))^\top = 0$ and the Kuhn-Tucker conditions simplify to

$$W_{ii}(\lambda) \Delta \gamma_i(a) \sigma^{-2} \Delta \gamma_i(a)^\top y_i(a)^\top \geq r \Delta u_i(a) \quad \text{and} \quad y_i(a) \geq 0$$

with complementary slackness. This needs $W_{ii}(\lambda) > 0$ if $\Delta u_i(a, b_i) > 0$ for some b_i .⁴⁶

In ppCE, the relevant dual variables are $y_i(a_i, b_i)$, which may only vary with a_i and not a_{-i} . As a result, as discussed in [Section 5.1](#), the order of inf and max is generally not interchangeable, and the standard Kuhn-Tucker conditions above no longer apply. Instead, values across action profiles are equated on the support of optimal correlated strategies. If optimal solutions of the welfare optimality equation for ppRE do not depend on others' actions then ppCE and ppRE payoffs coincide. As [Proposition 5.1](#) shows, this occurs if the game has sum and product structure.

Following [Theorem 4.1](#), the maximal fixed point of the welfare optimality equation above characterizes equilibrium payoffs in the frequent actions limit. Just as when $dt > 0$ is fixed, the relevant space for a fixed point remains that of positively homogeneous convex functions, which need not be differentiable everywhere. Thus, maximal fixed points may be viscosity solutions, where flats in the boundary of the equilibrium payoff set correspond to non-differentiability of (i.e., kinks in) the welfare potential.

⁴⁶Else, since $\Delta \gamma_i(a) \sigma^{-2} \Delta \gamma_i(a)^\top$ is positive semi-definite, if $\Delta \gamma_i(a) \sigma^{-2} (y_i(a) \Delta \gamma_i(a))^\top$ is a negative vector, $y_i(a) \Delta \gamma_i(a) \sigma^{-1} (y_i(a) \Delta \gamma_i(a) \sigma^{-1})^\top \leq 0$ by $y_i(a) \geq 0$, so $y_i(a) \Delta \gamma_i(a) \sigma^{-1} = 0$, a contradiction.

7.1.2. Poisson Signals

Suppose that $Z = \{\pm 1\}$ and let $\pi(+1|a) = \eta(a)dt > 0$, so $\pi(-1|a) = 1 - \eta(a)dt$, and $\Delta\eta(a, b_i) = \eta(b_i, a_{-i}) - \eta(a)$. As $dt \rightarrow 0$, this converges to a Poisson distribution. In particular, the density of arrival of the event $+1$ in an small interval of time $[t, t + dt)$ equals $\eta(a_t)e^{-\eta(a_t)dt}$. The limiting HJB equation for welfare in ppCE is as follows:

$$\begin{aligned} rW(\lambda) = \inf_{y \geq 0} \max_{a \in A} r \sum_{i \in N} [\lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a)] \\ + \eta(a)[W(\lambda'(a)) - W(\lambda)] + \sum_{i \in N} W_i(\lambda) y_i(a_i) \cdot \Delta \eta_i(a), \\ \text{where } \lambda'_i(a) = \lambda_i - y_i(a_i) \cdot \frac{\Delta \eta_i(a)}{\eta(a)}. \end{aligned}$$

The dual problem for ppRE is the same except that $y_i(a, b_i)$ may also depend on a_{-i} .

7.2. Other Extensions

We end this section with two extensions of our basic model: stochastic games and games with private monitoring. As will be seen, they constitute minor modifications.

7.2.1. Private Signals

Games with private monitoring are easy to incorporate into the framework above. Let $Z = Z_0 \times \prod_i Z_i$ with typical element $z = (z_0, z_1, \dots, z_n)$, where z_0 is the publicly observed signal and z_i is the signal observed only by player i . Assume for simplicity that players observe their private signal before the public signal. Players are assumed to be able to secretly report to the mediator their signal realizations, with the following timing: after players take their actions, they privately and independently submit a report to the mediator. After everyone has submitted their report, the public signal realizes, the mediator publicly announces the period's recommendation profile, and the period ends. A *reporting strategy* is any map $\rho_i : Z_i \rightarrow Z_i$. Let $R_i = \{\rho_i : Z_i \rightarrow Z_i\}$ be the set of all reporting strategies for player i . For instance, truthful reporting, denoted by τ_i , satisfies $\tau_i(z_i) = z_i$. Let

$$\pi(z|b_i, a_{-i}, \rho_i) = \sum_{\omega_i \in \rho_i^{-1}(z_i)} \pi(\omega_i, z_{-i}|b_i, a_{-i})$$

be the probability that the signal profile z is *reported* given that i played b_i , reported according to ρ_i , and everyone else played a_{-i} and reported truthfully. It is the sum of probabilities of signals ω_i that could have led i to report z_i when following ρ_i , i.e., ω_i belonging to $\rho_i^{-1}(z_i)$. Finally, write $\Delta\pi(z|a, b_i, \rho_i) = \pi(z|b_i, a_{-i}, \rho_i) - \pi(z|a)$.

The previous analysis can be undertaken with the following incentive constraints:

$$(1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) C(a; h_{t-1}) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i, \rho_i) V_i(a, z) \leq 0 \quad \forall (i, a_i, b_i, \rho_i)$$

in ppCE, with the corresponding version in ppRE (i.e., indexing constraints with respect to a_{-i} instead of summing across a_{-i}). All of the results above go through.

7.2.2. Stochastic Games

Let S be a finite set of *states*, with typical element s and initial state is $s_0 \in S$. Transitions from one state to another are assumed to be Markovian, governed by the transition probability $\pi(z, s'|s, a)$ of next period's state being s' whenever the previous state was s , the previous action profile played was a , and the realized public signal was z . The state realizes and is publicly observed at the end of each period.

Players' flow payoffs may depend on the state; let $u_i(a, s)$ be player i 's utility flow from action profile a in state s . A partial history is now defined as follows. At first, $H_0 = S$ is the set of initial histories, with typical element h_0 . Next, a 1-history is now any tuple $h_1 = (h_0, a, z, s') \in H_0 \times A \times Z \times S$, with H_1 the set of all such 1-histories. Continuing this way, let $h_t = (h_{t-1}, a, z, s') \in H_{t-1} \times A \times Z \times S$ be a t -history, with H_t the set of all such t -histories and $H = \bigcup_t H_t$ the set of all partial histories.

Now, the primal functional equation for ppCE is given for every $\lambda \in \mathbb{R}^N$ by

$$\begin{aligned} W(\lambda|s) &= \sup_{B \geq 0, V} \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a, s) B(a) + \delta \sum_{(a, z, s')} \pi(z, s'|s, a) V_i(a, z, s') \right] \text{ s.t.} \\ &\quad \sum_{a \in A} B(a) = 1, \quad [B(a)W]^*(V(a, z, s')|s') \leq 0 \quad \forall (a, y, s'), \\ &\quad (1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i, s) B(a) + \delta \sum_{(a_{-i}, z, s')} \Delta \pi(z, s'|a, b_i, s) V_i(a, z, s') \leq 0 \quad \forall (i, a_i, b_i), \end{aligned}$$

where $B \in \mathbb{R}^A$, $V \in \mathbb{R}^{N \times A \times Z \times S}$ and $[B(a)W]^*(w|s) = \sup_{\mu} \{\mu \cdot w - B(a)W(\mu|s)\}$ is the conjugate function of $B(a)W(\mu|s)$, with conjugation with respect to μ . (A similar functional equation corresponds to ppRE.) The dual functional equation becomes

$$\begin{aligned} W(\lambda|s) &= \inf_{y \geq 0} \max_{a \in A} \sum_{i \in N} \lambda_i u_i(a, s) - y_i(a_i) \cdot \Delta u_i(a, s) + \delta \sum_{(z, s')} W(\mu(a, z, s')) \text{ s.t.} \\ &\quad \mu_i(a, z, s') = \lambda_i \pi(z, s'|a, s) - y_i(a_i) \cdot \Delta \pi_i(z, s'|s, a) \quad \forall (i, a, z, s'), \end{aligned}$$

where $\Delta u_i(a, s) = (\Delta u_i(a, b_i, s))_{b_i \in A_i}$ and $\Delta \pi_i(z, s'|s, a) = (\Delta \pi_i(z, s'|a, b_i, s))_{b_i \in A_i}$. The same functional equation applies to ppRE with $y_i(a)$ instead of $y_i(a_i)$, as before.

8. CONCLUSION

The goal of this paper was to develop and analyze dual recursive methods for dynamic games. After defining two equilibrium concepts, we formulated and connected their primal and dual sequence problems as well as primal and dual functional equations for finding extremal equilibrium welfare. Developing an appropriate sequence problem applicable under general distributional conditions led to a linear program, with convex programs for the functional equations. Establishing interconnections between each of these parts required new analysis and results of theoretical and applied significance. Our main applications covered computing equilibrium payoffs, optimal contracting, and robustness of equilibrium to errors of incentive compatibility. We also observed the prevalence of kinks and flats in equilibrium payoff boundaries.

Although some extensions were initiated in [Section 7](#), many more come to mind, such as recursive utility, continuous action and signal spaces, and non-compact payoff sets. One avenue being pursued currently is that of extending and analyzing thoroughly the problem of endogenous costly monitoring in [Section 6.3.2](#). This is the subject of [Birkan and Rahman \(2025\)](#).

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APPENDIX A. ONLINE APPENDIX—PROOFS

Proof of Lemma 2.1

This result follows from a simple calculation:

$$\begin{aligned}
V_i(h_0) &= (1 - \delta) \sum_{t \geq 1} \delta^{t-1} \mathbb{E}[u_i(C_t)] = (1 - \delta) \left(\mathbb{E}[u_i(C_1)] + \sum_{t \geq 2} \delta^{t-1} \mathbb{E}[u_i(C_t)] \right) \\
&= (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_0) + \delta \sum_{(a, z)} \left((1 - \delta) \sum_{t \geq 2} \delta^{t-2} \mathbb{E}[u_i(C_t) | a, z] \right) \pi(z | a) C(a; h_0) \\
&= (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_0) + \delta \sum_{(a, z)} \pi(z | a) V_i(a, z),
\end{aligned}$$

where $V_i(a, z) = (1 - \delta) \sum_{t \geq 2} \delta^{t-2} \mathbb{E}[u_i(C_t) | a, z] C(a; h_0) = (1 - \delta) \sum_{t \geq 2} \delta^{t-2} \mathbb{E}[u_i(C_t); a, z]$. $\square$

Proof of Lemma 3.1

First, telescope (1) to obtain

$$V_i(h_{s-1}) = (1 - \delta) \sum_{t \geq s} \delta^{t-s} \mathbb{E}[u_i(C_t); h_{s-1}] + \lim_{t \rightarrow \infty} \delta^{t-s} \sum_{h_t \geq h_{s-1}} \prod_{\tau=s}^t \pi(z_\tau | a_\tau) V_i(h_t)$$

for each (i, h_{s-1}) . If $V \in \hat{\ell}_\infty^N$, the ‘bubble’ term equals zero: for all t and (i, h_{s-1}) ,

$$\begin{aligned}
\left| \sum_{h_t \geq h_{s-1}} \prod_{\tau=s}^t \pi(z_\tau | a_\tau) V_i(h_t) \right| &\leq \sum_{h_t \geq h_{s-1}} \prod_{\tau=s}^t \pi(z_\tau | a_\tau) |V_i(h_t)| \leq \sum_{a_s} \max_{z_s} \cdots \sum_{a_t} \max_{z_t} |V_i(h_t)| \\
&\leq \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} \max_{z_t} \sum_{i \in N} |V_i(h_t)| = |V|_t,
\end{aligned}$$

so

$$\left| \lim_{t \rightarrow \infty} \delta^{t-s} \sum_{h_t \geq h_{s-1}} \prod_{\tau=s}^t \pi(z_\tau | a_\tau) V_i(h_t) \right| \leq \lim_{t \rightarrow \infty} \delta^{t-s} |V|_t \leq \lim_{t \rightarrow \infty} \delta^{t-s} \|V\|_\infty = 0,$$

since $V \in \hat{\ell}_\infty^N$ implies $\|V\|_\infty = \sup_t |V|_{t-1} < +\infty$. Conversely, suppose that

$$V_i(h_{s-1}) = (1 - \delta) \sum_{t \geq s} \delta^{t-s} \mathbb{E}[u_i(C_t); h_{s-1}]$$

for all (i, h_{s-1}) . Letting $\|u\| = \max_a \sum_i |u_i(a)|$, it follows that

$$|V|_{s-1} = \sum_{a_1} \max_{z_1} \cdots \sum_{a_{s-1}} \max_{z_{s-1}} \sum_{i \in N} \left| (1 - \delta) \sum_{t \geq s} \delta^{t-s} \mathbb{E}[u_i(C_t); h_{s-1}] \right| \leq (1 - \delta) \sum_{t \geq s} \delta^{t-s} \|u\| = \|u\|$$

for every $s \in \mathbb{N}$, so $\|V\|_\infty = \sup_t |V|_{t-1} \leq \|u\| < +\infty$, therefore $V \in \hat{\ell}_\infty^N$. $\square$

Proof of Lemma 3.2

If $X \in \hat{\ell}_\infty^S$ and $Y \in \hat{\ell}_1^{*S}$ then, for each $t \in \mathbb{N}$,

$$\begin{aligned}
\left| \sum_{h_{t-1}} X(h_{t-1}) Y(h_{t-1}) \right| &= \left| \sum_{a^{t-1}} \sum_{z^{t-1}} X(a^{t-1}, z^{t-1}) Y(a^{t-1}, z^{t-1}) \right| \\
&\leq \sum_{a^{t-1}} \sum_{z^{t-1}} \left| X(a^{t-1}, z^{t-1}) Y(a^{t-1}, z^{t-1}) \right| \\
&\leq \sum_{a^{t-1}} \sum_{z^{t-1}} \left(\sum_{s \in S} |X(s; a^{t-1}, z^{t-1})| \right) \left(\max_{s \in S} |Y(s; a^{t-1}, z^{t-1})| \right) \\
&\leq \sum_{a^{t-2}} \sum_{z^{t-2}} \sum_{a_{t-1}} \left(\max_{z_{t-1}} \sum_{s \in S} |X(s; a^{t-1}, z^{t-1})| \right) \left(\sum_{z_{t-1}} \max_{s \in S} |Y(s; a^{t-1}, z^{t-1})| \right) \\
&\leq \sum_{a^{t-2}} \sum_{z^{t-2}} \left(\sum_{a_{t-1}} \max_{z_{t-1}} \sum_{s \in S} |X(s; a^{t-1}, z^{t-1})| \right) \left(\max_{a_{t-1}} \sum_{z_{t-1}} \max_{s \in S} |Y(s; a^{t-1}, z^{t-1})| \right) \\
&\leq \cdots \leq |X|_{t-1} |Y|_{t-1}^* < +\infty.
\end{aligned}$$

Therefore,

$$\begin{aligned}
|\langle X, Y \rangle| &\leq \sum_{t \in \mathbb{N}} \sum_{h_{t-1}} |X(h_{t-1}) Y(h_{t-1})| \leq \sum_{t \in \mathbb{N}} |X|_{t-1} |Y|_{t-1}^* \\
&\leq \left(\sup_{t \in \mathbb{N}} |X|_{t-1} \right) \sum_{t \in \mathbb{N}} |Y|_{t-1}^* = \|X\|_\infty \|Y\|_1^* < +\infty.
\end{aligned}$$

Next, let us prove that the families of semi-norms defined by $\{|\langle X, \cdot \rangle| : X \in \hat{\ell}_\infty^S\}$ and $\{|\langle \cdot, Y \rangle| : Y \in \hat{\ell}_1^{*S}\}$ separate the points of $\hat{\ell}_1^{*S}$ and $\hat{\ell}_\infty^S$, respectively. This is straightforward. Let X be a primal process such that $X(\hat{h}_{t-1}) \neq 0$ for some $\hat{h}_{t-1} \in H$. If Y is a dual process defined pointwise by $Y(\hat{h}_{t-1}) = X(\hat{h}_{t-1})$ and $Y(h_{s-1}) = 0$ for all other $h_{s-1} \in H$ then $\langle X, Y \rangle \neq 0$, so primal processes are separated. If Y is a dual process such that $Y(\hat{h}_{t-1}) \neq 0$ then, letting X be the primal process defined by $X(\hat{h}_{t-1}) = Y(\hat{h}_{t-1})$ and $X(h_{s-1}) = 0$ for all other $h_{s-1} \in H$, it follows that $\langle X, Y \rangle \neq 0$, thus dual processes are separated, too.

Next, we show that $(\hat{\ell}_1^{*S}, \|\cdot\|_1^*)^* = \hat{\ell}_\infty^S$. First, $(\hat{\ell}_1^{*S}, \|\cdot\|_1^*)^* \supset \hat{\ell}_\infty^S$ because, since $\langle \cdot, \cdot \rangle$ is bilinear and $|\langle X, Y \rangle| \leq \|X\|_\infty \|Y\|_1^*$ by the inequality above, every $X \in \hat{\ell}_\infty^S$ is a norm-bounded linear functional on $\hat{\ell}_1^{*S}$, hence norm-continuous. Conversely, note first that the norm-continuous linear functionals form a vector space, since the sum of finitely many continuous functionals is continuous. For any partial history $\hat{h}_{t-1}$ and $\hat{s} \in S$, let $\mathbf{1}_{\hat{h}_{t-1}}^{\hat{s}}$ be the indicator function of $(\hat{s}; \hat{h}_{t-1})$, equal to 1 if $(s; h_{t-1}) = (\hat{s}; \hat{h}_{t-1})$ and 0 otherwise. This indicator function is clearly a dual process, i.e., $\mathbf{1}_{\hat{h}_{t-1}}^{\hat{s}} \in \hat{\ell}_1^{*S}$.

For any continuous linear functional $f \in (\hat{\ell}_1^{*S}, \|\cdot\|_1^*)^*$, the evaluation $f(\mathbf{1}_{h_{t-1}}^s)$ gives a scalar $f_{h_{t-1}}^s$ for each $(s; h_{t-1}) \in S \times H$. Since f is norm-continuous, it is bounded by some $K < +\infty$, i.e., for any dual process Y , its inner product with f satisfies $|\langle f, Y \rangle| \leq K \|Y\|_1 < +\infty$. We will use this inequality to show that $f \in \hat{\ell}_\infty^S$.

Fix any $r \in \mathbb{N}$. For any node (a^{r-1}, z^{r-2}) , we will define iteratively the signal realizations that attain each maxima in the definition of the semi-norm of f at $r-1$. Specifically, let us define some objects and describe them immediately afterwards:

$$\begin{aligned} \bar{z}_{r-1}(a^{r-1}, z^{r-2}) &\in \arg \max_{z_{r-1}} \sum_{s \in S} |f_{(a^{r-1}, z^{r-1})}^s|, \\ \bar{z}_{r-2}(a^{r-2}, z^{r-3}) &\in \arg \max_{z_{r-2}} \sum_{a_{r-1}} \sum_{s \in S} |f_{(a^{r-1}, z^{r-2}, \bar{z}_{r-1}(a^{r-1}, z^{r-2}))}^s|, \\ \bar{z}_{r-3}(a^{r-3}, z^{r-4}) &\in \arg \max_{z_{r-3}} \sum_{a_{r-2}, a_{r-1}} \sum_{s \in S} |f_{(a^{r-1}, z^{r-3}, \bar{z}_{r-2}^{r-1}(a^{r-1}, z^{r-3}))}^s|, \\ \dots \quad \bar{z}_1(a_1) &\in \arg \max_{z_1} \sum_{a_2, \dots, a_{r-1}} \sum_{s \in S} |f_{(a^{r-1}, z_1, \bar{z}_2^{r-1}(a^{r-1}, z_1))}^s|, \end{aligned}$$

where

$$\begin{aligned} \bar{z}_{r-2}^{r-1}(a^{r-1}, z^{r-3}) &= (\bar{z}_{r-2}(a^{r-2}, z^{r-3}), \bar{z}_{r-1}(a^{r-1}, z^{r-3}, \bar{z}_{r-2}(a^{r-2}, z^{r-3}))), \\ \bar{z}_{r-3}^{r-1}(a^{r-1}, z^{r-4}) &= (\bar{z}_{r-3}(a^{r-3}, z^{r-4}), \bar{z}_{r-2}^{r-1}(a^{r-1}, z^{r-3})), \\ \dots \quad \bar{z}_3^{r-1}(a^{r-1}, z^2) &= (\bar{z}_3(a^3, z^2), \bar{z}_4^{r-1}(a^{r-1}, z^3)), \\ \bar{z}_2^{r-1}(a^{r-1}, z_1) &= (\bar{z}_2(a^2, z_1), \bar{z}_3^{r-1}(a^{r-1}, z^2)), \\ \bar{z}_1^{r-1}(a^{r-1}) &= (\bar{z}_1(a_1), \bar{z}_2^{r-1}(a^{r-1}, z_1)). \end{aligned}$$

We just defined two families of variables, $\bar{z}_{r-k}$ and $\bar{z}_{r-k}^{r-1}$. The first family, $\bar{z}_{r-k}$, gives the signal z at time $r-k$ that attains the maximum in the $(r-1)$ -semi-norm construction as a function of the history (a^{r-k}, z_{r-k-1}) . The second family, $\bar{z}_{r-k}^{r-1}$, collects these maximal z 's over time from $r-k$ to $r-1$ as a function the remaining components of an $(r-1)$ -history. Now, choose the dual process Y_r given by

$$Y_r(s; a^{t-1}, z^{t-1}) = \begin{cases} \operatorname{sgn} f_{(a^{r-1}, z^{r-1})}^s & \text{if } t = r \text{ and } (a^{r-1}, z^{r-1}) = (a^{r-1}, \bar{z}_1^{r-1}(a^{r-1})), \\ 0 & \text{otherwise.} \end{cases}$$

Clearly, $Y_r \in \hat{\ell}_1^{*S}$, since

$$\|Y_r\|_1^* = |Y_r|_{r-1}^* = \max_{a_1} \sum_{z_1} \cdots \max_{a_{r-1}} \sum_{z_{r-1}} \max_{s \in S} |Y_r(s; a^{r-1}, z^{r-1})| = 1.$$

Moreover, by construction (and since Y_r has finite support),

$$\begin{aligned}\langle f, Y_r \rangle &= \sum_{a^{r-1}} \sum_{z^{r-1}} \sum_{s \in S} f_{(a^{r-1}, z^{r-1})}^s Y_r(s; a^{r-1}, z^{r-1}) \\ &= \sum_{a_1} \max_{z_1} \cdots \sum_{a_{r-1}} \max_{z_{r-1}} \sum_{s \in S} |f_{(a^{r-1}, z^{r-1})}| = |f|_{r-1},\end{aligned}$$

so $|f|_{r-1} < +\infty$. It remains to show that $\sup_t |f|_{t-1} < +\infty$, since this implies $f \in \hat{\ell}_\infty^S$. If not, $|f|_{t-1} \rightarrow +\infty$ as $t \rightarrow +\infty$, but then for each $K \in \mathbb{R}$ there is $r \in \mathbb{N}$ such that $|\langle f, Y_r \rangle| = |f|_{r-1} \|Y_r\|_1^* > K \|Y_r\|_1^*$, contradicting boundedness of the linear functional f . Hence, $f \in \hat{\ell}_\infty^S$ and $(\hat{\ell}_1^{*S}, \|\cdot\|_1^*)^* \subset \hat{\ell}_\infty^S$. This completes the proof of [Lemma 3.2](#). $\square$

Proof of Theorem 3.1

The first part of [Theorem 3.1](#) follows from [Lemma 3.1](#) and the incentive constraints (2) for ppCE, (3) for ppRE. Next, we show that the primal sequence problem's feasible set is weak* compact and convex. This implies that the equilibrium payoff set is compact, convex, since the projection of a compact, convex set is compact, convex, completing the proof of [Theorem 3.1](#). Convexity of the feasible set follows by linearity. Weak* compactness is proved in three steps. First, by Alaoglu's Theorem, the unit ball of the space of choice variables is weak* compact. Next, the feasible set is shown to be weak* closed. Being contained in a weak* compact ball, it, too, is weak* compact.

Step 1 of 3: Alaoglu's Theorem

Start with $(\mathbf{E}, \|\cdot\|_1^*)$, where $\mathbf{E} = \hat{\ell}_1^{*A} \times \hat{\ell}_1^{*N}$ and, for any $D = (Y, \Lambda) \in \mathbf{E}$ with $Y \in \hat{\ell}_1^{*A}$ and $\Lambda \in \hat{\ell}_1^{*N}$, its norm is defined by $\|D\|_1^* = \|Y\|_1^* + \|\Lambda\|_1^*$, using the dual norm $\|\cdot\|_1^*$ on $\hat{\ell}_1^{*A}$ and $\hat{\ell}_1^{*N}$. Our choice space for the primal sequence problem consists of the space of pairs (C, V) in $\mathbf{P} = \hat{\ell}_\infty^A \times \hat{\ell}_\infty^N$ endowed with the weak* topology, since, by a trivial two-dimensional extension of [Lemma 3.2](#), $(\mathbf{E}, \|\cdot\|_1^*)^* = \mathbf{P}$. For any $P = (C, V) \in \mathbf{P}$ with $C \in \hat{\ell}_\infty^A$ and $V \in \hat{\ell}_\infty^N$, let $\|P\|_\infty = \max\{\|C\|_\infty, \|V\|_\infty\}$, using the primal norm $\|\cdot\|_\infty$ on C and V . First, we show that the unit ball of $\mathbf{P}$ is weak* compact.

LEMMA A.1. *The unit ball $\mathcal{B} = \{P \in \mathbf{P} : \|P\|_\infty \leq 1\}$ of $\mathbf{P}$ is weak* compact.*

Proof. By Alaoglu's Theorem (e.g., [Conway, 1990](#), p. 130), the unit ball of the dual space of a normed space is weak* compact. By [Lemma 3.2](#), $\hat{\ell}_1^{*S}$ is a normed space with dual space $\hat{\ell}_\infty^S$, so the unit ball of $\hat{\ell}_\infty^S$ is weak* compact. By Tychonoff's Theorem, the Cartesian product of two compact spaces is compact, so $\mathcal{B}$ is weak* compact. $\square$

By [Lemma A.1](#), the unit ball of $\mathbf{P}$ is weak* compact, so the Cartesian product $\hat{\mathcal{B}}$ of the unit ball of $\hat{\ell}_\infty^A$ and the unit ball of $\hat{\ell}_\infty^N$ scaled by $\|u\|$ is weak* compact, too.

Step 2 of 3: The Primal Feasible Set is Weak Closed*

Next, we show that the feasible set for the primal sequence problem is weak* closed. First, let $\mathbf{P}_+ = \hat{\ell}_{\infty+}^A \times \hat{\ell}_{\infty}^N$, where $\hat{\ell}_{\infty+}^A$ is the set of non-negative vectors in $\hat{\ell}_{\infty}^A$ and vectors in $\hat{\ell}_{\infty}^N$ need not be non-negative. Because $\mathbf{P}$ is a Banach lattice, the cone $\mathbf{P}_+$ is weak* closed. We first describe the sequence problem's feasibility constraints in terms of dual variables. We begin with the equality constraints that define a public correlated strategy. For each fixed $\hat{h} \in H$, define $Y^{\hat{h}} \in \hat{\ell}_1^{*A}$ as follows. If $\hat{h} = h_0$, i.e., the null history, then, for every $(a; h_{t-1}) \in A \times H$, let

$$Y^{h_0}(a; h_{t-1}) = \begin{cases} 1 & \text{if } h_{t-1} = h_0, \\ 0 & \text{otherwise.} \end{cases}$$

If $\hat{h} = (\hat{h}_{s-1}, \hat{a}, \hat{z}) \neq h_0$ then

$$Y^{\hat{h}}(a; h_{t-1}) = \begin{cases} 1 & \text{if } h_{t-1} = \hat{h}, \\ -1 & \text{if } (a; h_{t-1}) = (\hat{a}; \hat{h}_{s-1}), \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $Y^{\hat{h}}$ is either the indicator of the null history or a function that equals 1 at the non-null history $\hat{h}$ and -1 at its previous node. As in the previous paragraph, let $D^{\hat{h}} = (Y^{\hat{h}}, 0)$, where $0 \in \hat{\ell}_1^{*N}$ is the zero vector. By construction, it is clear that the operations $\langle P, D^{\hat{h}} \rangle$ realize the equality constraints for a public correlated strategy for each $\hat{h} \in H$. It is also clear that $D^{\hat{h}} \in \mathbf{E}$ for each $\hat{h} \in H$, since $\|D^{h_0}\|_1^* = |A| < +\infty$ and $\|D^{\hat{h}}\|_1^* = |A| + 1 < +\infty$. Therefore, each $D^{\hat{h}}$ is a weak* continuous linear functional on $\mathbf{P}$, with weak* closed kernel $\ker(D^{\hat{h}}) = \{P \in \mathbf{P} : \langle P, D^{\hat{h}} \rangle = 0\}$.

Continuing in this fashion, we describe the value recursion equations as stemming from linear functionals obtained from elements of $\mathbf{E}$. Given $\hat{g} = (\hat{i}, \hat{h}) \in N \times H$, let

$$Y^{\hat{g}}(a; h_{t-1}) = \begin{cases} (1 - \delta)u_i(a) & \text{if } (i, h_{t-1}) = \hat{g}, \\ 0 & \text{otherwise,} \end{cases}$$

$$\Lambda_i^{\hat{g}}(h_{t-1}) = \begin{cases} \delta\pi(z|a) & \text{if } (i, h_{t-1}) = (\hat{i}, \hat{h}, a, z), \\ -1 & \text{if } (i, h_{t-1}) = (\hat{i}, \hat{h}), \\ 0 & \text{otherwise.} \end{cases}$$

If $D^{\hat{g}} = (Y^{\hat{g}}, \Lambda^{\hat{g}})$, the norms of $Y^{\hat{g}}$ and $\Lambda^{\hat{g}}$ yield $\|D^{\hat{g}}\|_1^* = (1 - \delta)\|u_i\|_1 + 1 + \delta$, where $\|u_i\|_1 = \sum_a |u_i(a)|$. Hence, again, $D^{\hat{g}} \in \mathbf{E}$ and the kernel $\ker(D^{\hat{g}})$ is weak* closed.

Finally, consider the incentive constraints. For each $\hat{f} = (\hat{i}, \hat{h}, \hat{a}_i, \hat{b}_i)$, define

$$Y^{\hat{f}}(a; h_{t-1}) = \begin{cases} (1 - \delta)\Delta u_i(a, b_i) & \text{if } (i, h_{t-1}, a, b_i) = \hat{f}, \\ 0 & \text{otherwise,} \end{cases}$$

$$\Lambda_i^{\hat{f}}(h_{t-1}) = \begin{cases} \delta\Delta\pi(z|a, b_i) & \text{if } (i, h_{t-1}, a, b_i) = (\hat{i}, (\hat{h}, a, z), \hat{a}_i, \hat{b}_i), \\ 0 & \text{otherwise.} \end{cases}$$

By direct calculation, if $D^{\hat{f}} = (Y^{\hat{f}}, \Lambda^{\hat{f}})$ then $\|D^{\hat{f}}\|_1 \leq 2|A|[(1 - \delta)\|u_i\|_\infty + \delta] < +\infty$, where $\|u_i\|_\infty = \max_a |u_i(a)|$, so $D^{\hat{f}} \in \mathbf{E}$ and the kernel $\ker(D^{\hat{f}})$ is weak* closed.

Combining the previous arguments,

$$\mathcal{F} = \mathbf{P}_+ \cap \bigcap_{\hat{f}} D^{\hat{f}} \cap \bigcap_{\hat{g}} D^{\hat{g}} \cap \bigcap_{\hat{h}} D^{\hat{h}}$$

is weak* closed by virtue of being the intersection of weak* closed sets. This set $\mathcal{F}$ is the feasible set for the primal sequence problem.

Step 3 of 3: Weak Compactness of the Feasible Set*

The last step of the proof consists of combining the two previous steps. By the discussion immediately preceding the statement of [Lemma 3.1](#) and the proof of [Lemma 3.1](#) itself, any feasible $(C, V) \in \mathcal{F}$ must satisfy $\|C\|_\infty = 1$ and $\|V\|_\infty \leq \|u\|$. Therefore, $\mathcal{F} \subset \hat{\mathcal{B}}$, where $\hat{\mathcal{B}}$, defined at the end of Step 1, is weak* compact. Therefore, $\mathcal{F}$ is a weak* closed subset of a weak* compact set, hence weak* compact, too.

Finally, for ppRE we replace the dual-variable representations of incentive constraints in the obvious way to obtain the same result, i.e., for each $\hat{f} = (\hat{i}, \hat{h}, \hat{a}, \hat{b}_i)$, define

$$Y^{\hat{f}}(a; h_{t-1}) = \begin{cases} (1 - \delta)\Delta u_i(a, b_i) & \text{if } (i, h_{t-1}, a, b_i) = \hat{f}, \\ 0 & \text{otherwise,} \end{cases}$$

$$\Lambda_i^{\hat{f}}(h_{t-1}) = \begin{cases} \delta\Delta\pi(z|a, b_i) & \text{if } (i, h_{t-1}, a, b_i) = (\hat{i}, (\hat{h}, a, z), \hat{a}, \hat{b}_i), \\ 0 & \text{otherwise.} \end{cases}$$

The last step is to show that the set of ppCE is nonempty. But this is immediate, since every finite stage game has a correlated equilibrium, and the infinite repetition of a correlated equilibrium of the stage game is a ppCE. This non-emptiness does not follow for ppRE because a stage game need not have a pure-strategy Nash equilibrium.

This concludes the proof of [Theorem 3.1](#). $\square$

Proof of Proposition 3.1

The proof proceeds as follows. First, we describe the linear operator that defines the primal constraints, then take its adjoint, construct a Lagrangian and infer the dual.

Let $\mathbf{P} = \hat{\ell}_\infty^A \times \hat{\ell}_\infty^N$ be the space of *primal variables* with positive cone $\mathbf{P}_+ = \hat{\ell}_{\infty+}^A \times \hat{\ell}_\infty^N$, where $\hat{\ell}_{\infty+}^A$ is the set of non-negative vectors in $\hat{\ell}_\infty^A$ and vectors in $\hat{\ell}_\infty^N$ need not be non-negative. Let $\mathbf{Q} = \hat{\ell}_\infty \times \hat{\ell}_\infty^N \times \hat{\ell}_\infty^{\Pi_i A_i \times A_i}$ be the space of *primal constraints*. Let $M : \mathbf{P} \rightarrow \mathbf{Q}$ be the linear operator that maps primal variables (C, V) in $\mathbf{P}$ to primal constraints $M(C, V) = (I, J, K)$ in $\mathbf{Q}$ where, for all $(i, h_{t-1}, a_t, z_t, a_i, b_i)$,

- (i) $I(h_0) = \sum_a C(a; h_0)$ and $I(h_{t-1}, a_t, z_t) = \sum_a C(a; h_{t-1}, a_t, z_t) - C(a_t; h_{t-1})$,
- (ii) $J_i(h_{t-1}) = V_i(h_{t-1}) - (1 - \delta) \sum_a u_i(a) C(a; h_{t-1}) - \delta \sum_{(a,z)} \pi(z|a) V_i(h_{t-1}, a, z)$,
- (iii) $K_i(h_{t-1}, a_i, b_i) = (1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) C(a; h_{t-1}) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(h_{t-1}, a, z)$.

Thus, M realizes the left-hand side of the primal sequence problem's constraints.

LEMMA A.2. M is a bounded, hence norm-continuous, linear operator.

Proof. Clearly, M is linear. For boundedness, let $P = (C, V) \in \mathbf{P}$, with norm given by $\|P\|_\infty = \max\{\|C\|_\infty, \|V\|_\infty\} < +\infty$. We will show that there exists $c \in [0, +\infty)$ such that $\|M(C, V)\|_\infty = \max\{\|I\|_\infty, \|J\|_\infty, \|K\|_\infty\} \leq c \|(C, V)\|_\infty$. Consider first I . Of course, $|I|_0 = |\sum_a C(a; h_0)| \leq \sum_a |C(a; h_0)| = |C|_0$. For $t > 0$,

$$\begin{aligned} |I|_t &= \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} \max_{z_t} \left| \sum_{a \in A} C(a; h_{t-1}, a_t, z_t) - C(a_t; h_{t-1}) \right| \\ &\leq \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} \max_{z_t} \left| \sum_{a \in A} C(a; h_{t-1}, a_t, z_t) \right| + |C(a_t; h_{t-1})| \\ &\leq \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} \max_{z_t} \left| \sum_{a \in A} C(a; h_{t-1}, a_t, z_t) \right| + \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} |C(a_t; h_{t-1})| = |C|_t + |C|_{t-1}. \end{aligned}$$

Hence, $\|I\|_\infty = \sup_t |I|_{t-1} \leq \sup_t |C|_{t-1} + |C|_{t-2} \leq 2\|C\|_\infty \leq 2\|(C, V)\|_\infty$, where $|C|_{-1} = 0$. Now consider J . Adding with respect to i the absolute value of $J_i(h_{t-1})$,

$$\begin{aligned} \sum_{i \in N} |J_i(h_{t-1})| &= \sum_{i \in N} \left| V_i(h_{t-1}) - (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_{t-1}) - \delta \sum_{(a,z)} \pi(z|a) V_i(h_{t-1}, a, z) \right| \\ &\leq \sum_{i \in N} |V_i(h_{t-1})| + (1 - \delta) \sum_{i \in N} \sum_{a \in A} |u_i(a)| |C(a; h_{t-1})| + \sum_{i \in N} \delta \sum_{(a,z)} \pi(z|a) |V_i(h_{t-1}, a, z)|. \end{aligned}$$

Therefore, taking the max-sum-...-max-sum with respect to (h_{t-1}, a, z) on each side of this inequality and letting $\|u\| = \sum_i \max_a |u_i(a)|$, it follows that, for every $t \in \mathbb{N}$,

$$|J|_{t-1} \leq |V|_{t-1} + (1 - \delta) \|u\| |C|_{t-1} + \delta |V|_t.$$

Finally, taking suprema, this clearly implies that

$$\begin{aligned}\|J\|_\infty &\leq \|V\|_\infty + (1 - \delta) \|u\| \|C\|_\infty + \delta \|V\|_\infty = (1 + \delta) \|V\|_\infty + \|u\| \|C\|_\infty \\ &\leq (1 + \delta + (1 - \delta) \|u\|) \max\{\|C\|_\infty, \|V\|_\infty\} = (1 + \delta + (1 - \delta) \|u\|) \|(C, V)\|_\infty.\end{aligned}$$

A similar argument applies to $K \in \hat{\ell}_\infty^{\prod_i A_i \times A_i}$:

$$\begin{aligned}\sum_{(i, a_i, b_i)} |K_i(h_{t-1}, a_i, b_i)| &\leq (1 - \delta) \sum_{(i, a_i, b_i)} \sum_{a-i} |\Delta u_i(a, b_i) C(a; h_{t-1})| + \delta \sum_{(a-i, z)} |\Delta \pi(z|a, b_i) V_i(h_{t-1}, a, z)| \\ &\leq (1 - \delta) \sum_{(i, b_i)} 2 \|u\| \sum_{a \in A} |C(a; h_{t-1})| + \delta \sum_{(i, b_i)} \sum_{(a, z)} 2 |V_i(h_{t-1}, a, z)|.\end{aligned}$$

Taking the max-sum-...-max-sum of the left-hand side gives $|K|_{t-1}$. If $\alpha = \sum_i |A_i|$ is the sum across players of the number of actions $|A_i|$ available to each i , the first term on the right-hand side equals $2 \|u\| \alpha |C|_{t-1}$. For the second term, notice that

$$\begin{aligned}\sum_{(i, b_i)} \sum_{(a, z)} |V_i(h_{t-1}, a, z)| &\leq \alpha \sum_{i \in N} \sum_{(a, z)} |V_i(h_{t-1}, a, z)| \\ &= \alpha \sum_{(a, z)} \sum_{i \in N} |V_i(h_{t-1}, a, z)| \\ &\leq \alpha |Z| \sum_{a_t} \max_{z_t} \sum_{i \in N} |V_i(h_{t-1}, a_t, z_t)| \\ &\leq \alpha |Z| \sum_{a_1} \max_{z_1} \cdots \sum_{a_t} \max_{z_t} \sum_{i \in N} |V_i(h_{t-1}, a_t, z_t)| = \alpha |Z| |V|_t.\end{aligned}$$

Therefore, for every $t \in \mathbb{N}$, it follows that $|K|_{t-1} \leq (1 - \delta) 2 \|u\| \alpha |C|_{t-1} + 2\delta \alpha |Z| |V|_t$. Now, taking suprema with respect to t ,

$$\|K\|_\infty \leq (1 - \delta) 2 \|u\| \alpha \|C\|_\infty + 2\delta \alpha |Z| \|V\|_\infty \leq 2\alpha((1 - \delta) \|u\| + \delta |Z|) \|(C, V)\|_\infty.$$

Putting together the inequalities for I , J and K yields

$$\begin{aligned}\|M(C, V)\|_\infty &= \max\{\|I\|_\infty, \|J\|_\infty, \|K\|_\infty\} \\ &\leq \max\{2, 1 + \delta + (1 - \delta) \|u\|, 2\alpha((1 - \delta) \|u\| + \delta |Z|)\} \|(C, V)\|_\infty.\end{aligned}$$

Finally, $\max\{2, 1 + \delta + (1 - \delta) \|u\|, 2\alpha((1 - \delta) \|u\| + \delta |Z|)\} < +\infty$, so M is a bounded, hence norm-continuous, linear operator. This concludes the proof of [Lemma A.2](#). $\square$

Next, we define the adjoint M^* of M . First, let $\mathbf{D} = \hat{\ell}_1^* \times \hat{\ell}_1^{*N} \times \hat{\ell}_1^{*\prod_i A_i \times A_i}$ be the space of *dual variables* and positive cone $\mathbf{D}_+ = \hat{\ell}_1^* \times \hat{\ell}_1^{*N} \times \hat{\ell}_{1+}^{*\prod_i A_i \times A_i}$. (Only the third family of variables is non-negative.) The space of *dual constraints* is $\mathbf{E} = \hat{\ell}_1^{*A} \times \hat{\ell}_1^{*N}$, with positive cone $\mathbf{E}_+ = \hat{\ell}_{1+}^{*A} \times \{0\}$, where $0 \in \hat{\ell}_1^{*N}$.

The operator $M^* : \mathbf{D} \rightarrow \mathbf{E}$ takes a vector of dual variables $(W, \Lambda, Y) \in \mathbf{D}$ and maps it into a vector of dual constraints $(F, G) \in \mathbf{E}$ such that, for every (i, h_{t-1}, a) ,

- (i) $F(h_{t-1}, a) = W(h_{t-1}) - (1 - \delta) \sum_i [\Lambda_i(h_{t-1}) u_i(a) - Y_i(h_{t-1}, a_i) \Delta u_i(a)] - \sum_z W(h_{t-1}, a, z)$,
- (ii) $G_i(h_0) = \Lambda_i(h_0)$, $G_i(h_{t-1}, a, z) = \Lambda_i(h_{t-1}, a, z) - \delta [\Lambda_i(h_{t-1}) \pi(z|a) - Y_i(h_{t-1}, a_i) \Delta \pi_i(z|a)]$.

LEMMA A.3. *The operator M^* is the adjoint of M , i.e.,*

$$\langle MP, D \rangle = \langle P, M^* D \rangle$$

for every $P \in \mathbf{P}$ and $D \in \mathbf{D}$, where $\langle \cdot, \cdot \rangle$ is the usual inner product, as in [Lemma 3.2](#).

Proof. If $P = (C, V)$ and $D = (W, \Lambda, Y)$ are any vectors of primal and dual variables,

$$\begin{aligned} \langle MP, D \rangle &= W(h_0) \sum_{a \in A} C(a; h_0) + \sum_{(h_{t-1}, a_t, z_t)} W(h_{t-1}, a_t, z_t) \left[\sum_{a \in A} C(a; h_{t-1}, a_t, z_t) - C(a_t; h_{t-1}) \right] \\ &\quad + \sum_{i \in N} \sum_{h_{t-1}} \Lambda_i(h_{t-1}) [V_i(h_{t-1}) - (1 - \delta) \sum_{a \in A} u_i(a) C(a; h_{t-1}) - \delta \sum_{(a, z)} \pi(z|a) V_i(h_{t-1}, a, z)] \\ &\quad + \sum_{i \in N} \sum_{(h_{t-1}, a_i, b_i)} Y_i(h_{t-1}, a_i, b_i) (1 - \delta) \left[\sum_{a-i} \Delta u_i(a, b_i) C(a; h_{t-1}) + \delta \sum_{(a-i, z)} \Delta \pi(z|a, b_i) V_i(h_{t-1}, a, z) \right] \\ &= \sum_{i \in N} V_i(h_0) \Lambda_i(h_0) + \sum_{(h_{t-1}, a)} C(a; h_{t-1}) \left[W(h_{t-1}) - \sum_{z \in Z} W(h_{t-1}, a, z) \right. \\ &\quad \left. - (1 - \delta) \sum_{i \in N} \left[\Lambda_i(h_{t-1}) u_i(a) - \sum_{b_i \in A_i} Y_i(h_{t-1}, a_i, b_i) \Delta u_i(a, b_i) \right] \right] \\ &\quad + \sum_{i \in N} \sum_{(h_{t-1}, a, z)} V_i(h_{t-1}, a, z) \left[\Lambda_i(h_{t-1}, a, z) - \delta \left[\Lambda_i(h_{t-1}) \pi(z|a) - \sum_{b_i \in A_i} Y_i(h_{t-1}, a_i, b_i) \Delta \pi(z|a, b_i) \right] \right] \\ &= \langle (C, V), M^*(W, \Lambda, Y) \rangle = \langle P, M^* D \rangle, \end{aligned}$$

as required. $\square$

Define right-hand side primal and dual constraints $\overline{Q}$ and $\overline{E}$, respectively, as follows. First, let $\overline{Q} = (\overline{I}, \overline{J}, \overline{K})$ satisfy (i) $\overline{I}(h_0) = 1$, $\overline{I}(h_{t-1}, a, z) = 0$ for all (h_{t-1}, a, z) , (ii) $\overline{J} \equiv 0$, and (iii) $\overline{K} \equiv 0$. Next, given $\lambda \in \mathbb{R}^N$, define $\overline{E} = (\overline{F}, \overline{G})$ by (i) $\overline{F} \equiv 0$, and (ii) $\overline{G}(h_0) = \lambda$, $\overline{G}(h_{t-1}, a, z) = 0$ for all (h_{t-1}, a, z) .

In the notation above, the primal of [Definition 3.3](#) can be rewritten succinctly as

$$(P) \quad \sup_{P \in \mathbf{P}_+} \{ \langle P, \overline{E} \rangle : MP \leq \overline{Q} \}.$$

The dual of this linear programming problem is given by

$$(D) \quad \inf_{D \in \mathbf{D}_+} \{ \langle D, \overline{Q} \rangle : M^* D \geq \overline{E} \},$$

where $M^* D \geq \overline{E}$ means $M^* D - \overline{E} \in \mathbf{E}_+$. Substituting yields the dual problem

$$\begin{aligned} & \inf_{Y \geq 0, \Lambda, W} W(h_0) \quad \text{s.t.} \quad \Lambda_i(h_0) = \lambda_i \quad \forall i \in N, \\ W(h_{t-1}) & \geq (1 - \delta) \sum_{i \in N} [\Lambda_i(h_{t-1}) u_i(a) - Y_i(h_{t-1}, a_i) \Delta u_i(a)] + \sum_{z \in Z} W(h_{t-1}, a, z) \quad \forall (h_{t-1}, a), \\ \Lambda_i(h_{t-1}, a, z) & = \delta [\Lambda_i(h_{t-1}) \pi(z|a) - Y_i(h_{t-1}, a_i) \Delta \pi_i(z|a)] \quad \forall (i, h_{t-1}, a, z), \end{aligned}$$

with $(W, \Lambda, Y) \in \mathbf{D}$ such that $Y \geq 0$ (i.e., $Y \in \hat{\ell}_{1+}^{\prod_i A_i \times A_i}$). To show that the dual is feasible, let $W(h_{t-1}) = \delta^{t-1} \|u\|_\infty \prod_{s=1}^{t-1} \pi(z_s|a_s) \sum_i |\lambda_i|$ for all h_{t-1} , where $\|u\|_\infty = \max_{i,a} \{u_i(a)\}$, $\Lambda_i(h_{t-1}) = \delta^{t-1} \lambda_i \prod_{s=1}^{t-1} \pi(z_s|a_s)$, and $Y_i(h_{t-1}, a_i, b_i) = 0$ for all (i, h_{t-1}, a_i, b_i) . Clearly, this (W, Λ, Y) is feasible for the dual problem. The proof for ppRE is identical, hence omitted. This completes the proof of [Proposition 3.1](#). $\square$

Proof of Theorem 3.2

By convention, $\underline{W}(\lambda) = -\infty$ if the primal feasible set is empty. In the notation of [Proposition 3.1](#), if $P = (C, V)$ is feasible for the primal and $D = (W, \Lambda, Y)$ for the dual,

$$\sum_{i \in N} \lambda_i V_i(h_0) = \langle P, \overline{E} \rangle \leq \langle P, M^* D \rangle = \langle MP, D \rangle \leq \langle D, \overline{Q} \rangle = W(h_0).$$

This is known as weak duality. To establish stronger duality, we will apply the following result due to [Gale \(1967\)](#), where, for any $(\zeta, \xi) \in \hat{\ell}_1^A \times \hat{\ell}_1^N$,

$$\begin{aligned} (*) \quad \overline{W}(\lambda; \zeta, \xi) & = \inf_{Y \geq 0, \Lambda, W} W(h_0) \quad \text{s.t.} \quad -\xi_{i-}(h_0) \leq \Lambda_i(h_0) - \lambda_i \leq \xi_{i+}(h_0), \\ W(h_{t-1}) & \geq \sum_{i \in N} (1 - \delta) [\Lambda_i(h_{t-1}) u_i(a) - Y_i(h_{t-1}, a_i) \Delta u_i(a)] + \sum_{z \in Z} W(h_{t-1}, a, z) + \zeta(a; h_{t-1}), \\ -\xi_{i-}(h_{t-1}, a, z) & \leq \Lambda_i(h_{t-1}, a, z) - \delta [\Lambda_i(h_{t-1}) \pi(z|a) - Y_i(h_{t-1}, a_i) \Delta \pi_i(z|a)] \leq \xi_{i+}(h_{t-1}, a, z). \end{aligned}$$

PROPOSITION A.1. *The following statements are equivalent: (i) both the primal has an optimal solution and there is no duality gap, so $\underline{W}(\lambda) = \overline{W}(\lambda)$, (ii) $\overline{W}(\lambda; \zeta, \xi)$ is subdifferentiable with respect to (ζ, ξ) at $(\zeta, \xi) = 0$, (iii) $\overline{W}(\lambda; \zeta, \xi)$ has bounded steepness with respect to (ζ, ξ) :*

$$\sup_{(\zeta, \xi) \neq 0} \frac{\overline{W}(\lambda) - \overline{W}(\lambda; \zeta, \xi)}{\|(\zeta, \xi)\|} < +\infty.$$

See also [Gretsky et al. \(2002\)](#). We will verify (iii), implying that there is no duality gap. (By [Theorem 3.1](#) the primal already has a solution, since it has a weak* continuous objective on a non-empty weak* compact set whenever $\overline{W}(\lambda) > -\infty$.)

First, we assume that $\overline{W}(\lambda) > -\infty$. By weak duality, a sufficient condition for this is that the primal constraint set be non-empty, that is, equilibrium exists. This holds automatically for ppCE, but may or may not hold for ppRE, depending on the game. To show bounded steepness, we need only consider the case $\xi_+ \geq -\xi_-$, since otherwise the constraint set is empty and $\overline{W}(\lambda; \zeta, \xi) = +\infty$, hence $\overline{W}(\lambda) - \overline{W}(\lambda; \zeta, \xi) = -\infty$, so $[\overline{W}(\lambda) - \overline{W}(\lambda; \zeta, \xi)] / \|(\zeta, \xi)\|$ is bounded above. Thus, without loss of generality, let $\xi_+ \geq -\xi_-$. Moreover, by revealed preference, clearly $\overline{W}(\lambda; \zeta, \xi) \geq \overline{W}(\lambda; -|\zeta|, \xi)$, where $|\zeta|(a; h_{t-1}) = |\zeta(a; h_{t-1})|$ is the pointwise absolute value of ζ . Hence, without loss, let $\zeta \leq 0$, since $[\overline{W}(\lambda) - \overline{W}(\lambda; \zeta, \xi)] / \|(\zeta, \xi)\| \leq [\overline{W}(\lambda) - \overline{W}(\lambda; -|\zeta|, \xi)] / \|(\zeta, \xi)\|$.

We now show bounded steepness in three steps. As a first step, we find an equivalent expression for $\overline{W}(\lambda; \zeta, \xi)$. Clearly, without loss some welfare constraints bind. Indeed, reducing a collection of slack welfare constraints indexed by $(a; h_{t-1})$ for each $a \in A$ by reducing $W(h_{t-1})$ does not increase the value of the objective and possibly reduces it. Moreover, assuming for now that $(\zeta, \xi) = 0$ and unfolding the multiplier constraints in $\overline{W}(\lambda)$ into the welfare constraints with left-hand side denoted by $W_0(h_0)$ yields

$$\begin{aligned} W_0(h_0) &\geq (1 - \delta) \sum_{i \in N} \lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1) + \sum_{z_1} W(h_0, a_1, z_1) \quad \forall a_1, \\ &\geq (1 - \delta) \sum_{i \in N} \lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1) + \sum_{z_1} \delta [\lambda_i \pi(z_1 | a_1) - Y_i(h_0, a_1) \Delta \pi_i(z_1 | a_1)] u_i(a_2) \\ &\quad - Y_i(h_1, a_{i2}) \Delta u_i(a_2) + \sum_{z_2} W(h_0, a_1, z_1, a_2, z_2) \quad \forall (a_1, a_2(h_1)), \\ &\dots \geq \sum_{i \in N} \lambda_i V_i(\mathbf{a}; h_0) - \sum_{h_{t-1}} Y_i(h_{t-1}, \mathbf{a}_{it}(h_{t-1})) [\Delta u_i(\mathbf{a}_t(h_{t-1})) \\ &\quad + \delta \sum_{z_t} \Delta \pi_i(z_t | \mathbf{a}_t(h_{t-1})) V_i(\mathbf{a}; h_{t-1}, \mathbf{a}_t(h_{t-1}), z_t)] \quad \forall \mathbf{a} \in A^H, \end{aligned}$$

where $V_i(\mathbf{a}; h_0) = (1 - \delta) \sum_{t \geq 1} \sum_{h_{t-1}} \delta^{t-1} \prod_{\tau=1}^{t-1} \pi(z_\tau | a_\tau) u_i(\mathbf{a}(h_{t-1}))$ and

$$V_i(\mathbf{a}; h_{s-1}) = (1 - \delta) \sum_{t \geq s} \sum_{h_{t-1} \geq h_{s-1}} \delta^{t-s} \prod_{\tau=s}^{t-1} \pi(z_\tau | a_\tau) u_i(\mathbf{a}(h_{t-1})),$$

since $W \in \hat{\ell}_1^*$, so $\|W\|_1 = \sum_t |W|_{t-1} < +\infty$ and as a result $|W|_{t-1} \rightarrow 0$ as $t \rightarrow \infty$, hence $W(h_{t-1}) \rightarrow 0$, too, where $|W|_{t-1} = \max_{a_1} \sum_{z_1} \dots \max_{a_{t-1}} \sum_{z_{t-1}} |W(h_{t-1})|$.

Our second step is to show that, given (ζ, ξ) and feasible (W, Λ, Y) of problem $(*)$ with value $\bar{W}(\lambda; \zeta, \xi)$, the value of the objective $W(h_0)$ is bounded by

$$(**) \quad W(h_0) \geq W_0(h_0) - \|\zeta\|_1 - \|u\|_\infty \|\xi\|_1,$$

where $W_0(h_0)$ was defined in the previous step. We derive this inequality in several steps of simple but tedious algebra. Expanding $W(h_0)$, the objective in $(*)$,

$$\begin{aligned} W(h_0) &\geq \max_{a_1} \sum_{i \in N} (1 - \delta) [\Lambda_i(h_0) u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] + \sum_{z_1} W(h_0, a_1, z_1) + \zeta(a_1; h_0) \\ &\geq \max_{a_1} \sum_{i \in N} (1 - \delta) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] + \sum_{z_1} W(h_0, a_1, z_1) \\ &\quad + \zeta(a_1; h_0) - \sum_{i \in N} (1 - \delta) \xi_i(h_0) |u_i(a_1)| \\ &\geq \max_{a_1} \sum_{i \in N} (1 - \delta) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] + \sum_{z_1} W(h_0, a_1, z_1) \\ &\quad - |\zeta|_0 - (1 - \delta) \|u\|_\infty |\xi|_0, \end{aligned}$$

where $\xi_i(h_0) = \max\{\xi_{i+}(h_0), \xi_{i-}(h_0)\}$; recall that $\zeta \leq 0$. Now, unfolding each $W(h_0, a_1, z_1)$ and using the fact that $\sum_{z_1} \pi(z_1|a_1) = 1$,

$$\begin{aligned} W(h_0) &\geq \max_{a_1} \sum_{z_1} \sum_{i \in N} (1 - \delta) \pi(z_1|a_1) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] \\ &\quad + W(h_0, a_1, z_1) - |\zeta|_0 - (1 - \delta) \|u\|_\infty |\xi|_0 \\ &\geq \max_{a_1} \sum_{z_1} \max_{a_2} \sum_{i \in N} (1 - \delta) \pi(z_1|a_1) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] \\ &\quad + \sum_{i \in N} (1 - \delta) [\Lambda_i(a_1, z_1) u_i(a_2) - Y_i(a_1, z_1, a_{i2}) \Delta u_i(a_2)] + \sum_{z_2} W(a_1, z_1, a_2, z_2) \\ &\quad + \zeta(a_2; a_1, z_1) - |\zeta|_0 - (1 - \delta) \|u\|_\infty |\xi|_0 \\ &\geq \max_{a_1} \sum_{z_1} \max_{a_2} \sum_{i \in N} (1 - \delta) \pi(z_1|a_1) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] \\ &\quad + \sum_{i \in N} (1 - \delta) \delta [\Lambda_i(h_0) \pi(z_1|a_1) - Y_i(h_0, a_{i1}) \Delta \pi_i(z_1|a_1)] u_i(a_2) \\ &\quad - (1 - \delta) \xi_i(a_1, z_1) |u_i(a_2)| - (1 - \delta) Y_i(a_1, z_1, a_{i2}) \Delta u_i(a_2) + \sum_{z_2} W(a_1, z_1, a_2, z_2) \\ &\quad + \zeta(a_2; a_1, z_1) - |\zeta|_0 - (1 - \delta) \|u\|_\infty |\xi|_0. \end{aligned}$$

Therefore, continuing in this fashion, we obtain that

$$\begin{aligned}
W(h_0) &\geq \max_{a_1} \sum_{z_1} \max_{a_2} \sum_{i \in N} (1 - \delta) \pi(z_1|a_1) [\lambda_i u_i(a_1) - Y_i(h_0, a_{i1}) \Delta u_i(a_1)] \\
&\quad + \sum_{i \in N} (1 - \delta) \delta [\lambda_i \pi(z_1|a_1) - Y_i(h_0, a_{i1}) \Delta \pi_i(z_1|a_1)] u_i(a_2) \\
&\quad - (1 - \delta) [\delta \xi_i(h_0) \pi(z_1|a_1) + \xi_i(a_1, z_1)] |u_i(a_2)| - (1 - \delta) Y_i(a_1, z_1, a_{i2}) \Delta u_i(a_2) \\
&\quad + \sum_{z_2} W(a_1, z_1, a_2, z_2) + \zeta(a_2; a_1, z_1) - |\zeta|_0 - (1 - \delta) \|u\|_\infty |\xi|_0.
\end{aligned}$$

A little further algebra yields

$$\begin{aligned}
W(h_0) &\geq \max_{a_1} \sum_{z_1} \max_{a_2} \sum_{i \in N} (1 - \delta) \pi(z_1|a_1) \lambda_i [u_i(a_1) + \delta u_i(a_2)] \\
&\quad - \sum_{i \in N} (1 - \delta) [Y_i(h_0, a_{i1}) (\Delta u_i(a_1) \pi(z_1|a_1) + \delta \Delta \pi_i(z_1|a_1) u_i(a_2)) + Y_i(a_1, z_1, a_{i2}) \Delta u_i(a_2)] \\
&\quad + \sum_{z_2} W(a_1, z_1, a_2, z_2) - |\zeta|_0 - |\zeta|_1 - (1 - \delta) \|u\|_\infty [(1 + \delta) |\xi|_0 + |\xi|_1].
\end{aligned}$$

Multiply $\pi(z_2|a_2)$, unfold $W(a_1, z_1, a_2, z_2)$, rearrange, multiply $\pi(z_3|a_3)$, etc., to get

$$\begin{aligned}
W(h_0) &\geq \lim_{t \rightarrow \infty} \max_{a_1} \sum_{z_1} \cdots \max_{a_t} \sum_{z_t} \sum_{i \in N} (1 - \delta) \prod_{s=1}^t \pi(z_s|a_s) \lambda_i \sum_{s=1}^t \delta^{s-1} u_i(a_s) \\
&\quad - \sum_{i \in N} (1 - \delta) \sum_{s=1}^t Y_i(h_{s-1}, a_{is}) \prod_{r=s}^{t-1} \pi(z_{r+1}|a_{r+1}) [\Delta u_i(a_s) \pi(z_s|a_s) \\
&\quad + \Delta \pi_i(z_s|a_s) \sum_{r=s}^{t-1} \delta^r u_i(a_{r+1})] \\
&\quad - \sum_{s=0}^{t-1} |\zeta|_s + \|u\|_\infty (1 - \delta^{t-s}) |\xi|_s.
\end{aligned}$$

The first two terms converge to $W_0(h_0)$ and the last summation to $\|\zeta\|_1 + \|u\|_\infty \|\xi\|_1$, therefore $W(h_0) \geq W_0(h_0) - \|\zeta\|_1 - \|u\|_\infty \|\xi\|_1$. Substituting into (**),

$$\frac{\overline{W}(\lambda) - \overline{W}(\lambda; \zeta, \xi)}{\|(\zeta, \xi)\|} \leq \frac{\|\zeta\|_1 + \|u\|_\infty \|\xi\|_1}{\|\zeta\|_1 + \|\xi\|_1} \leq 1 + \|u\|_\infty < +\infty.$$

This proves bounded steepness. By [Proposition A.1](#), the primal has a solution and there is no duality gap, i.e., $\underline{W}(\lambda) = \overline{W}(\lambda)$. Finally, if $\overline{W}(\lambda) = -\infty$, steepness is not well defined and subdifferentiability fails. By weak duality, $\underline{W}(\lambda) = -\infty$, the primal feasible set is empty. Thus, $\underline{W}(\lambda) = \overline{W}(\lambda)$ also when the primal feasible set is empty.

This completes the proof of [Theorem 3.2](#).

Proof of Theorem 4.1

The proof of Theorem 4.1 has four parts. First, we introduce preliminaries. Next, we define a Bellman operator for ppCE, and then relate it to the functional equation of Theorem 4.1. The proof for ppRE is applied verbatim. We end by arguing that the supremum in the functional equation is attained.

Step 1: Preliminaries

A *poset* is a pair $(L, \leq)$ such that L is a set and $\leq$ is a partial order, i.e., a reflexive ($a \leq a$), antisymmetric ($a \leq b \leq a$ implies $a = b$), and transitive ($a \leq b \leq c$ implies $a \leq c$) binary relation. A poset $(L, \leq)$ is a *complete lattice* if every subset A of L has a least upper bound and a greatest lower bound in L (hence L is not empty).

THEOREM A.1 (TARSKI, 1955). *If $(L, \leq)$ is a complete lattice and $f : L \rightarrow L$ a monotone operator then the set of fixed points of f in L forms a complete lattice with respect to $\leq$. In particular, the set of fixed points is nonempty and has a maximal element.*

We will apply Tarski's Theorem above to prove Theorem 4.1, in several steps. We start by constructing two related lattices. Let $\mathcal{C}$ be the collection of all closed, convex subsets of $\mathbb{R}^N$. For any subset $\mathcal{S} \subset \mathcal{C}$, define the operations $\bigwedge$ and $\bigvee$, called *meet* and *join*, respectively:

$$\bigwedge_{F \in \mathcal{S}} F = \bigcap_{F \in \mathcal{S}} F \quad \text{and} \quad \bigvee_{F \in \mathcal{S}} F = \overline{\text{conv}}\left(\bigcup_{F \in \mathcal{S}} F\right),$$

where $\overline{\text{conv}}$ stands for the closure of the convex hull of a set.

CLAIM A.1. *The pair $(\mathcal{C}, \subset)$ is a complete lattice, where $\subset$ denotes weak set inclusion.*

Proof. Clearly, $(\mathcal{C}, \subset)$ is a poset. Let $\mathcal{S} \subset \mathcal{C}$ be a collection of closed convex sets. If $A \subset F$ for all $F \in \mathcal{S}$ then $A \subset \bigcap\{F \in \mathcal{S}\} = \bigwedge\{F \in \mathcal{S}\}$, therefore $\bigwedge\{F \in \mathcal{S}\} \in \mathcal{C}$ is the greatest lower bound of $\mathcal{S}$. If $F \subset G$ for all $F \in \mathcal{S}$ and $G \in \mathcal{C}$ then $\bigcup\{F \in \mathcal{S}\} \subset G$. Since convex hulls and closures are monotone operations on sets,

$$\bigvee\{F \in \mathcal{S}\} = \overline{\text{conv}}\left(\bigcup\{F \in \mathcal{S}\}\right) \subset \overline{\text{conv}} G = G,$$

so $\bigvee\{F \in \mathcal{S}\} \in \mathcal{C}$ is the least upper bound of $\mathcal{S}$. □

For any $F \in \mathcal{C}$, the *indicator* function of F is defined as

$$\delta_F(v) = \begin{cases} 0 & \text{if } v \in F, \\ +\infty & \text{otherwise.} \end{cases}$$

Thus, $\delta_F(\cdot)$ is a closed, convex function.⁴⁷ There is a bijection between closed convex subsets and their indicator functions (obviously) and their support functions (dually), where the *support* function of $F \in \mathcal{C}$ is the conjugate of δ_F :

$$\delta_F^*(\lambda) = \sup_{v \in \mathbb{R}^N} \{\lambda \cdot v - \delta_F(v)\} = \sup_{v \in F} \lambda \cdot v.$$

Clearly, $\delta_F^*(0) = 0$ whenever $F \neq \emptyset$ and $\delta_F^*(p\lambda) = p\delta_F^*(\lambda)$ whenever $p \geq 0$, so δ_F^* is positively homogeneous. (By convention, $\delta_F^*(\lambda) = -\infty$ for all $\lambda \in \mathbb{R}^N$ if $F = \emptyset$.) Clearly, if $F, G \in \mathcal{C}$ are closed convex sets and $F \subset G$ then $\delta_F^* \leq \delta_G^*$. A function $f : \mathbb{R}^N \rightarrow [-\infty, \infty]$ is *proper* if $f(v) < +\infty$ for some $v \in \mathbb{R}^n$ and $f(v) > -\infty$ for all $v \in \mathbb{R}^n$. If F is nonempty, δ_F^* is clearly proper and convex, and if F is closed, so is δ_F^* . The next result identifies the defining properties of support functions.

LEMMA A.4 (ROCKAFELLAR, 1970, §13). *There is a bijection between nonempty, closed, convex sets and closed, proper, convex, positively homogeneous functions.*

Recall that $U = \text{conv}\{u(a) : a \in A\}$ is the convex hull of payoff profiles of Γ and $\delta_U^*(\lambda) = \sup\{\lambda \cdot v : v \in U\}$ is the support function of U . This corresponds to the maximum attainable welfare with play in the game for every vector of welfare weights.

DEFINITION A.1. Let $\mathcal{P}_U$ be the set of all possibly extended-real-valued functions $f : \mathbb{R}^N \rightarrow [-\infty, \infty]$ that satisfy the following four conditions:

- (1) f is feasible: $f(\lambda) \leq \delta_U^*(\lambda)$ for every $\lambda \in \mathbb{R}^N$.
- (2) f is positively homogeneous: $f(p\lambda) = pf(\lambda)$ for every $p \geq 0$ and $\lambda \in \mathbb{R}^N$.
- (3) f is subadditive: $f(\lambda + \mu) \leq f(\lambda) + f(\mu)$ for every $\lambda, \mu \in \mathbb{R}^N$.
- (4) f is closed: the epigraph $\text{epi} f = \{(\lambda, y) \in \mathbb{R}^{N+1} : f(\lambda) \leq y\}$ is a closed set.

Endow $\mathcal{P}_U$ with the pointwise partial order, so for any two functions $f, g \in \mathcal{P}_U$, we have $f \leq g$ if and only if $f(\lambda) \leq g(\lambda)$ for every $\lambda \in \mathbb{R}^N$. Thus, $(\mathcal{P}_U, \leq)$ is a poset.

It is easy to see that a positively homogeneous, subadditive function is convex, therefore every function $f \in \mathcal{P}_U$ is positively homogeneous, closed, convex and feasible, i.e., less than or equal to δ_U^* . For f to also be proper, it must satisfy $f(\lambda) < +\infty$ for some λ and $f(\lambda) > -\infty$ for all λ . By definition, $\delta_U^*(\lambda) < +\infty$ for all λ , hence $f(\lambda) \leq \delta_U^*(\lambda) < +\infty$, too. Therefore, if $f(\lambda) = -\infty$ for some λ , then by subadditivity $f(\mu) \leq f(\lambda) + f(\mu - \lambda) = -\infty$. This means that $f(\lambda) = -\infty$ for some λ implies $f(\lambda) = -\infty$ for all λ . Using Lemma A.4, this observation implies the following result.

⁴⁷Convexity is obvious. Closedness of a convex function means that its epigraph, defined to be the set $\text{epi } \delta_F = \{(v, y) \in \mathbb{R}^N \times \mathbb{R} : \delta_F(v) \leq y\}$, is closed. This clearly holds when F is closed.

LEMMA A.5. $\mathcal{P}_U$ is the set of all the support functions of closed, convex subsets of U .

Note that $\mathcal{P}_U$ has only one improper function, $\delta_\emptyset^*$, the support function of the empty set. We now turn $(\mathcal{P}_U, \leq)$ into a complete lattice by defining suprema and infima as follows. By Lemma A.5, every $f \in \mathcal{F}$ is the support function of some closed, convex set $F(f)$. For any subset of functions $\mathcal{F} \subset \mathcal{P}_U$, define its *meet* by

$$\left(\bigwedge_{f \in \mathcal{F}} f \right) = \delta_{\bigcap_{f \in \mathcal{F}} F(f)}^*, \quad \text{where} \quad \bigwedge F(\mathcal{F}) = \bigcap_{f \in \mathcal{F}} F(f),$$

and its *join* similarly by

$$\left(\bigvee_{f \in \mathcal{F}} f \right) = \delta_{\overline{\text{conv} \bigcup_{f \in \mathcal{F}} F(f)}}^*, \quad \text{where} \quad \bigvee F(\mathcal{F}) = \overline{\text{conv} \bigcup_{f \in \mathcal{F}} F(f)}.$$

In words, the meet of a set of support functions is the support function of the intersection of the sets supported by the functions, and the join is the support function of the closed convex hull of the supported sets.

It is easy to see that $(\mathcal{P}_U, \leq)$ is a complete lattice, with maximal element δ_U^* and minimal element $\delta_\emptyset^*$, the function always mapping to $-\infty$. Indeed, for any set of functions $\mathcal{F} \subset \mathcal{P}_U$, clearly its meet and join both belong to $\mathcal{P}_U$, and the lattice operations are easily seen to be consistent with the pointwise order. Clearly, by construction the lattice structure of $\mathcal{P}_U$ is inherited from that of $\mathcal{C}$. This lattice structure is used below in order to apply Tarski's Theorem. First, let us define our Bellman operator.

Step 2: Bellman Operator

In this step, we define the Bellman operator associated with Theorem 4.1 and show that it has a fixed point. In the following step, we show that its maximal fixed point coincides with $\underline{W}$. Given the functional equation of Definition 4.1, its natural Bellman operator $\mathbf{B}$ is given by

$$\begin{aligned} \mathbf{B}W(\lambda) = \sup_{B \geq 0, V} \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] \quad \text{s.t.} \quad \sum_{a \in A} B(a) = 1, \\ (1 - \delta) \sum_{a_i} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(a, z) \leq 0 \quad \forall (i, a_i, b_i), \\ [B(a)W]^*(V(a, z)) \leq 0 \quad \forall (a, z), \end{aligned}$$

Let us view the operator $\mathbf{B}$ as mapping each function $W \in \mathcal{P}_U$ to some function $\mathbf{B}W$, with the equation above defining such function pointwise for each $\lambda \in \mathbb{R}^N$.

LEMMA A.6. $\mathbf{BW} \in \mathcal{P}_U$ whenever $W \in \mathcal{P}_U$.

Proof. Clearly, $\mathbf{BW}$ is positively homogeneous. It is also feasible: the conjugate constraint $[B(a)W]^*(V(a, z)) \leq 0$ for all (a, z) implies that the objective above satisfies

$$\begin{aligned} & \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] \\ & \leq (1 - \delta) \sum_{(i, a)} \lambda_i u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) B(a) W(\lambda) \\ & \leq (1 - \delta) \delta_U^*(\lambda) + \delta \sum_{(a, z)} \pi(z|a) B(a) \delta_U^*(\lambda) = \delta_U^*(\lambda), \end{aligned}$$

since $W \in \mathcal{P}_U$ is feasible. For subadditivity, if the feasible set is empty, we're done. For any feasible $(B^{\lambda+\mu}, V^{\lambda+\mu})$, there exist feasible (B^λ, V^λ) and (B^μ, V^μ) such that

$$\begin{aligned} & \sum_{i \in N} (\lambda_i + \mu_i) \left[(1 - \delta) \sum_{a \in A} u_i(a) B^{\lambda+\mu}(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^{\lambda+\mu}(a, z) \right] \\ & = \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B^{\lambda+\mu}(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^{\lambda+\mu}(a, z) \right] \\ & \quad + \sum_{i \in N} \mu_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B^{\lambda+\mu}(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^{\lambda+\mu}(a, z) \right] \\ & \leq \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B^\lambda(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^\lambda(a, z) \right] \\ & \quad + \sum_{i \in N} \mu_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B^\mu(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^\mu(a, z) \right], \end{aligned}$$

since $(B^\lambda, V^\lambda) = (B^\mu, V^\mu) = (B^{\lambda+\mu}, V^{\lambda+\mu})$ could be chosen. Subadditivity follows by taking suprema in turn with respect to (B^λ, V^λ) , (B^μ, V^μ) and $(B^{\lambda+\mu}, V^{\lambda+\mu})$.

Next, we show that $\mathbf{BW}$ is closed, i.e., it is lower semi-continuous. This follows easily because the constraint set is closed, as we argue next. First, the incentive constraints are closed, being linear in (B, V) : let $L_i(a_i, b_i) = (1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(a, z)$ be the left-hand side of the (i, a_i, b_i) th incentive constraint. The set $\{(B, V) : L_i(a_i, b_i) \leq 0\}$ is clearly closed and convex. Since the intersection of closed, convex sets is closed and convex, so is the set $L = \bigcap_{(i, a_i, b_i)} L_i(a_i, b_i)$. If the constraint set is empty, clearly $\mathbf{BW}$ is closed because $\mathbf{BW}(\lambda) = -\infty$ for all λ . Let $L' = L \cap \{(B, V) : B \geq 0, \sum_a B(a) = 1\}$. Again, this set is clearly closed and convex, too, being the intersection of closed, convex sets. Finally, let $L_\mu(a, z) = \{(B, V) : \sum_i \mu_i V_i(a, z) \leq B(a) W(\mu)\}$ be the constraint that, for a given

μ , welfare promised does not exceed welfare delivered. This is clearly a closed convex set, therefore $L'' = L' \cap \bigcap_{(\mu, a, z)} L_\mu(a, z)$ is a closed, convex subset of $\mathbb{R}^N$. But this L'' is the feasible set of the Bellman operator. Therefore, the Bellman operator satisfies

$$\mathbf{BW}(\lambda) = \sup_{(B, V) \in L''} \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right].$$

Moreover, letting $\mu = \pm \mathbf{1}_i$ for every player i , it follows that $V_i(a, z)$ is bounded above and below for every (i, a, z) . Therefore, since B is also bounded above and below, L'' is a compact set. Now, lower semi-continuity follows easily: for every λ there exists $(B, V) \in L''$ that attains the supremum of $\mathbf{BW}(\lambda)$. Let $\lambda^m \rightarrow \lambda$, and pick $(B^m, V^m), (B, V) \in L''$ that attain the corresponding supremum for each λ^m and λ , respectively. By revealed preference,

$$\begin{aligned} \mathbf{BW}(\lambda^m) &= \sum_{i \in N} \lambda_i^m \left[(1 - \delta) \sum_{a \in A} u_i(a) B^m(a) + \delta \sum_{(a, z)} \pi(z|a) V_i^m(a, z) \right] \\ &\geq \sum_{i \in N} \lambda_i^m \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] \\ &\rightarrow \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] = \mathbf{BW}(\lambda), \end{aligned}$$

therefore $\mathbf{BW}$ is lower semi-continuous, completing the proof of the lemma. $\square$

LEMMA A.7. *The operator $\mathbf{B} : \mathcal{P}_U \rightarrow \mathcal{P}_U$ is monotone.*

Proof. The proof of this result is elementary: let $L''(W)$ be the feasible set corresponding to the Bellman operator of the proof of Lemma A.6 with welfare function W . If $W \leq W'$ then clearly $L''(W) \subset L''(W')$, since, for any $\mu \in \mathbb{R}^N$ and $(a, z) \in A \times Z$,

$$\sum_{i \in N} \mu_i V_i(a, y) \leq B(a) W(\mu) \quad \Rightarrow \quad \sum_{i \in N} \mu_i V_i(a, y) \leq B(a) W'(\mu).$$

Therefore, by revealed preference, $\mathbf{BW} \leq \mathbf{BW}'$. $\square$

PROPOSITION A.2. *$\mathbf{B}$'s fixed points form a nonempty complete sub-lattice of $\mathcal{P}_U$.*

Proof. Follows from Tarski's Theorem (Theorem A.1), Lemmata A.6 and A.7. $\square$

Step 3: Back to the Sequence Problem

We now show that the maximal fixed point of the Bellman operator equals the sequence problem's value function $\underline{W}$, which, by Theorem 3.2, is the same for both primal and dual. To do so, first we show that every fixed point of $\mathbf{B}$ is bounded by $\underline{W}$, and then that $\underline{W}$ is a fixed point of $\mathbf{B}$, hence $\underline{W}$ is the maximal fixed point of $\mathbf{B}$.

LEMMA A.8. *If W is a fixed point of $\mathbf{B}$ then $W \leq \underline{W}$.*

Proof. Let $W \in \mathcal{P}_U$ be a fixed point of $\mathbf{B}$, so $W(\lambda) = \mathbf{B}(W)(\lambda)$ for all $\lambda \in \mathbb{R}^N$. If $W \equiv -\infty$, there is nothing to prove. If not, i.e., $W(\lambda) > -\infty$ for some λ , then W is proper. Hence, by Lemma A.4 and Footnote 17, W is subdifferentiable at 0, i.e., $\partial W(0)$ is nonempty. For any $\hat{V}^{h_0} \in \partial \mathbf{B}(W)(0) = \partial W(0)$, there exists $(B^{h_0}, V^{h_0}) \in \Delta A \times \mathbb{R}^{N \times A \times Z}$ feasible for the functional equation and satisfying $\hat{V}^{h_0} = (1 - \delta) \sum_a u(a) B^{h_0}(a) + \delta \sum_{(a,z)} \pi(z|a) V^{h_0}(a, z)$. For each 1-history $h_1 = (a, z)$, let

$$\hat{V}^{h_1} = \begin{cases} V^{h_0}(a, z)/B^{h_0}(a) & \text{if } B^{h_0}(a) > 0, \text{ and} \\ \hat{V}^{h_0} & \text{if } B^{h_0}(a) = 0. \end{cases}$$

Now, $\hat{V}^{h_1} \in \partial \mathbf{B}(W)(0) = \partial W(0)$ because (i) $V^{h_0}(a, z)/B^{h_0}(a) \in \partial W(0)$ when $B^{h_0}(a) > 0$, since $V^{h_0}(a, z) \in \partial(B^{h_0}(a)W)(0)$ by the conjugate constraint, and (ii) $\hat{V}^{h_0} \in \partial W(0)$ by hypothesis. Proceeding inductively, given $\hat{V}^{h_t} \in \partial \mathbf{B}(W)(0)$, there exists (B^{h_t}, V^{h_t}) feasible for the functional equation such that $\hat{V}^{h_t} = (1 - \delta) \sum_a u(a) B^{h_t}(a) + \delta \sum_{(a,z)} \pi(z|a) V^{h_t}(a, z)$, which leads to defining

$$\hat{V}^{h_{t+1}} = \begin{cases} V^{h_t}(a, z)/B^{h_t}(a) & \text{if } B^{h_t}(a) > 0, \text{ and} \\ \hat{V}^{h_t} & \text{if } B^{h_t}(a) = 0. \end{cases}$$

Again, $\hat{V}^{h_{t+1}} \in \partial W(0)$ for every partial history h_{t+1} . We now build a ppCE with the objects above. Start with $V(h_0) = \hat{V}^{h_0}$ and $C(a; h_0) = B^{h_0}(a)$. Next, $V(a_1, z_1) = B^{h_0}(a_1) V^{h_0}(a_1, z_1)$ and $C(a; a_1, z_1) = B^{h_0}(a_1) B^{h_0}(a_1, z_1)$. Finally, for any h_t , let $V(h_t) = C(a_t; h_{t-1}) \hat{V}^{h_t}$ and $C(a; h_t) = C(a_t; h_{t-1}) B^{h_t}(a)$. Clearly, C is a public correlated strategy with continuation values V , by value recursion. Incentive compatibility holds by feasibility for the functional equation after every partial history, so C is a ppCE. Hence, $\lambda \cdot V(h_0) \leq \underline{W}(\lambda)$, since $\underline{W}(\lambda)$ takes the supremum over all ppCE. $\square$

LEMMA A.9. *$\underline{W}$ is a fixed point of $\mathbf{B}$.*

Proof. $\underline{W} \equiv -\infty$ is clearly a fixed point of $\mathbf{B}$. If $\underline{W}(\lambda) > -\infty$ for some λ then, as was argued, there exists $V(h_0) \in \partial \underline{W}(0)$, so there exists a public correlated strategy C together with probability-weighted continuation values V that attain the value $V(h_0)$, that is, $V(h_0) = (1 - \delta) \sum_a u(a) C(a; h_0) + \delta \sum_{(a,z)} \pi(z|a) V(a, z)$ and both the value recursion equation (1) as well as incentive compatibility (2) hold for (C, V) .

We first show that $V(h_0) \in \partial \mathbf{B}(\underline{W})(0)$. For every partial history $h_1 = (a, z)$, if $C(a; h_0) = 0$ then $V(a, z) = 0$ by Lemma 2.1. If $C(a; h_0) > 0$ then $V(a, z) \in \partial(C(a; h_0) \underline{W})(0)$. To see this, define the public correlated strategy C' pointwise by

$C'(a'; h) = C(a'; a, z, h)/C(a; h_0)$ and the probability-weighted continuation values V' pointwise by $V'(h) = V(h, a, z)/C(a; h_0)$. The pair (C', V') is feasible for the sequence problem, since it clearly satisfies all of its constraints, therefore $V'(h_0) \in \partial \overline{W}(0)$. Hence, $V(a, z) \in \partial(C(a; h_0)\underline{W})(0)$ for all (a, z) , and, since clearly incentive compatibility holds for (C, V) , it follows that $V(h_0) \in \partial \mathbf{B}(\underline{W})(0)$.

Conversely, let $\hat{V}^{h_0} \in \partial \mathbf{B}(\underline{W})(0)$. We will show that $\hat{V}^{h_0} \in \partial \underline{W}(0)$. Since $\hat{V}^{h_0} \in \partial \mathbf{B}(\underline{W})(0)$, there is $(B^{h_0}, V^{h_0}) \in \Delta A \times \mathbb{R}^{N \times A \times Z}$ feasible for the functional equation with welfare function $\underline{W}$, so, by the conjugate constraint, $V^{h_0}(a, z) \in \partial(B^{h_0}(a)\underline{W})(0)$. For each $h_1 = (a, z)$, let

$$\hat{V}^{h_1} = \begin{cases} V^{h_0}(a, z)/B^{h_0}(a) & \text{if } B^{h_0}(a) > 0, \text{ and} \\ \hat{V}^{h_0} & \text{if } B^{h_0}(a) = 0. \end{cases}$$

Since $V^{h_0}(a, z)/B^{h_0}(a) \in \partial \underline{W}(0)$, by the previous paragraph $V^{h_0}(a, z)/B^{h_0}(a) \in \partial \mathbf{B}(\underline{W})(0)$, therefore we may “unfold” $\hat{V}^{h_1}$ as in the proof of [Lemma A.8](#). Just like that proof, all the incentive constraints hold along the way, so we end up constructing a ppCE with associated continuation payoff profiles, therefore $\hat{V}^{h_0} \in \partial \underline{W}(0)$. $\square$

Step 4: ppRE and Supremum Attainment

The above arguments prove [Theorem 4.1](#) for ppCE. For ppRE, the proof above applies verbatim after replacing the incentive constraints for ppCE with those for ppRE. To prove that the supremum is attained in the primal functional equation, note that any proper W in $\mathcal{P}_U$ is the support function of some nonempty compact, convex subset of U , therefore the V 's belong to a compact, convex set. But the B 's do, too, since they belong to the unit simplex. Linearity of incentive compatibility implies that the feasible set is compact, convex. Continuity of the objective yields a maximum.

This completes the proof of [Theorem 4.1](#).

Proof of [Proposition 4.1](#)

We begin by writing the primal functional equation as a linear program:

$$\begin{aligned} \mathbf{B}(W)(\lambda) = \sup_{B \geq 0, V} \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right] \quad \text{s.t.} \quad \sum_{a \in A} B(a) = 1, \\ (1 - \delta) \sum_{a \sim i} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a \sim i, z)} \Delta \pi(z|a, b_i) V_i(a, z) \leq 0 \quad \forall (i, a_i, b_i), \\ \sum_{i \in N} \mu_i V_i(a, z) - B(a) W(\mu) \leq 0 \quad \forall (a, z, \mu \in \mathbb{R}^N). \end{aligned}$$

This linear program is infinite dimensional because it has infinitely many constraints, i.e., infinitely many welfare-weight vectors μ . Formally, this problem is equivalent to the primal functional equation, since the conjugate constraint there is equivalent to the infinite collection of linear promise-keeping constraints above. There are several ways to obtain the dual of the primal functional equation. One way is to take the dual of the linear programming version above; we follow this approach below.

The right-hand side of the primal constraints above belongs to the real vector space $\mathbb{R} \times \mathbb{R}^{\prod_i A_i \times A_i} \times \mathbb{R}^{A \times Z \times M}$, where $M = \mathbb{R}^N$. From the Lagrangian we get the dual:

$$\begin{aligned} \inf_{y, \varepsilon \geq 0, \mu, \kappa} \quad & \kappa \quad \text{s.t.} \quad \sum_{\mu \in M} \varepsilon(\mu, a, z) \mu_i = \delta [\lambda_i \pi(z|a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta \pi(z|a, b_i)] \quad \forall (i, a, z), \\ \kappa \geq (1 - \delta) \sum_{i \in N} & \left[\lambda_i u_i(a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta u_i(a, b_i) \right] + \sum_{z \in Z} \sum_{\mu \in M} \varepsilon(\mu, a, z) W(\mu) \quad \forall a \in A, \end{aligned}$$

where $\kappa \in \mathbb{R}$, $y \in \mathbb{R}^{\prod_i A_i \times A_i}$ and $\varepsilon \in \mathbb{R}_+^{A \times Y \times (M)}$, where $\mathbb{R}^{(M)}$ denotes the subspace of real-valued functions of M with finitely many non-zero entries. The duality between $\mathbb{R}^M$ and $\mathbb{R}^{(M)}$ is standard (see, e.g., [Conway, 1990](#), p. 115). Since W is positively homogeneous and convex, for any $\hat{\varepsilon} \in \mathbb{R}_+^{(M)}$,

$$\sum_{\mu \in M} \hat{\varepsilon}(\mu) W(\mu) \geq W\left(\sum_{\mu \in M} \hat{\varepsilon}(\mu) \mu\right) = W(\hat{\mu}),$$

where $\hat{\mu} = \sum_{\mu} \hat{\varepsilon}(\mu) \mu \in M$. I.e., since $W \in \mathcal{P}_U$, $\hat{\varepsilon}$ with singleton support has lower value than with a support of $\hat{\varepsilon}$ having more than one vector of welfare weights with the same expectation. Hence, without loss, the infimum above chooses a single vector of welfare weights $\mu(a, z) \in M$ instead of $\varepsilon(a, z) \in \mathbb{R}^{(M)}$ for each (a, z) , so the dual becomes (by positive homogeneity, μ is replaced by $\delta \mu$ and δ is pulled from $W(\delta \mu)$)

$$\begin{aligned} \inf_{y \geq 0, \mu, \kappa} \quad & \kappa \quad \text{s.t.} \quad \mu_i(a, z) = \lambda_i \pi(z|a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta \pi(z|a, b_i) \quad \forall (i, a, z), \\ \kappa \geq (1 - \delta) \sum_{i \in N} & \left[\lambda_i u_i(a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta u_i(a, b_i) \right] + \delta \sum_{z \in Z} W(\mu(a, z)) \quad \forall a \in A. \end{aligned}$$

Finally, since κ is unsigned, without loss this infimum chooses κ equal to the maximum right-hand side with respect to $a \in A$. Hence, the optimization problem becomes

$$\begin{aligned} \inf_{y \geq 0} \max_{a \in A} (1 - \delta) \sum_{i \in N} & \left[\lambda_i u_i(a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta u_i(a, b_i) \right] + \delta \sum_{z \in Z} W(\mu(a, z)) \quad \text{s.t.} \\ \mu_i(a, z) &= \lambda_i \pi(z|a) - \sum_{b_i \in A_i} y_i(a_i, b_i) \Delta \pi(z|a, b_i) \quad \forall (i, a, z). \end{aligned}$$

This is precisely the problem stated in the theorem.

Proof of Theorem 4.2

Assume ppCE first. Given $W \in \mathcal{P}_U$, let the ‘partial domain’ of (B, V) be $\mathcal{K} = \{(B, V) : B \in \Delta A, [B(a)W]^*(V(a, z)) \leq 0 \forall (a, z)\}$. The primal functional equation can be written as the following saddle-point problem given $\lambda \in \mathbb{R}^N$:

$$\begin{aligned} & \max_{(B, V) \in \mathcal{K}} \inf_{y \geq 0} \mathbf{L}(B, V, y), \\ \text{where } \mathbf{L}(B, V, y) &= (1 - \delta) \sum_{(i, a)} \lambda_i u_i(a) B(a) + \delta \sum_{(i, a, z)} \lambda_i \pi(z|a) V_i(a, z) \\ & - \sum_{(i, a_i, b_i)} y_i(a_i, b_i) \left[(1 - \delta) \sum_{a-i} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a-i, z)} \Delta \pi(z|a, b_i) V_i(a, z) \right]. \end{aligned}$$

The function $\mathbf{L}$ is clearly continuous and bi-linear in (B, V) and y . Moreover, the set $\mathcal{K}$ is compact and convex and the domain $\mathbb{R}_+^{\prod_i A_i \times A_i}$ of y is a convex set. Therefore, by the Minimax Theorem (e.g., [Sion, 1958](#)),

$$(*) \quad \max_{(B, V) \in \mathcal{K}} \inf_{y \geq 0} \mathbf{L}(B, V, y) = \inf_{y \geq 0} \max_{(B, V) \in \mathcal{K}} \mathbf{L}(B, V, y).$$

The left-hand side above is equivalent to the right-hand side of the primal functional equation, with value $\underline{\mathbf{B}}(W)(\lambda)$, since it chooses $(B, V) \in \mathcal{K}$ to maximize welfare subject to incentive compatibility, with multipliers y . To complete the proof, we will now show that the right-hand side above is equivalent to the right-hand side of the dual functional equation, with value $\overline{\mathbf{B}}(W)(\lambda)$. To this end, first rearrange $\mathbf{L}$ as

$$\mathbf{L}(B, V, y) = (1 - \delta) \sum_{(i, a)} [\lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a)] B(a) + \delta \sum_{(i, a, z)} [\lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)] V_i(a, z),$$

using the notation of [Proposition 4.1](#). Now, given $y \geq 0$, consider maximizing $\mathbf{L}$ with respect to (B, V) . If W is the support function of the empty set then $\mathcal{K}$ is empty and both sides of $(*)$ take the value of $-\infty$. Otherwise, W is the support function of a nonempty, closed, convex subset of U . Maximizing $\mathbf{L}$ with respect to V gives

$$(1 - \delta) \sum_{(i, a)} [\lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a)] B(a) + \delta \sum_{(i, a, z)} W(\lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)) B(a).$$

Now maximizing with respect to B yields

$$\max_{a \in A} (1 - \delta) \sum_{(i, a)} [\lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a)] + \delta \sum_{(a, z)} W(\mu_i(a, z)),$$

where $\mu_i(a, z) = \lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)$ for every player i . Plugging this back into the RHS of $(*)$ shows that this is the RHS of the dual functional equation, completing the proof of the proposition for ppCE. The proof for ppRE is identical, so omitted.

Proof of Proposition 4.2

We proceed in three steps. First, we use [Assumption 4.1](#) to find a lower bound on welfare. Next, we argue that the welfare potential is monotone in the discount factor. Finally, we put these two steps together to argue that, for δ high enough, any y of unbounded size leads to unbounded value—a contradiction to [Theorem 4.2](#)—because deviation gains become small and welfare changes relatively big when δ is high.

Step 1: Lower Bound on Welfare

We will focus on ppCE; the argument for ppRE is very close, hence omitted. Let $y \geq 0$. By [Assumption 4.1](#), the equilibrium payoff set has nonempty interior for $\delta = \underline{\delta}$. Hence, there is a closed ball $\bar{B}_\rho(\hat{V})$ of radius $\rho > 0$ with center $\hat{V}$ inside $\partial W(0)$. Given welfare weights λ and a dual vector $y \geq 0$, letting $\mu_i(a, z) = \lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)$ for every (i, a, z) , define

$$\underline{W}(\lambda; a, y) := (1 - \delta) \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a) + \delta \sum_{z \in Z} W(\mu(a, z)).$$

Since support functions are monotone (i.e., $F \subset G$ implies $\delta_F^*(\lambda) \leq \delta_G^*(\lambda)$ for all λ),

$$\underline{W}(\lambda; a, y) \geq (1 - \delta) \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a) + \delta \sum_{z \in Z} \delta_{\bar{B}_\rho(\hat{V})}^*(\mu(a, z)).$$

It is easy to see that the support function applied to $\bar{B}_\rho(\hat{V})$ satisfies

$$\delta_{\bar{B}_\rho(\hat{V})}^*(\lambda) = \lambda \cdot \hat{V} + \rho \|\lambda\|$$

for every vector of welfare weights λ , where $\|\lambda\| = (\sum_i \lambda_i^2)^{1/2}$. This is because welfare in the direction λ is maximized on the unit ball at the point $\lambda / \|\lambda\|$. Therefore, the welfare attained at that point equals $\lambda \cdot \lambda / \|\lambda\| = \|\lambda\|$. This implies that

$$\underline{W}(\lambda; a, y) \geq (1 - \delta) \sum_{i \in N} \lambda_i u_i(a) - y_i(a_i) \cdot \Delta u_i(a) + \delta \sum_{z \in Z} [\mu(a, z) \cdot \hat{V} + \rho \|\mu(a, z)\|].$$

Substituting for $\mu(a, z)$, since $\sum_z y_i(a_i) \cdot \Delta \pi_i(z|a) = 0$, clearly

$$\underline{W}(\lambda; a, y) \geq (1 - \delta) \sum_{i \in N} \lambda_i [u_i(a) + \hat{V}_i] - y_i(a_i) \cdot \Delta u_i(a) + \delta \rho \sum_{z \in Z} \|\mu(a, z)\|,$$

where $\|\mu(a, z)\| = (\sum_i [\lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)]^2)^{1/2}$.

Step 2: δ -Monotonicity

Let $\underline{W}_\delta$ be the welfare potential with discount factor δ . In this step, we simply show that $\underline{W}_\delta$ is monotone in δ . Even though this argument is not new, we include it

here for completeness. Let $0 \leq \delta < \delta' < 1$ be two discount factors. Let V_0 be any equilibrium payoff profile with discount factor δ . We will show that if (B, V) is feasible for the right-hand side of the primal functional equation with discount factor δ with value V_0 , there exists (B, V') with the same value that is feasible with discount factor δ' . Indeed, suppose that

$$V_{i0} = (1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a,z)} \pi(z|a) V_i(a, z)$$

for every player i . Consider V' defined pointwise by

$$V'_i(a, z) = \frac{1 - \delta'}{\delta'} \frac{\delta}{1 - \delta} V_i(a, z) + \left(1 - \frac{1 - \delta'}{\delta'} \frac{\delta}{1 - \delta}\right) B(a) V_{i0}.$$

We will show that (B, V') as defined satisfies the prerequisites stipulated above. First, notice that $V'_i(a, z)$ is a convex combination of $V_i(a, z)$ and $B(a) V_{i0}$, since clearly $0 \leq \frac{1 - \delta'}{\delta'} \frac{\delta}{1 - \delta} < 1$. Now, both of these vectors satisfy the conjugate constraints, i.e., $[B(a) \underline{W}_\delta]^*(V_i(a, z)) \leq 0$ and $[B(a) \underline{W}_\delta]^*(B(a) V_{i0}) \leq 0$, therefore their convex combination $V'_i(a, z)$ does, too. Hence, (B, V') satisfies the conjugate constraints with welfare function $\underline{W}_\delta$. Moreover,

$$\begin{aligned} & (1 - \delta') \sum_{a \in A} u_i(a) B(a) + \delta' \sum_{(a,z)} \pi(z|a) V'_i(a, z) \\ &= (1 - \delta') \sum_{a \in A} u_i(a) B(a) + \delta' \sum_{(a,z)} \pi(z|a) \left[\frac{1 - \delta'}{\delta'} \frac{\delta}{1 - \delta} V_i(a, z) + \left(1 - \frac{1 - \delta'}{\delta'} \frac{\delta}{1 - \delta}\right) B(a) V_{i0} \right] \\ &= \frac{1 - \delta'}{1 - \delta} (1 - \delta) \sum_{a \in A} u_i(a) B(a) + \frac{1 - \delta'}{1 - \delta} \delta \sum_{(a,z)} \pi(z|a) V_i(a, z) + \frac{\delta' - \delta}{1 - \delta} V_{i0} \\ &= \frac{1 - \delta'}{1 - \delta} V_{i0} + \frac{\delta' - \delta}{1 - \delta} V_{i0} = V_{i0}. \end{aligned}$$

Finally, (B, V') satisfies incentive compatibility when the discount factor is δ' :

$$\begin{aligned} \sum_{a_{-i}} \Delta u_i(a, b_i) B(a) &\leq \frac{\delta}{1 - \delta} \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(a, z) \\ &= \frac{\delta'}{1 - \delta'} \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V'_i(a, z), \end{aligned}$$

where the inequality follows because (B, V) is incentive compatible for δ by hypothesis and the equality follows by definition of $V'_i(a, z)$ and the fact that $u(a) B(a)$ doesn't depend on z , so $\sum_z \Delta \pi(z|a, b_i) u(a) B(a) = 0$. To recap, we just showed that any feasible pair (B, V) of the right-hand side of the primal functional equation given δ

and $\underline{W}_\delta$ implies a feasible pair (B, V') given δ' with the same welfare function $\underline{W}_\delta$ for the conjugate constraints. Unfolding the primal functional equation indefinitely as in [Theorem 4.1](#)—that is, re-applying the argument above to each $V'(a, z)$ by finding a feasible (B', V'') for the right-hand side of the primal functional equation with value V' , and then also for each V'' , etc.—implies that any feasible solution of the primal sequence problem with δ is also feasible with δ' . Hence, the δ -equilibrium payoff profile V_0 is also a δ' -equilibrium payoff profile, i.e., the set of equilibrium payoff profiles is monotone in δ , thus welfare functions, too.

Step 3: Coerciveness

At the end of Step 1, we concluded that, given weights λ and a dual vector $y \geq 0$,

$$\underline{W}(\lambda; a, y) \geq (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] + \delta \rho \sum_{z \in Z} \|\mu(a, z)\|$$

for every action profile a . By Step 2, this holds for every discount factor $\delta > \underline{\delta}$.

Since the 1-norm is bounded above by $\sqrt{n}$ times the 2-norm,⁴⁸ it follows that

$$\begin{aligned} \underline{W}(\lambda; a, y) &\geq (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] + \frac{\delta \rho}{\sqrt{n}} \sum_{(i, z)} |\mu_i(a, z)| \\ &= (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] + \frac{\delta \rho}{\sqrt{n}} \sum_{(i, z)} |\lambda_i \pi(z|a) - y_i(a_i) \cdot \Delta \pi_i(z|a)| \\ &\geq (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] + \frac{\delta \rho}{\sqrt{n}} \sum_{(i, z)} |y_i(a_i) \cdot \Delta \pi_i(z|a)| - |\lambda_i \pi(z|a)|. \end{aligned}$$

Letting $\underline{W}(\lambda; y) = \max_a \underline{W}(\lambda; a, y)$, it follows that

$$\underline{W}(\lambda; y) \geq \max_{a \in A} (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] - \delta \rho \|\lambda\| + \frac{\delta \rho}{\sqrt{n}} \sum_{(i, z)} |y_i(a_i) \cdot \Delta \pi_i(z|a)|,$$

where we used the fact that $\sum_{(i, z)} |\lambda_i \pi(z|a)| = \|\lambda\|_1 \leq \|\lambda\|_2 \sqrt{n}$ (see [Footnote 48](#)).

By weak detectability ([Assumption 4.1](#)),

$$\max_{a_{-i}} \sum_{z \in Z} |y_i(a_i) \cdot \Delta \pi_i(z|a)| \geq \Delta y_i(a_i) K_i(a_i),$$

where $K_i(a_i)$ is the “ K ” in [Assumption 4.1](#) for each (i, a_i) . Therefore

$$\underline{W}(\lambda; y) \geq \max_{a \in A} (1 - \delta) \sum_{i \in N} [\lambda_i(u_i(a) + \hat{V}_i) - y_i(a_i) \cdot \Delta u_i(a)] - \delta \rho \|\lambda\| + \frac{\delta \rho}{\sqrt{n}} \sum_{i \in N} \Delta y_i(a_i) K_i(a_i).$$

⁴⁸By the Cauchy-Schwartz inequality, clearly $\|x\|_1 = |x| \cdot \mathbf{1} \leq \|x\|_2 \|\mathbf{1}\|_2 = \|x\|_2 \sqrt{n}$.

Collecting terms, since $y_i(a_i) \cdot \Delta u_i(a) \leq \Delta y_i(a_i) 2 \|u\|_\infty$, where $\|u\|_\infty = \max_{(i,a)} |u_i(a)|$,

$$\underline{W}(\lambda; y) \geq \max_{a \in A} (1 - \delta) \sum_{i \in N} \lambda_i [u_i(a) + \hat{V}_i] - \delta \rho \|\lambda\| + \sum_{i \in N} \Delta y_i(a_i) \left[\frac{\delta \rho}{\sqrt{n}} \underline{K} - (1 - \delta) 2 \|u\|_\infty \right],$$

where $\underline{K} = \min_{(i,a_i)} K_i(a_i) > 0$. Now, clearly there exists $\delta < 1$ large enough that

$$\frac{\delta \rho}{\sqrt{n}} \underline{K} > (1 - \delta) 2 \|u\|_\infty.$$

Therefore, $\underline{W}(\lambda; y) \rightarrow +\infty$ as $\Delta y_i(a_i) \rightarrow +\infty$ for some (i, a_i) at such δ .⁴⁹ Denote this δ by δ_λ . At this δ_λ , there exists $\bar{Y}$ such that $\underline{W}(\lambda; y) > \delta_U^*(\lambda)$ whenever $\Delta y_i(a_i) > \bar{Y}$ for some (i, a_i) . This value of $\underline{W}(\lambda; y)$ contradicts, of course, optimality of the right-hand side of the dual functional equation. Therefore, without loss of optimality, we may restrict the set of feasible $y \geq 0$ to ones for which $\Delta y_i(a_i) \leq \bar{Y}$ for all (i, a_i) . In this case, the set of feasible y becomes a compact set (ignoring $y_i(a_i, a_i)$, which of course is irrelevant). Since the dual objective is continuous, by Weierstrass' Theorem the right-hand side of the dual functional equation has an optimal solution, as required. The proof for ppRE is almost identical, so omitted. This completes the proof of [Proposition 4.2](#).

Proof of [Proposition 4.3](#)

Let Q be a compact set of welfare weights. Being compact, it belongs to a closed ball of radius $\eta > 0$ centered at the origin. By the proof of [Proposition 4.2](#),

$$\underline{W}(\lambda; y) \geq \max_{a \in A} (1 - \delta) \sum_{i \in N} \lambda_i [u_i(a) + \hat{V}_i] - \delta \rho \|\lambda\| + \sum_{i \in N} \Delta y_i(a_i) \left[\frac{\delta \rho}{\sqrt{n}} \underline{K} - (1 - \delta) 2 \|u\|_\infty \right]$$

for some $\underline{K} > 0$. Recall that $\|u\|_\infty = \max_{(i,a)} |u_i(a)|$; let $\|\hat{V}\|_\infty = \max_i |\hat{V}_i|$. Since clearly $\lambda_i [u_i(a) + \hat{V}_i] \geq -|\lambda_i| (\|u\|_\infty + \|\hat{V}\|_\infty)$ for each i and $\|\lambda\| \leq \eta$, it follows that

$$\underline{W}(\lambda; y) \geq \max_{a \in A} -\eta \left[(1 - \delta) (\|u\|_\infty + \|\hat{V}\|_\infty) + \delta \rho \right] + \sum_{i \in N} \Delta y_i(a_i) \left[\frac{\delta \rho}{\sqrt{n}} \underline{K} - (1 - \delta) 2 \|u\|_\infty \right].$$

Again, following the proof of [Proposition 4.2](#), by [Assumption 4.1](#) there is a large enough $\bar{Y}$ that for $\Delta y_i(a_i) > \bar{Y}$ the right-hand side of this inequality exceeds $\max_{\|\lambda\| \leq \eta} \delta_U^*(\lambda)$, implying suboptimality outside of $\Delta y_i(a_i) \leq \bar{Y}$. Since this $\bar{Y}$ only depends on λ through η , it follows that, for every $\lambda \in Q$, the set of optimal dual solutions belong to the compact set of $y \geq 0$ such that $\sum_{b_i} y_i(a_i, b_i) \leq \bar{Y}$ for all (i, a_i) . Hence, the set of dual solutions is uniformly bounded for all λ in the compact set Q , as claimed.

⁴⁹This phenomenon is known as “coerciveness.”

Proof of Lemma 4.1

By Proposition 4.3, if Assumption 4.1 holds then for any compact set of welfare weights λ , as long as players are sufficiently patient, $y^*(\lambda)$ is uniformly bounded with respect to λ . By the proof of Proposition 4.3, if the compact set of welfare weights is the unit ball (i.e., $\|\lambda\|_u = 1$, where $\|\lambda\|_u = \max_i |\lambda_i|$), then there exists $M < +\infty$ such that $\Delta y_i^*(a_i; \lambda) \leq M$ for all (i, a_i) , where $\Delta y_i^*(a_i; \lambda)$ was defined in Section 4.3. By positive homogeneity, if $\|\lambda\|_u \neq 0$ then $y^*(\lambda) = \|\lambda\|_u y^*(\lambda / \|\lambda\|_u)$, so, in general, $\Delta y_i^*(a_i; \lambda) \leq \|\lambda\|_u M$. Hence, $|Y^*|_0^* = \max_{(i, a_i, b_i)} Y_i^*(h_0; a_i, b_i) \leq \|\lambda\|_u M$, where, without loss of optimality, we normalize $y_i^*(a_i, a_i; \mu) = 0$ for all (i, a_i) and welfare weights μ . Since by definition $\Lambda_i^*(a, z) = \delta[\Lambda_i^*(h_0)\pi(z|a) - Y_i^*(h_0, a_i) \cdot \Delta\pi_i(z|a)]$,

$$\begin{aligned} |\Lambda^*|_1^* &= \max_{a \in A} \sum_{z \in Z} \max_{i \in N} \delta |\Lambda_i^*(h_0)\pi(z|a) - Y_i^*(h_0, a_i) \cdot \Delta\pi_i(z|a)| \\ &\leq \max_{a \in A} \sum_{z \in Z} \delta \max_{i \in N} |\lambda_i| \pi(z|a) + \|\lambda\|_u M \sum_{b_i \neq a_i} [\pi_i(z|b_i, a_{-i}) + \pi(z|a)] \\ &\leq \delta \max_{a \in A} \max_{i \in N} |\lambda_i| + 2 \|\lambda\|_u M (|A_i| - 1) \\ &\leq \delta[1 + 2M(|A| - 1)] |\Lambda^*|_0^* \end{aligned}$$

where $|A_i|$ and $|A|$ are the cardinality of the sets A_i and A , respectively. Similarly,

$$\begin{aligned} |Y^*|_1^* &= \max_{a_1} \sum_{z_1} \max_{(i, a_i, b_i)} Y_i^*(a_1, z_1; a_i, b_i) \leq \max_{a_1} \sum_{z_1} \|\Lambda^*(a_1, z_1)\| M \\ &\leq \max_{a_1} \sum_{z_1} \max_{i \in N} |\Lambda_i^*(a_1, z_1)| M \leq M |\Lambda^*|_1^*, \end{aligned}$$

therefore, by the same argument as above,

$$|\Lambda^*|_2^* \leq \delta[1 + 2M(|A| - 1)] |\Lambda^*|_1^*.$$

Continuing in this fashion gives the dual semi-norm bounds $|Y^*|_{t-1}^* \leq M |\Lambda^*|_{t-1}^*$ and $|\Lambda^*|_t^* \leq \delta[1 + 2M(|A| - 1)] |\Lambda^*|_{t-1}^*$ for all $t \in \mathbb{N}$. Letting $g = 1/[1 + 2M(|A| - 1)]$, which is clearly in $(0, 1]$, it follows that $|\Lambda^*|_{t-1}^* \leq (\delta/g)^{t-1} \|\lambda\|_u$ for all $t \in \mathbb{N}$, so

$$\|\Lambda^*\|_{1,g}^* = \sum_{t \in \mathbb{N}} g^{t-1} |\Lambda^*|_{t-1}^* \leq \sum_{t \in \mathbb{N}} g^{t-1} (\delta/g)^{t-1} \|\lambda\|_u = \|\lambda\|_u / (1 - \delta) < +\infty.$$

The above inequalities imply $|Y^*|_{t-1}^* \leq (\delta/g)^{t-1} M \|\lambda\|_u$. Hence,

$$\|Y^*\|_{1,g}^* = \sum_{t \in \mathbb{N}} g^{t-1} |Y^*|_{t-1}^* \leq \sum_{t \in \mathbb{N}} g^{t-1} (\delta/g)^{t-1} M \|\lambda\|_u = M \|\lambda\|_u / (1 - \delta) < +\infty.$$

Finally, consider W^* . First, let $|W^{**}| = \sup_{\mu} \{|\underline{W}(\mu)| : \|\mu\|_u = 1\}$. By construction, $|W^*|_0^* = |\underline{W}(\lambda)| = \|\lambda\|_u |\underline{W}(\lambda/\|\lambda\|_u)| \leq \|\lambda\|_u |\underline{W}^{**}| < +\infty$, where now $\lambda/\|\lambda\|_u$ belongs to the unit ball of welfare weights. Next,

$$\begin{aligned}
|W^*|_{t-1}^* &= \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} |W^*(h_{t-1})| \\
&= \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} |\underline{W}(\Lambda^*(h_{t-1}))| \\
&\leq \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} |\underline{W}(\Lambda^*(h_{t-1}))| \\
&\leq \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} |\underline{W}(\Lambda^*(h_{t-1})/\max_i |\Lambda_i^*(h_{t-1})|)| \max_i |\Lambda_i^*(h_{t-1})| \\
&\leq \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} |\underline{W}^{**}| \max_i |\Lambda_i^*(h_{t-1})| \\
&\leq |\underline{W}^{**}| \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} \max_i |\Lambda_i^*(h_{t-1})| \\
&= |\underline{W}^{**}| |\Lambda^*|_{t-1}^* = |\underline{W}^{**}| (\delta/g)^{t-1} \|\lambda\|_u.
\end{aligned}$$

Therefore,

$$\|W^*\|_{1,g}^* = \sum_{t \in \mathbb{N}} g^{t-1} |W^*|_{t-1}^* \leq \sum_{t \in \mathbb{N}} g^{t-1} (\delta/g)^{t-1} |\underline{W}^{**}| \|\lambda\|_u = \frac{|\underline{W}^{**}| \|\lambda\|_u}{1 - \delta} < +\infty.$$

This completes the proof. $\square$

Proof of Theorem 4.3

The proof proceeds in two steps. First, we derive the g -dual sequence problem. Then we establish the claimed duality ‘trifecta’ (strong duality, dual attainment). In this step, we rely on the notation developed for the proof of Proposition 3.1. There, for the primal sequence problem, the space of primal variables was denoted by $\mathbf{P}$ and the space of right-hand side constraints by $\mathbf{Q}$. The operator $M : \mathbf{P} \rightarrow \mathbf{Q}$ described its linear constraints. There we showed that M was a continuous linear operator in the weak* topologies with respect to the ‘unweighted’ spaces $\mathbf{D}$ and $\mathbf{E}$ of dual variables and right-hand side dual constraints.

First, let $\mathbf{D}_g = \hat{\ell}_{1,g}^* \times \hat{\ell}_{1,g}^{*N} \times \hat{\ell}_{1,g}^{*\prod_i A_i \times A_i}$ be the space of *dual variables* and positive cone $\mathbf{D}_{g+} = \hat{\ell}_{1,g}^* \times \hat{\ell}_{1,g}^{*N} \times \hat{\ell}_{1,g+}^{*\prod_i A_i \times A_i}$. (Only the third family of variables is non-negative.) The space of *dual constraints* is $\mathbf{E}_g = \hat{\ell}_{1,g}^{*A} \times \hat{\ell}_{1,g}^{*N}$, with positive cone $\mathbf{E}_{g+} = \hat{\ell}_{1,g+}^{*A} \times \{0\}$, where $0 \in \hat{\ell}_{1,g}^{*N}$. We place the spaces $\mathbf{P}$ and $\mathbf{E}_g$ as well as the spaces $\mathbf{Q}$ and $\mathbf{D}_g$

in dual pairs with respect to $\langle \cdot, \cdot \rangle_g$. Thus, $\mathbf{P}$ inherits the weak* topology from $\mathbf{E}_g$ under $\langle \cdot, \cdot \rangle_g$, and similarly with $\mathbf{Q}$ and $\mathbf{D}_g$. We endow each quotient space in $\mathbf{D}_g$ and $\mathbf{E}_g$ their respective norm topology, and the product with the absolute sum norm. Therefore, $\|D\|_{1,g}^* = \|W\|_{1,g}^* + \|\Lambda\|_{1,g}^* + \|Y\|_{1,g}^*$ whenever $D = (W, \Lambda, Y) \in \mathbf{D}_g$. Similarly, $\|E\|_{1,g}^* = \|F\|_{1,g}^* + \|G\|_{1,g}^*$ whenever $E = (F, G) \in \mathbf{E}_g$.

Next, we define the adjoint M_g^* of M relative to $\langle \cdot, \cdot \rangle_g$. The operator $M_g^* : \mathbf{D}_g \rightarrow \mathbf{E}_g$ takes a vector of dual variables $(W, \Lambda, Y) \in \mathbf{D}_g$ and maps it into a vector of dual constraints $(F, G) \in \mathbf{E}_g$ such that, for every (i, h, a) ,

$$\begin{aligned} F(h, a) &= W(h) - (1 - \delta) \sum_{i \in N} [\Lambda_i(h) u_i(a) - Y_i(h, a_i) \Delta u_i(a)] - g \sum_{z \in Z} W(h, a, z), \\ G_i(h_0) &= \Lambda_i(h_0), \quad G_i(h, a, z) = \Lambda_i(h, a, z) - (\delta/g) [\Lambda_i(h) \pi(z|a) - Y_i(h, a_i) \Delta \pi_i(z|a)]. \end{aligned}$$

The operator $M_g^* : \mathbf{D}_g \rightarrow \mathbf{E}_g$ is defined pointwise by $M_g^*(W, \Lambda, Y) = (F, G)$.

LEMMA A.10. *The operator M_g^* is the adjoint of M with respect to $\langle \cdot, \cdot \rangle_g$, i.e.,*

$$\langle MP, D \rangle_g = \langle P, M_g^* D \rangle_g$$

for every $P \in \mathbf{P}$ and $D \in \mathbf{D}_g$.

Proof. If $P = (C, V) \in \mathbf{P}$ and $D = (W, \Lambda, Y) \in \mathbf{D}_g$ are vectors of primal and g -dual variables, and $\tau(h_{t-1}) = t - 1$ for every $h_{t-1} \in H$,

$$\begin{aligned} \langle MP, D \rangle_g &= W(h_0) \sum_{a \in A} C(a; h_0) + \sum_{(h, a_t, z_t)} g^{\tau(h)+1} W(h, a_t, z_t) \left[\sum_{a \in A} C(a; h, a_t, z_t) - C(a_t; h) \right] \\ &\quad + \sum_{i \in N} \sum_{h \in H} g^{\tau(h)} \Lambda_i(h) [V_i(h) - (1 - \delta) \sum_{a \in A} u_i(a) C(a; h) - \delta \sum_{(a, z)} \pi(z|a) V_i(h, a, z)] \\ &\quad + \sum_{i \in N} \sum_{(h, a_i, b_i)} g^{\tau(h)} Y_i(h, a_i, b_i) (1 - \delta) \left[\sum_{a-i} \Delta u_i(a, b_i) C(a; h) + \delta \sum_{(a-i, z)} \Delta \pi(z|a, b_i) V_i(h, a, z) \right] \\ &= \sum_{i \in N} V_i(h_0) \Lambda_i(h_0) + \sum_{(h, a)} g^{\tau(h)} C(a; h) \left[W(h) - g \sum_{z \in Z} W(h, a, z) \right. \\ &\quad \left. - (1 - \delta) \sum_{i \in N} \left[\Lambda_i(h) u_i(a) - \sum_{b_i \in A_i} Y_i(h, a_i, b_i) \Delta u_i(a, b_i) \right] \right] \\ &\quad + \sum_{i \in N} \sum_{(h, a, z)} g^{\tau(h)+1} V_i(h, a, z) \left[\Lambda_i(h, a, z) - \delta g^{-1} \left[\Lambda_i(h) \pi(z|a) - \sum_{b_i \in A_i} Y_i(h, a_i, b_i) \Delta \pi(z|a, b_i) \right] \right] \\ &= \langle (C, V), M_g^*(W, \Lambda, Y) \rangle_g = \langle P, M_g^* D \rangle_g. \end{aligned}$$

Hence, the g -dual sequence problem is the linear program of [Definition 4.4](#). $\square$

To establish the duality trifecta, it remains to show that M is a (weak*-to-weak*) continuous linear operator. For this, it suffices to show that M_g^* (norm-to-norm) continuous.⁵⁰ This is relatively straightforward.

LEMMA A.11. M_g^* is bounded.

Proof. The proof below is close to that of [Lemma A.2](#). Given $(W, \Lambda, Y) \in \mathbf{D}_g$, consider $(F, G) = M_g^*(W, \Lambda, Y)$. Letting $\|u\| = \max_{i,a} |u_i(a)|$ and $\|\Delta u\| = \max_{i,a,b_i} |\Delta u_i(a, b_i)|$,

$$\begin{aligned} |F|_{t-1}^* &= \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} \max_{a \in A} |F(h_{t-1}, a)| \\ &= \max_{a_1} \sum_{z_1} \cdots \max_{a_{t-1}} \sum_{z_{t-1}} \max_{a \in A} |W(h_{t-1}) - \\ &\quad (1 - \delta) \sum_{i \in N} [\Lambda_i(h_{t-1}) u_i(a) - Y_i(h_{t-1}, a_i) \Delta u_i(a)] - g \sum_{z \in Z} W(h_{t-1}, a, z)| \\ &\leq |W|_{t-1}^* + (1 - \delta)n(|\Lambda|_{t-1}^* \|u\| + |Y|_{t-1}^* \|\Delta u\|) + g |W|_t^* \end{aligned}$$

for every $t \in \mathbb{N}$, therefore

$$\begin{aligned} \|F\|_{1,g}^* &\leq \sum_{t \in \mathbb{N}} g^{t-1} [|W|_{t-1}^* + (1 - \delta)n(|\Lambda|_{t-1}^* \|u\| + |Y|_{t-1}^* \|\Delta u\|) + g |W|_t^*] \\ &\leq 2 \|W\|_{1,g}^* + (1 - \delta)n \|u\| \|\Lambda\|_{t-1}^* + \|\Delta u\| \|Y\|_{1,g}^* \\ &\leq K \|(W, \Lambda, Y)\|_{1,g}^* < +\infty, \end{aligned}$$

where $K = \max\{2, (1 - \delta)n \|u\|, \|\Delta u\|\}$. Hence, the F -coordinate of M_g^* is bounded.

For the G -coordinate, by definition $|G|_0^* = \|\lambda\|_u$ and

$$\begin{aligned} |G|_t^* &= \max_{a_1} \sum_{z_1} \cdots \max_{a_t} \sum_{z_t} \max_{i \in N} |G_i(h_t)| \\ &= \max_{a_1} \sum_{z_1} \cdots \max_{a_t} \sum_{z_t} \max_{i \in N} |\Lambda_i(h_{t-1}, a_t, z_t) - \\ &\quad (\delta/g)[\Lambda_i(h_{t-1})\pi(z_t|a_t) - Y_i(h_{t-1}, a_{it})\Delta\pi_i(z_t|a_t)]| \\ &\leq |\Lambda|_t^* + (\delta/g)(|\Lambda|_{t-1}^* + 2|Y|_{t-1}^*) \end{aligned}$$

⁵⁰See, e.g., [Conway \(1990, Proposition VI.1.3, p. 167\)](#).

for every $t \in \mathbb{N}$, therefore

$$\begin{aligned} \|G\|_{1,g}^* &\leq |\Lambda|_0^* + \sum_{t \in \mathbb{N}} g^t [|\Lambda|_t^* + (\delta/g)(|\Lambda|_{t-1}^* + 2|Y|_{t-1}^*)] \\ &\leq (1 + \delta) \|\Lambda\|_{1,g}^* + 2\delta \|Y\|_{1,g}^* \\ &\leq (1 + \delta) \|(W, \Lambda, Y)\|_{1,g}^* < +\infty, \end{aligned}$$

so the G -coordinate is bounded, too. Finally,

$$\|M_g^*(W, \Lambda, Y)\|_{1,g}^* \leq (1 + \delta + K) \|(W, \Lambda, Y)\|_{1,g}^*,$$

so M_g^* is a bounded operator, as was claimed. $\square$

By [Lemma A.11](#), M_g^* is bounded, therefore M is a continuous linear operator. This implies that the standard duality theory applies to the primal sequence problem and its g -dual counterpart. For example, both the g -dual has a solution and there is no gap if and only if the primal sequence problem is subdifferentiable. (See, e.g., [Gretsky et al. \(2002\)](#).) In particular, if a solution to the primal sequence problem exists, by weak duality the g -dual is bounded, and by the hypotheses of [Lemma 4.1](#) the g -dual has a feasible solution, D^* . This D^* is in fact a solution of the g -dual, since its value coincides with that of the primal sequence problem by duality of the functional equations and the construction of D^* . Therefore, in addition, there is no duality gap, so the primal sequence problem is subdifferentiable. $\square$

Proof of [Proposition 5.1](#)

Recall the incentive constraints for ppCE:

$$(1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) B(a) + \delta \sum_{(a_{-i}, z)} \Delta \pi(z|a, b_i) V_i(a, z) \leq 0 \quad \forall (i, a_i, b_i).$$

Since the game has sum structure, without loss of generality the first term on the left-hand side may be rewritten as $(1 - \delta) \Delta u_i(a_i, b_i) B(a_i)$, where $B(a_i) = \sum_{a_{-i}} B(a)$. Moreover, since the game has product structure, the second term may be rewritten as $\delta \sum_{z_i} \Delta \pi_i(z_i|a_i, b_i) V_i(a_i, z_i)$, where $V_i(a_i, z_i) = \sum_{(a_{-i}, z_{-i})} \pi_{-i}(z_{-i}|a_{-i}) V_i(a, z)$ and $\pi_{-i}(z_{-i}|a_{-i}) = \prod_{j \neq i} \pi_j(z_j|a_j)$. Therefore, making B a public randomization versus a private randomization does not affect incentives. This implies that ppRE outcomes coincide with ppCE outcomes. $\square$

Proof of Theorem 6.1

We prove the result in two steps. First, we recall Kleene's Fixed Point Theorem. Then we established its key continuity assumption to deduce the theorem.

Step 1: Kleene's Fixed Point Theorem

I follow Baranga (1991) in stating Kleene's Fixed Point Theorem (Kleene, 1952) as Theorem A.2 below. Let $(P, \leq)$ be a partially ordered set (or poset) and

$$\bigwedge_{n \in \mathbb{N}} W_n$$

the greatest lower bound, if it exists, of an increasing sequence $\{W_n\}$ in P .

P is *sequentially complete* if every decreasing sequence $\{W_n\}$ of elements in P has a greatest lower bound in P . Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be two posets. A function $f : P \rightarrow Q$ is *sequentially continuous* if for every decreasing sequence $\{W_n\}$ in P with $\bigwedge_n W_n$ in P , the infimum $\bigwedge_n f(W_n)$ exists in Q and $f(\bigwedge_n W_n) = \bigwedge_n f(W_n)$. Clearly, a sequentially continuous function is monotone.

THEOREM A.2 (KLEENE). *Let $(P, \leq)$ be a sequentially complete poset and $f : P \rightarrow P$ a sequentially continuous function. If $W \in P$ and $f(W) \leq W$ then $\underline{W} := \bigwedge_n f^n(W)$ is a fixed point of f . Moreover, $\underline{W} \leq W$, and for all $W' \in P$, if $W' \leq W$ and $W' \leq f(W')$, then $W' \leq \underline{W}$. That is, $\underline{W}$ is the greatest fixed point of f in $\{W' \in P : W' \leq W\}$.*

Step 2: Proving Sequential Continuity

The proof relies on results from the proof of Theorem 4.1. By Lemma A.7, the Bellman operator $\underline{\mathbf{B}} : \mathcal{P}_U \rightarrow \mathcal{P}_U$ is monotone, therefore $W_n \leq W_{n-1}$ for all $n \in \mathbb{N}$. By Theorem A.2, to prove Theorem 6.1 it suffices to show that $\underline{\mathbf{B}}$ is sequentially continuous. Let $\mathcal{F}_n = \mathcal{F}(W_n)$ denote the feasible set of the right-hand side of the primal functional equation when the conjugate constraint uses the support function W_n . As was argued in the proof of Theorem 4.1, the set $\mathcal{F}_n$ is compact and convex. From the promise-keeping constraints, $\sum_i \mu_i V_i(a, z) \leq B(a)W_n(\mu)$ for every (a, z, μ) , therefore, since $\{W_n\}$ is decreasing, so is $\mathcal{F}_n$, i.e., $\mathcal{F}_n \subset \mathcal{F}_{n-1}$ for all $n \in \mathbb{N}$. Let $\underline{W} = \bigwedge_n W_n$ and $\underline{\mathcal{F}} = \mathcal{F}(\underline{W})$ be the feasible set implied by $\underline{W}$. By definition, $W(\mu) = \inf_n W_n(\mu)$ for every vector μ of welfare weights. But $\sum_i \mu_i V_i(a, z) \leq B(a)\underline{W}(\mu)$ if and only if $\sum_i \mu_i V_i(a, z) \leq B(a)W_n(\mu)$ for every $n \in \mathbb{N}$, therefore $\underline{\mathcal{F}} = \bigcap_n \mathcal{F}_n$, since the other constraints stay the same. Now, since each $\mathcal{F}_n$ is compact, their intersection $\underline{\mathcal{F}}$ is compact, too. If the intersection is empty, then—since the intersection of nested nonempty compact sets is nonempty and compact—there must be some n such that $\mathcal{F}$ is empty, in which case the equation $\underline{\mathbf{B}}(\bigwedge_n W_n) = \bigwedge_n \underline{\mathbf{B}}(W_n) = -\infty$ holds.

Next, assume that the intersection is nonempty. The objective of the primal functional equation is linear in (B, V) , hence continuous. Denote the objective by w , thus

$$w(B, V) = \sum_{i \in N} \lambda_i \left[(1 - \delta) \sum_{a \in A} u_i(a) B(a) + \delta \sum_{(a, z)} \pi(z|a) V_i(a, z) \right].$$

It is easy to see that

$$\max_{(B, V) \in \underline{\mathcal{F}}} w(B, V) = \lim_{n \rightarrow \infty} \max_{(B, V) \in \mathcal{F}_n} w(B, V).$$

Indeed, on the one hand $\underline{\mathcal{F}} \subset \mathcal{F}_n$ for each n implies that $\max_{\underline{\mathcal{F}}} w \leq \max_{\mathcal{F}_n} w$ for each n . Since each right-hand side problem entails a continuous objective on a compact set, it has a solution; let (B_n, V_n) be such a solution. By compactness of $\mathcal{F}_1$, the sequence $\{(B_n, V_n)\}$ has a convergent subsequence with limit $(\underline{B}, \underline{V})$. Since the sequence $\{\mathcal{F}_n\}$ is nested, every cluster point belongs to every $\mathcal{F}_n$, hence to $\underline{\mathcal{F}}$, too. By continuity of w , the values of the right-hand side objective converge: $w(B_n, V_n) \rightarrow w(\underline{B}, \underline{V})$. Since λ was arbitrary, sequential continuity is proved. $\square$

Proof of Claim 6.1

We will show something more general than the claim: the sum of players' payoffs is less than or equal to 1 in a minor generalization of ppRE, along the lines of [Fudenberg et al. \(1994\)](#). A set $\mathcal{W} \subset U$ is *self generated* if for every payoff profile $V_0 \in \mathcal{W}$ there exists $B \in \Delta(A)$ and continuation payoffs $V \in \mathbb{R}^{N \times Z}$ such that (1) (B, V) attains the value v : for every player i , $V_{i0} = (1 - \delta)u_i(B) + \delta[p(B)V_{i+} + q(B)V_{i-}]$, and (2) w enforces B : $(1 - \delta) \sum_{a_{-i}} \Delta u_i(a, b_i) B(a) + \delta \sum_{a_{-i}} \Delta q(a, b_i) B(a) [V_{i+} - V_{i-}] \leq 0$ for every player i and $a_i, b_i \in \{c_i, d_i\}$, where $p(B) = \sum_a p(a) B(a)$, $q = 1 - p$, and $p(a)$ is the probability of the positive signal ($p(c_1, c_2) = \frac{3}{4}$, etc.), therefore $\Delta p(a, b_i) = p(b_i, a_{-i}) - p(a) = -\Delta q(a, b_i)$. Note that in this proof we use continuation values, not probability-weighted ones.

For a contradiction, let $\mathcal{E}$ be the set of equilibrium payoff profiles (i.e., the largest self generated set) and say $\gamma = \max\{v_1 + v_2 : (v_1, v_2) \in \mathcal{E}\} > 1$. In any equilibrium payoff profile with welfare γ , there must be mutual cooperation among players at some point, thus $B(c_1, c_2) > 0$. Letting $\Delta q(a_i) = \sum_{a_{-i}} \Delta q(a, b_i) B(a) / B(a_i)$, by hypothesis $\Delta q(c_i) = \frac{1}{4}[B(c_i, c_j) + B(c_i, d_j)] / B(c_i) = \frac{1}{4}$, so incentive compatibility implies that

$$(0) \quad V_{i+} - V_{i-} \geq 4 \frac{1 - \delta}{\delta}, \quad \text{or} \quad V_{i-} \leq V_{i+} - 4 \frac{1 - \delta}{\delta}.$$

By definition, γ is the greatest equilibrium welfare. By self generation, continuation values are equilibrium payoffs, too, so $V_{1+} + V_{2+} \leq \gamma$. Using (0),

$$\begin{aligned} \gamma = v_1 + v_2 &= (1 - \delta)[u_1(B) + u_2(B)] + \delta[p(B)(V_{1+} + V_{2+}) + q(B)(V_{1-} + V_{2-})] \\ &\leq u_1(B) + u_2(B) - 4q(B). \end{aligned}$$

From the stage game payoffs, $u_1(B) + u_2(B) \leq 2$. Since $q(B) \geq 1/4$ for all B , it follows that $\gamma \leq 1$, a contradiction. $\square$

Proof of Proposition 6.1

Ignoring the retirement option for the moment, the Lagrangian for the right-hand side of the principal's problem equals

$$\begin{aligned} \mathcal{L} &= (1 - \delta)(\mu - c) + \delta[pV(U_+) + qV(U_-)] + y[\delta(p - \tfrac{1}{2})(U_+ - U_-) - (1 - \delta)h] \\ &\quad + \lambda[(1 - \delta)[u(c) - h] + \delta(pU_+ + qU_-) - U] \\ &= (1 - \delta)[\mu + \lambda(u(c) - c/\lambda) - (\lambda + y)h] \\ &\quad + \delta[p[V(U_+) + (\lambda + y^{\frac{p-1/2}{p}})U_+] + q[V(U_-) + (\lambda - y^{\frac{p-1/2}{q}})U_-]] - \lambda U. \end{aligned}$$

The planner's problem first minimizes $\mathcal{L}$ with respect to $y \geq 0, \lambda$ and then maximizes it with respect to $c, U_{\pm} \geq 0$. Maximizing first gives

$$\begin{aligned} \mathcal{L}^* &= (1 - \delta)[\mu + \lambda[u^*(-1/\lambda) - (\lambda + y)h] \\ &\quad + \delta[pV^*(\lambda + y^{\frac{p-1/2}{p}}) + q[V^*(\lambda - y^{\frac{p-1/2}{q}})]] - \lambda U. \end{aligned}$$

Adding λU to both sides of the planner's problem, the left-hand side clearly becomes $V(U) + \lambda U$. Instead of choosing λ and taking U as given, we take λ as given and choose $U^*(\lambda) \in \partial V^*(\lambda)$, therefore $V^*(\lambda) = V(U^*(\lambda)) + \lambda U^*(\lambda)$. This gives the left-hand side of the dual functional equation as stated in the proposition. Finally, $\hat{V}_0^*(\lambda) = \sup\{V_0(U) + \lambda U : U \in \partial V^*(\lambda)\}$ emerges from the optimization in $V^*(\lambda)$ and having added λU to both sides of the planner's problem, hence, too, to the value of retiring the agent. $\square$

Proof of Lemma 6.1

Follows immediately from Gale (1967) and Gretskey et al. (2002). $\square$

Proof of Proposition 6.2

By definition of the subdifferential (using the notation of the proof of Theorem 4.3)

$$\partial W(\lambda; 0) = \{E \in \mathbf{E}_g : W(\lambda, \varepsilon) \leq W(\lambda) + \langle \varepsilon, E \rangle_g \ \forall \varepsilon \in \hat{\ell}_{\infty}^{\prod_i A_i \times A_i}\},$$

where we are assuming ppCE. Basic duality logic implies that if $D^* = (W^*, \Lambda^*, Y^*)$ solves the g -dual sequence problem then Y^* belongs to this subdifferential, since

$$\begin{aligned} \underline{W}(\lambda; \varepsilon) &\leq \sum_{i \in N} \lambda_i V_i^\varepsilon(h_0) + \langle D^*, Q^\varepsilon - M(C^\varepsilon, V^\varepsilon) \rangle_g \\ &\leq \sum_{i \in N} \lambda_i V_i^0(h_0) + \langle D^*, Q^0 - M(C^0, V^0) \rangle_g + \langle D^*, Q^\varepsilon - Q^0 \rangle_g \\ &= \underline{W}(\lambda) + \langle Y^*, \varepsilon \rangle_g, \end{aligned}$$

where $(C^\varepsilon, V^\varepsilon)$ are solutions of the perturbed sequence problem with right-hand side Q^ε and (C^0, V^0) are solutions of the primal sequence problem with the usual right-hand side Q^0 . Indeed, the first inequality follows because D^* is not chosen to minimize the g -dual of the perturbed sequence problem, and the second inequality follows because (C^0, V^0) is optimal when $\varepsilon = 0$. Therefore,

$$\underline{W}(\lambda; \varepsilon) - W(\lambda) \leq \langle Y^*, \varepsilon \rangle_g \leq \|\varepsilon\|_\infty \|Y^*\|_{1,g}^*.$$

This proves the result for ppCE. The proof for ppRE is identical, hence omitted. $\square$

Proof of Proposition 7.1

From the martingale equation of Proposition 4.1, for each $dt > 0$, it follows that $d\lambda_i(a, z) = \lambda'_i(a, z) - \lambda_i = -y_i(a_i) \cdot \Delta\pi_i(z|a)/\pi(z|a)$. Substitute the random walk representation and take a second-order Taylor series expansion:

$$\begin{aligned} \frac{\Delta\pi(z|a, b_i)}{\pi(z|a)} &= \frac{2^{-d}[1 + \sum_j (\gamma_j(b_i, a_{-i})/\sigma_j^2) z_j dh_j] - 2^{-d}[1 + \sum_j (\gamma_j(a)/\sigma_j^2) z_j dh_j]}{2^{-d}[1 + \sum_j (\gamma_j(a)/\sigma_j^2) z_j dh_j]} \\ &= \frac{\sum_j (\Delta\gamma_j(a, b_i)/\sigma_j^2) z_j dh_j}{1 + \sum_j (\gamma_j(a)/\sigma_j^2) z_j dh_j} \\ &\approx \sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} z_j dh_j - \left(\sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} z_j dh_j \right) \left(\sum_{j=1}^d \frac{\gamma_j(a)}{\sigma_j^2} z_j dh_j \right) \\ &= \sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} dS_j - \sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} \frac{\gamma_j(a)}{\sigma_j^2} \sigma_j^2 dt - \sum_{j \neq k} \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} \frac{\gamma_k(a)}{\sigma_k^2} dS_j dS_k \\ &= \sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} [dS_j - \gamma_j(a) dt] - \sum_{j \neq k} \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} \frac{\gamma_k(a)}{\sigma_k^2} dS_j dS_k. \end{aligned}$$

To complete the derivation of the law for welfare weights, we will show that the second term in the last equation above vanishes asymptotically and $\Delta\pi(z|a, b_i)/\pi(z|a)$ converges to the first term in distribution.

First, it is easy to see that $E[dS_j dS_k] = 0$ whenever $j \neq k$, since

$$\begin{aligned} E[dS_j dS_k] &= P(z_j = z_k) - P(z_j \neq z_k) \\ &= \frac{1}{4} \left([1 + (\gamma_j(a)/\sigma_j^2)dh_j + (\gamma_k(a)/\sigma_k^2)dh_k] \right. \\ &\quad + [1 - (\gamma_j(a)/\sigma_j^2)dh_j - (\gamma_k(a)/\sigma_k^2)dh_k] \\ &\quad - [1 + (\gamma_j(a)/\sigma_j^2)dh_j - (\gamma_k(a)/\sigma_k^2)dh_k] \\ &\quad \left. - [1 - (\gamma_j(a)/\sigma_j^2)dh_j + (\gamma_k(a)/\sigma_k^2)dh_k] \right). \end{aligned}$$

Since $E[dS_j] = dh_j(\frac{1}{2}[1 + (\gamma_j(a)/\sigma_j^2)dh_j] - \frac{1}{2}[1 - (\gamma_j(a)/\sigma_j^2)dh_j]) = \gamma_j(a)dt$, it follows that $E[\Delta\pi(z|a, b_i)/\pi(z|a)] = 0$, so $\text{Var}(\Delta\pi(z|a, b_i)/\pi(z|a)) = E[(\Delta\pi(z|a, b_i)/\pi(z|a))^2]$.

Expanding this variance term yields

$$\begin{aligned} \text{Var} \left(\frac{\Delta\pi(z|a, b_i)}{\pi(z|a)} \right) &= E \left[\left(\sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} [dS_j - \gamma_j(a)dt] \right)^2 \right. \\ &\quad - 2 \left(\sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} [dS_j - \gamma_j(a)dt] \right) \left(\sum_{j \neq k} \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} \frac{\gamma_k(a)}{\sigma_k^2} dS_j dS_k \right) \\ &\quad \left. + \left(\sum_{j \neq k} \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} \frac{\gamma_k(a)}{\sigma_k^2} dS_j dS_k \right)^2 \right]. \end{aligned}$$

Up to order dt , most of the terms in the expression above are higher-order, thus negligible as $dt \rightarrow 0$. The only terms that survive asymptotically are the ones multiplying $(dS_j)^2$, thus

$$\text{Var} \left(\frac{\Delta\pi(z|a, b_i)}{\pi(z|a)} \right) \approx \sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)^2}{\sigma_j^2} dt = \text{Var} \left(\sum_{j=1}^d \frac{\Delta\gamma_j(a, b_i)}{\sigma_j^2} [dS_j - \gamma_j(a)dt] \right).$$

Asymptotic normality of the random walk as $dt \rightarrow 0$ implies that $\Delta\pi(z|a, b_i)/\pi(z|a)$ has the same asymptotic distribution as $\sum_j (\Delta\gamma_j(a, b_i)/\sigma_j^2)[dS_j - \gamma_j(a)dt]$. Collecting terms for each b_i and multiplying by $y_i(a_i)$, the claimed law for λ_i now follows.

It remains to establish the HJB equation for welfare. To do so, we begin with the dual functional equation of [Proposition 4.1](#), which holds for all $dt > 0$:

$$W(\lambda) = \inf_{y \geq 0} \max_{a \in A} (1 - \delta) \sum_{i \in N} [\lambda_i u_i(a) - y_i(a_i) \Delta u_i(a)] + \delta \sum_{z \in Z} \pi(z|a) W(\lambda'(a, z)),$$

where $\lambda'_i(a, z) = \lambda_i - y_i(a_i) \Delta \pi_i(z|a)/\pi(z|a)$ for each (i, a, z) . By [Theorem 4.1](#), this equation has a maximal solution W for each $dt > 0$. Subtracting $\delta W(\lambda)$ from both

sides gives

$$(1 - \delta)W(\lambda) = \inf_{y \geq 0} \max_{a \in A} (1 - \delta) \sum_{i \in N} [\lambda_i u_i(a) - y_i(a_i) \Delta u_i(a)] + \delta E[dW(\lambda)|a, y],$$

where $dW(\lambda) = W(\lambda'(a, z)) - W(\lambda)$. Fix a and y . By Ito's Lemma,

$$dW = \sum_{i \in N} W_i d\lambda_{it} + \frac{1}{2} \sum_{(i,j)} W_{ij} d\lambda_{it} d\lambda_{jt},$$

where $W_i = \frac{\partial W}{\partial \lambda_i}$ and $W_{ij} = \frac{\partial^2 W}{\partial \lambda_i \partial \lambda_j}$. Substituting for the law of motion for λ yields

$$dW = - \sum_{i \in N} W_i y_i(a_i) \Delta \gamma_i(a) \sigma^{-2} [dS - \gamma(a) dt] + \frac{1}{2} \sum_{(i,j)} W_{ij} y_i(a_i) \Delta \gamma_i(a) \sigma^{-2} (y_j(a_j) \Delta \gamma_j(a))^\top dt,$$

since $(dS)(dS)^\top = \sigma^2 dt$. Since $E[dS - \gamma(a) dt] = 0$, the first term above vanishes and

$$E[dW] = \frac{1}{2} \sum_{(i,j)} W_{ij} y_i(a_i) \Delta \gamma_i(a) \sigma^{-2} (y_j(a_j) \Delta \gamma_j(a))^\top dt.$$

Substituting this into the modified expression above, approximating $1 - \delta$ with $r dt$ and δ with 1 for small dt , and dividing by dt gives the claimed HJB equation. $\square$